

A Low Cost and Power Efficient Real-time Traffic based Urban Street Lighting System using Smart Sensors and IoT

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Abstract—Both in highway and urban scenarios, street lighting is essential to improve visibility and lower the accident rate at night. However, it uses a lot of energy, with recent increases in global utilization from 13% to 15%. This enormous worldwide energy use is a result of excessive energy consumption, which annually produces roughly 3900 tons of CO₂ emissions. Additionally, standard constant illumination lighting systems raise electricity costs and may be responsible for as much as 60% of them. Further, according to the India-USAID statistics, by 2027, the usage of streetlights globally is expected to reach about 363 million. Thus, may lead to an increase in power consumption and Carbon components in the future with existing approaches. Therefore, in this paper, an Internet of Things Enabled Smart Street Lighting System for Highway and Urban infrastructures is proposed. To this end, the proposed model makes use of a low-cost, low-power Passive Infrared (PIR) sensor for effective lighting control and has dimming capabilities during any movement of humans or vehicles in or out of sensor range. In addition, a Light Dependent Resistor (LDR) sensor is incorporated into the system for automatic detection of the environment brightness levels and turning on/off the streetlights on roads. For energy-saving lighting options, Sodium Lamps are replaced with Light-Emitting Diodes to lower the environmental effect while ensuring safety. Further, sensor data is analyzed and exchanged with each other over the internet. Finally, the values are stored in the cloud through Thingspeak.

Keywords—PIR Sensor LDR Sensor, Dimming Circuit, LED

I. INTRODUCTION

Automation plays an increasingly very important role in the world economy and in daily life Automatic systems are being preferred over any kind of manual systems. We can also call it an “Auto Intensity Control of Street Light”. Intelligent light sensing refers to public street lighting that takes the motion of vehicles and pedestrians. It is also known as smart adaptive model that switches off in absence of moving objects. Smart LED based street lighting systems are entirely different from classical approaches. The research work shows automatic control of street lights as a result of

which power is saved to some extent. In the scope of industrialization, automation is a step beyond mechanization. Whereas mechanization provided human operators within machinery to assist the users with muscular requirements of work, automation greatly decreases the need for human sensory and mental requirements as well.

At the beginning, street lamps were operated physically by human power, where a control push button is arranged to each of the street lamps known as first generation of the original street light. After that, another method done using high pressure sodium lamp in their system.

Now-a-days, it is seen that the above method is widely used in the country. The method operates by set up an optical control circuit, change the resistance by using of sensitive device to control street lamps light up automatically at dusk and turn off automatically after dawn in the morning. Most of the existing models work on locally independent street lighting systems, i.e., dimming action of the lights is not taken by the pole available. To this end, by considering the recent literature, authors in this paper have proposed decentralized (locally dependent) street lighting system to improve energy efficiency. The major contributions of the proposed model are listed below:

1. Select the different sensors like, integrate LDR, PIR and Dimming circuit. with Arduino controller.
2. Configure LEDs to adjust the lighting system based on ambient light, movement of vehicles and any objects through dynamic programming.
3. Real-Time monitoring of LEDs using the IoT devices and the cloud data.

The remaining part of the paper is categorized into multiple sections. Section II illustrates the complete study of the street lighting systems. Proposed street lighting system is explained in section III. Section IV, demonstrates the results and performance analysis. Finally, section v concludes the proposed model.

II. LITERATURE

There exist numerous studies on street lighting systems that are indispensable in making smart cities [1] [2]. Street

light is one of the main energy consuming factors in any metropolis. Street lights are frequently left on even after day break, squandering a significant amount of electricity. By implementing an automatic system that switches on and off the street lights when the light falls below a predetermined intensity. To this end, authors in [3], have presented and intensity controlled system for street lights using LEDs. Prasad et.al [4], have presented an energy efficient based solar-powered LED streetlight system. It responds to changes in ambient light by automatically adjusting intensity through PWM using a microcontroller and real-time clock. Adaptive street lighting is intended to reduce seasonal effects and maximize energy efficiency. An infrared (IR) sensors is used to detect motion and a programmable logic controller (PLC) to control intensity.

In [5], authors have proposed an autonomous street lighting system based on vehicle movement. In which, the IR sensor activates the LED to glow at full brightness (300mA) when a person or car passes by the street light, and when the obstruction has passed, it switches to a faded mode (40mA), assuring energy-efficient operation. A traffic light control system that recognizes automobiles through image processing. Traditional electronic sensors on sidewalks are replaced by taking a series of photos with a nearby camera. Traffic signal timing is determined by a digital image processing analysis of the sequence and the state of the roads [6]. A Raspberry Pi 3 model is used to construct a street light control system. With the aid of LDR, they employed Python code to regulate the amount of light. A ZigBee remote module was also utilized. Although the architecture of this system is sophisticated, it minimizes power usage [7]

Authors in [8], have presented a solar-powered streetlight control system based on traffic movement. The amount of energy used was reduced by three times, thanks to the use of an Arduino Wi-Fi module IOT module, IR sensors. The microcontroller is instructed to turn on and off the lights in response to the movements of the vehicles via sensors placed on either side of the streets. The solar panel and all other components make this system's installation relatively expensive.

Rudrawar, Omkar, et al [9], have proposed a power electronics-based system for controlling street lights. In this, TRIAC regulates the applied voltage in order to control the light intensity. Authors in [10], have presented a graphene coated LED based street lighting system. The proposed model consists of different sensors and each sensor uses an Arduino-UNO microcontroller to control whether the lighting column is ON or OFF. Therefore, by employing graphene-coated LED and managing the layout, it assists us by increasing energy conservation. In [11], authors have explained an energy efficient street lighting system by replacing conventional sodium lamps with LEDs. Younis et.al [12], have illustrated an automatic IoT enabled street control system in place of conventional manual system. The system includes LDR and IR sensors for signaling, movement of obstacles respectively. Later, the data is sent to controller for analysis and stored in cloud. In [13] [14], authors have designed a Long Range (LoRa) enabled centralized wireless technology based street lighting system. It enables communication up to 5Km in urban areas. However, the LoRa enabled system allows very low data rate and thus increases energy efficiency of the battery.

In [15-16], researchers have proposed a ZigBee and IoT based street lighting model. The sensors used in the design are connected to exchange valuable data with each other over internet. However, the short range of communication is one of the limitation of the model. From the literature [1-16], most of the street lighting systems the communication speed is slow or low and consuming high battery energy. This is due to the fact that, some of the models are not autonomous and few models incorporated multiple sensors consuming most of the battery high energy. In addition, the street lighting models designed particularly for highway or urban or semirural areas. Highway and Urban areas need high speed of communication among the sensors over sensors used for semirural. To this end, to address the speed of communication and energy consumption issues, an IoT enabled street lighting system is proposed in this paper.

III. PROPOSED WORK

In this section, the proposed design model is explained. The model works by employing sensors and a microcontroller (Arduino) to dynamically adjust the intensity lights based on ambient light conditions and detected motion. Here's a step-by-step explanation of how the system operates is mentioned with the help of block diagram shown in figure.1:

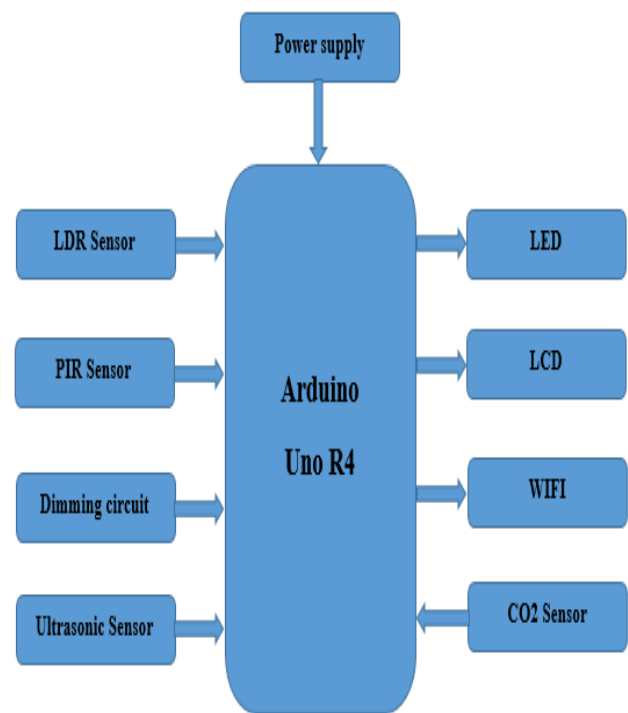


Fig.1. Block diagram of the proposed model

The proposed real time street lighting system makes use of Arduino Uno R4 WiFi, a 32 bit controller. The controller comes with a ESP32 module for Wi-Fi and Bluetooth connectivity. It has distinctive 12X8 LED matrix, enables the entire prototype visuals directly on the board. In addition, the controller offers 48 MHz clock speed and 256 kB flash memory. Further, the Arduino UNO R4 is compatible with cloud platform and IoT devices. It operates on 5V DC supply.

The proposed model consists of LEDs to light streets as an alternative to traditional tungsten and halogen, which require higher power. Then, for high speed switching operation of LEDs, the system is characterized by dimming circuit. LED

with Dimming circuit has the ability to dim light intensity based on traffic conditions on the streets.

From the block diagram, the proposed model includes a photo resistor or LDR sensor to measure the ambient light levels in the environment. LDRs change their resistance based on the intensity of the light falling on them. Low cost and low power consumptions are its advantages. The LDR is connected to one of the analog input pins of the Arduino. The Arduino reads the analog voltage output from the LDR, which is proportional to the ambient light intensity. The Arduino processes the LDR readings and applies a logic algorithm to determine the appropriate intensity level. For the street lights. The algorithm is designed to increase the brightness of the lights during low ambient conditions (e.g., nighttime) and decreases or turn off lights during well-lit conditions.

In addition, an Ultrasonic sensor is also used to get the traffic density information. It detects the vehicles and records its density for every 1 hour. A delay of 5sec is provided for cumulative entry of the vehicle. Further, it detects the distance of the vehicle to the LED through sound waves. Then, noting the time taken for the soundwaves generated and returned. The moment sensor detects the objects, sensor changes its output and makes it to greater than logic one. Then, the vehicle (object) count incremented by 1. Table 1. Shows the total number of components required for the proposed model with the operating voltages.

TABLE I. REQUIRED COMPONENTS OF THE MODEL

S.No	Components required		
	Active component name	Operating voltage	Used for
1	Arduino UNO R4	5V-12V DC	Integrates several sensors
2	Infrared Sensor	8V	Detects the infrared radiation
3	LDR Sensor	8V	For lighting control
4	LED & Resistor	1.8V-3.3V	Works as a street light and resistor to control the flow of electricity
5	Dimming circuit	3V-32V	LED lightening

In addition, for ambient sensing, the proposed model includes PIR sensors to detect the motion in the surroundings. It is made up of a pyroelectric sensor to identify various levels of infrared radiation. Thus, PIR sensors can identify changes in infrared radiation caused by the movement of objects. The IR sensors are connected to digital input pins on the Arduino. When motion is detected, the Arduino receives a digital signal from the IR sensors. Then, The Arduino combines information from both the LDR and IR sensors to dynamically adjust the brightness of the street lights. If motion is detected, it may temporarily increase the intensity of the lights for improved visibility in the detected area.

The Arduino controls the street lights using a relay module. A relay is an electromechanical switch that can turn on or off the high-voltage street lights based on the low-voltage signals from the Arduino. Depending on the logic applied in the Arduino code, the system sends control signals to the relay module to adjust the street light intensity. For

example, when ambient light is low and motion is detected, the Arduino can command the relay module to increase the brightness of the lights. The entire process occurs in real-time, allowing the street lights to adapt dynamically to changing environmental conditions. This real-time adjustment enhances energy efficiency, safety, and overall system responsiveness. Finally, for the data analysis and visualization at real time, ThingSpeak is used. It is a cloud and IoT analytical platform.

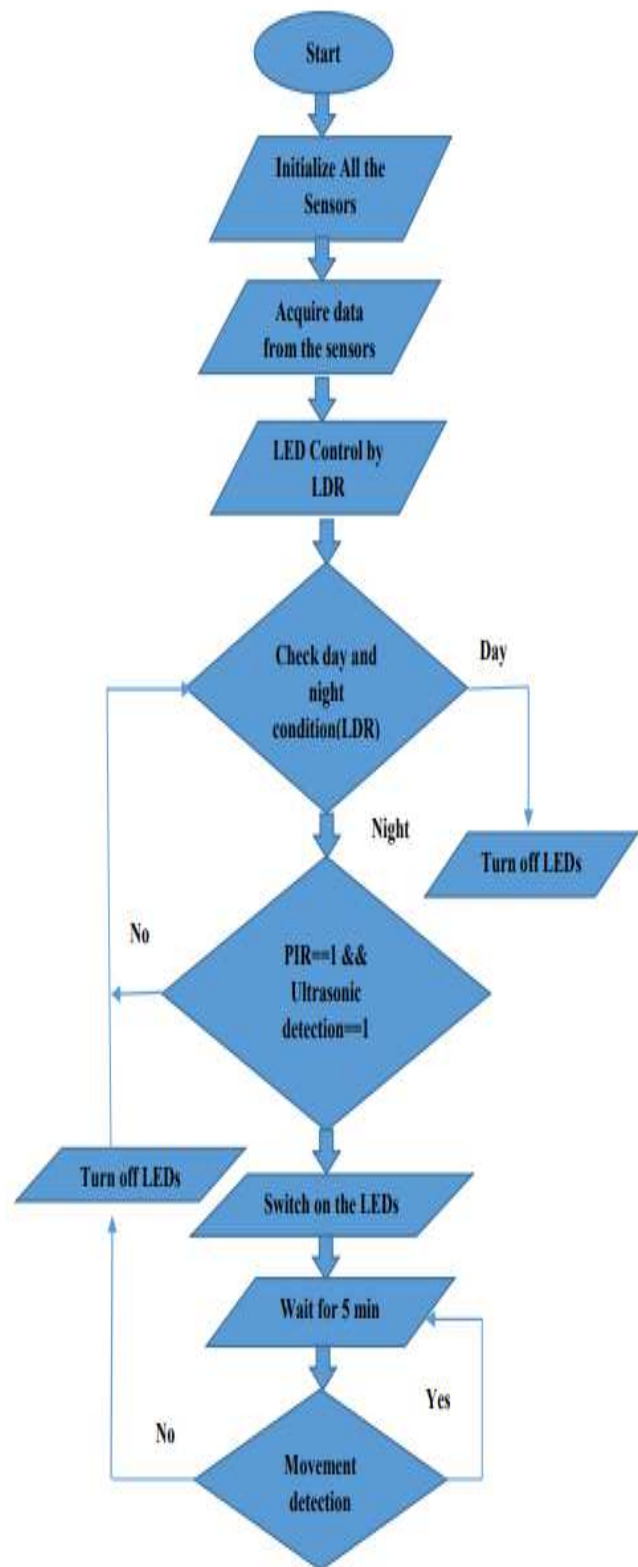


Fig.2. Workflow of the proposed model

IV. RESULTS AND DISCUSSION

This section exemplifies the performance of experimental setup. The proposed model uses 12 V DC supply and produces an output of 13-15W. However, LEDs require 7-10W to run and costs around 0.17 in Indian rupees, which is quite sufficient to turn on LEDs. Therefore, the proposed traffic lighting system available at low cost and economically viable. Secondly, in order to prove the power efficiency of the model, determine the maximum power dissipated in watts for single LED using the formula given below.

$$\text{Power dissipated}_{LED} = \text{Voltage (V)} * \text{Current (I)} \quad (1)$$

$$\text{Power dissipated}_{LED} = 3.5 \text{ V} * 0.035 \text{ A} = 0.1225 \text{ W}$$

Therefore, based on the information obtained for one LED, it is recommended to calculate for the power consumption for 100 LEDs by considering 100% intensity per hour. Mathematically it is obtained as shown in equation (2).

$$\text{Power dissipated}_{LED} = 100 \text{ LEDs} * 0.1225 \text{ W} * 1 \text{ hr} \quad (2)$$

$$\text{Power dissipated}_{LED} = 12.25 \text{ Wh} \quad (3)$$

Repeat the same and calculate the power dissipated for 100 LEDs at different times in a day and at different intensity. Table II. Shows the test cases examined for different intensities in a day.

TABLE II. INTENSITY LEVELS AND POWER DISSIPATION

S.No	Timings	Intensity measured using LDR	Power dissipated in W
1	7 AM- 5PM	100%	12.25W
2	6 PM- 12 AM	80%	9.8W
3	1 AM- 6 AM	50%	6.17W
Total Power dissipated with intensity control per day			28.22W

Power dissipated without intensity control systems for 100 LEDs is 40.89W in a day. Then, total power saved is equal to 12.67W per day. Thus, 30.98% of power is saved per day. In terms of rupees, 0.20 rupees saved per day.

From the Table II, it is considered that the 6 PM to 6AM is the active period and 7AM to 5PM as inactive. During this time, the power dissipation by LEDs is calculated based on intensity and it is 9.8W, 12.25W respectively. In cloudy environment, the intelligent lights will function based on intensity levels assumed. The intensity levels are kept 100% and 80% in the active period. This is based on the traffic density calculated using Ultrasonic sensor and data is stored in the cloud during the times mentioned. The traffic density information can be retrieved from the cloud before setting the intensity levels. In addition, the carbon components emitted by the model is also comes down as the LEDs are switched on for discrete times rather continuously. In general, an LED emits 451lbs carbon components per year for 100% intensity. Thus, the proposed intelligent LED street lighting model not only reduces cost, power dissipation but also has its impact in reducing CO2 emissions and useful for ecofriendly environment over existing approaches [3] [5].

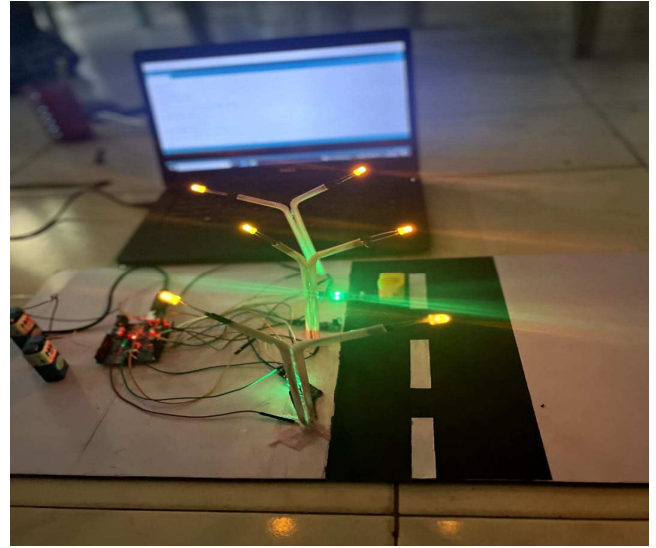


Fig.3. Working model of the proposed streetlighting system during night

Figure 3. illustrates the working model of the proposed work during night time. The firmware is meant to work in tandem with the signals provided by the sensors, causing the colors appear greater intensity in certain code-defined scenarios.

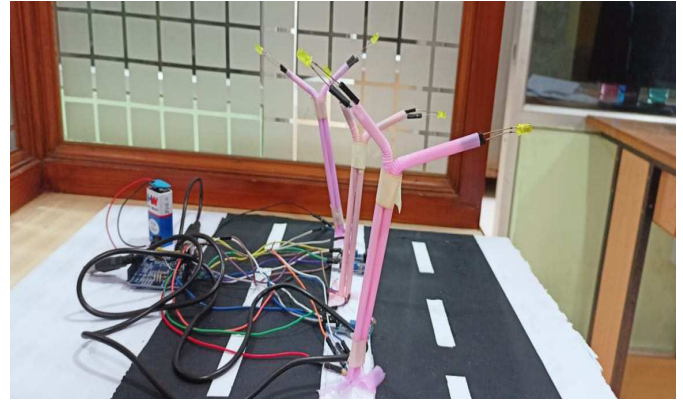


Fig.4. Working model of the proposed streetlighting system during day

Figure 4. shows the working model functioning using automated entrance the system's on or off during day time. Since it is daylight, as indicated by the numbers, the LEDs aren't lighting. Figure 5. Explains the variation in power dissipation for different intensity levels in case of 100LEDs. Further, sensor data is analyzed and exchanged with each other over the internet. Finally, the values are stored in the cloud through Thingspeak.

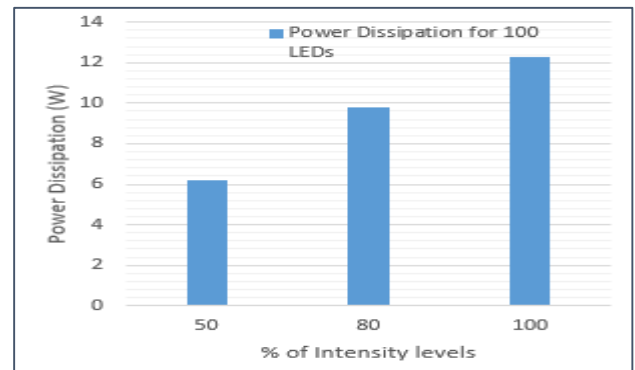


Fig.5. Characteristics of Power dissipation for different intensity levels

V. CONCLUSION

In this paper, an IoT enabled real-time street lighting system is proposed using low cost sensors for reducing overall power consumption. Further, the proposed model has automatic intensity control LEDs with dimming circuit makes it simple and power efficient. In addition, the present work also offers to store the health condition of the LEDs in the cloud via Wi-Fi module. The carbon components emission is also reduced using the proposed model as the LEDs switch on and off for discrete time rather continuously. Further, the proposed model is indorsed as it eliminates the requirement of manual operation to switch on/off LEDs due to the presence of LDR. Although the goals of this experiment were satisfactorily met, as a part of future work, this model can be enhanced by integrating surveillance cameras for tracking to improve security for the environment.

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A Street lighting Control system based on holonic structures and traffic system

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Abstract— Street lighting system is one of the main sectors of energy consumers in the area of the electrical energy consumption in the country. Since suitable lighting system plays an important role in terms of individual and social security in the city, providing an optimal management plan in line with the aim of saving consumption of electrical energy while providing appropriate lighting levels, can be a great help to the field of energy consumption. In this article we have tried to provide a comprehensive and overall plan for controlling and intelligent managing the street lighting system based on passages traffic using holonic multi agent systems. This approach controls the lighting system by intelligent agents in a holonic organization mounted on the traffic infrastructure. This control is done at different levels in centralized and distributed form. We can see two levels in our holonic multi agent system. In the high level of the system, there are geographical holons and their heads with strategic tasks. In the low level of the structure, there are learning agents with tactical tasks in geographical holons. They are main controllers installed in the streets. They form high consumption and low consumption holons dynamically, according to their decision for controlling the street lighting. The head of geographical holon also supervise the learning agents and formation of the consumption holons. We show that our approach can control the street lighting dynamically and efficiently in different conditions. The results of implementing this approach on a simulated network of streets and their traffic, shows that this method can lead to significant decreases in electrical energy consumption.

Keywords- *Smart control; street lighting; holonic multi agent system; traffic*

I. INTRODUCTION

One of the significant cases in consuming electrical energy is the consumption of lighting system in the streets. So considering the significant energy consumption in this system and the importance of individual and social security provided by proper lighting system in the city, it can be inferred that the smart management of energy in this area can be helpful to for both targets [1]. On the other hand, since the lighting systems are used in the peak hours of consumption, we can prevent from problems and tensions in the network by providing a suitable model for consumption management in this field. In this work we have tried to provide a new plan for controlling and intelligent managing of street lighting

system based on passages traffic using holonic multi agent systems. This plan will consist of distributed and centralized controlling in different levels.

Today intelligent multi agent systems can be used in many industrial and scientific fields. One agent is an entity that can sense the environment by some sensors and acts upon the environment using some actuators. Holonic structure is a kind of hierarchical structure and organization in multi agent systems. The holonic structure is one of the suitable structures for modeling many complicated systems such as education, transportation, traffic control systems, etc. The holonic structures are adaptive to the environmental changes, cooperative, self-organizing, flexible and reconfigurable. The fundamental units of this structure are holons. The holons may consist of subholons and also be a component of a super holon. So a tree structure called holarchy is made by holons. There are a lot of examples of holarchy such as human body. The holarchy structure can be also considered for many social organizations. Giving an education case such as university, a view of holarchy structure is shown in Fig. 1 [2, 3, 4, 5].

The smart control of lighting system will be discussed by giving a special method in this paper based on the traffic rate in the streets and using multi agent systems. In this method the traffic system is used as the default infrastructure of the street lightning control. It is worth noting that the traffic control system is itself a suitable platform for applying holonic structures. Therefore if the traffic control system is also controlled based on the holonic multi agent systems, the performance of two systems can simultaneously be suitable in the different layers.

II. RELATED WORKS

In the field of street lighting automation, we can mostly mention to the equipments and devices produced in this field. These equipments are installed on the poles or in the central control unit. None of these works, however, won't give us a comprehensive and general design based on smart controlling the whole system in different layers and creating an interactive, flexible and smart system. One sample of these equipments is the control device made in SNR company that using the light sensors. In fact, SNR device is used as a substitution for fast turning off/on the lamps, instead of the human operators. This device is capable of turning off/on the lamps after sunset and before sunrise [6].

These are several instances made in china, same as the above-mentioned device in the market. There is another example of these devices such as SLC-02-25-GSM-insuindia made in India. Starting hours of midnight the lighting lamps power are decreased by this device, thus saving energy will happen [7]. In Iran for example we can point out performing centralized lighting control project in North Khorasan. In this research project, turning off/on the street lights can be done by remote control systems, in addition to controlling by the operators. Performing this project, in a short period of time street lights can be turned off and turned on [8]. Smart street lighting control is another project has been done in an Iranian Company (Karizan Company). In this project, it is attempted to turn off/on the lamps by remote control system, GSM network and sending message [9].

III. HOLONIC MULTI AGENT SYSTEMS

A multi agent system consists of several intelligent entity called agents. These systems are used widely for the problems that centralized and integrated systems aren't capable of solving them. Since these systems have many capabilities such as being distributable, it is an efficient substitution for the classic methods in artificial intelligence. On the other hand, in some recent decades, since there are some more complicated and greater problems and consequently increasing the number of agents and interaction among them, it is difficult to solve the problems with the simple multi agent systems. In recent years, some solutions such as holonic multi agent systems have been proposed. For the first time a Hungarian philosopher used the word "holon" to introduce the regressive and self-similar structures in the social and biologic organization. The bases of the holonic structures are holons. The holon could consist of some sub holons and also be a part of a super holon. A tree structure which its members cooperate together to reach a complex goal, is made by these holons. The tree structure or holarchy has unique advantages in terms of stability and resistance to internal and external injuries and disorders. These structures are efficient in using sources and have a high adaptability against the environment changes. The idea of the holonic structures is a combination of top-down hierarchical structures and distributed control systems. In holonic multi agent systems, each holon could be a group of some intelligent agents or only consists of one agent. Since this structure is a regressive one, each holon is also considered as one agent and is shown by a representative agent called "head". The head can be chosen among the agents in the holon or a new agent is made for this purpose as long as holon is alive. It states the intensions of the holon. The head (and only it) communicates with the outside of the holon and can negotiate with the agent's in the holon environment. A view of the holarchy is shown in Fig. 2. There are some interesting discussions about the features of the agents such as autonomy, goals, communications and etc in holonic multi agent systems. The more concepts about holonic muti agent systems for further reading are in [2, 4, 5, 10].

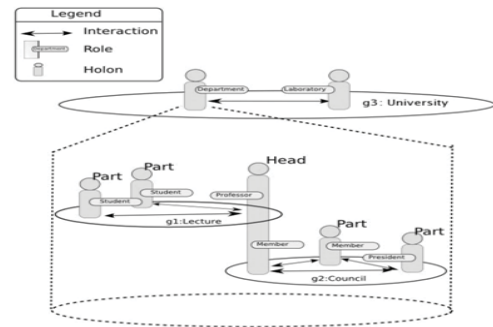


Figure 1. A view of holarchy structure in an education system

IV. SMART CONTROL OF STREET LIGHTING SYSSYM BY HOLONIC MULTI AGENT SYSTEM

In this paper we propose an architecture to control the street lighting using holonic multi agent systems and a simple learning for agents. We show that our approach can control the street lighting dynamically and efficiently in different conditions. In this approach learning agents are the main controllers installed in the streets. These controllers control the lighting of the street lamps. In the proposed approach holons are formed in two distinctive levels. Some of them are static and some ones change dynamically. This feature can cause increasing the flexibility and efficiency of the approach. In this method in the high level, a simple static structure according to the street geographical adjacency is chosen. In another word the studied area is divided in to some geographical holons. Each holon consists of some streets approximately situated in one geographical district. The streets of each holon in terms of some effective parameters such as peak hours and off- peak hours limit are somewhat similar. One example of geographical separation of Tehran districts is shown in Fig. 3. It can be used for holonification based on geographical districts. A responsible agent called "head" is considered for each geographical holon. The head of geographical holon calculate the time of turning on or off the street lamps of the holon in different seasons based on the Latitude and longitude of the district, climatic conditions and atmospheric brightness. Based on this calculation it sends the command of turning on/off the lamps to body agents in the holon. In the low level of the architecture, there are learner agents, simple inferring agents can be installed along with the traffic controller agents and use the information obtained by the traffic sensors installed in the streets. (Traffic controller agents for example can be installed in the junctions and use a kind of reinforcement learning for their operations). These inferring agents have a fuzzy inference engine consisting of some fuzzy rules based on the present state of the street lamps and the present traffic volume in the street. Table 1 shows the fuzzy rules of the inference engine. These controller agents in the low level of the architecture after receiving the turning-on command from the head in the geographical holon, start their learning and using the data received from camera sensors and based on inducing rules, control the lighting of the street lamps

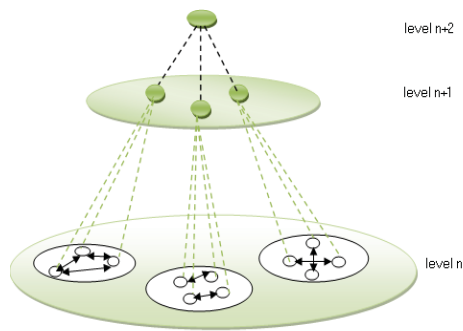


Figure 2. structure of a holonic system

appropriately. The heads of the geographical holons classifies the low-level learner agents into groups of “high” and “low” consumption and according to the streets adjacency and kind of their consumption form “high” and “low” consumption holons of the agents. This process happens every some minutes. So the consumption holons form dynamically for a period of time and after that according to the new situation, some holons may be destroyed or changed and also some new holons may be created. So a new arrangement of consumption holons will forms. The holonification of the streets in the low level of the structure is as follows:

The adjacent streets having the same consumption rate will be in one holon. Thus the head can supervise the consumption rate of holons and sometimes apply some additional necessary actions such as: increasing “low consumption” holons, stopping the dynamic changes of holons, alternatively turning off the lamps in high consumption holons in special and compulsive times. In this method if the traffic control system is also based on a holonic structure, this proposed holonic structure can be mounted on the traffic system. Thus simultaneously and efficiently the traffic and lighting system can be controlled by using a good holonic multi agent system and applying the learning methods in the different layers.

Fig. 4 shows an instance of forming high and low consumption holons in a geographical holon (geographical holon of district #1 in Tehran). In this picture, the districts with red lines represent high consumption holons and the districts with black lines represent low consumption holons. (The colorful dots in districts are just for showing the area). Fig. 5 also shows a general view of the proposed structure. Surely, this method will be more efficient and useful when it is used with a smart traffic control system, because the traffic volume is used as a parameter for controlling the lighting in this method. So a smart and efficient traffic control system can prevent from traffic jam and forming a lot of “high consumption” holons.

Table 1. Fuzzy rules

Lamps state	Traffic Volume	Lighting level/ mode
On	Low	Low / low consumption
On	high	High / high consumption
Off	-	-

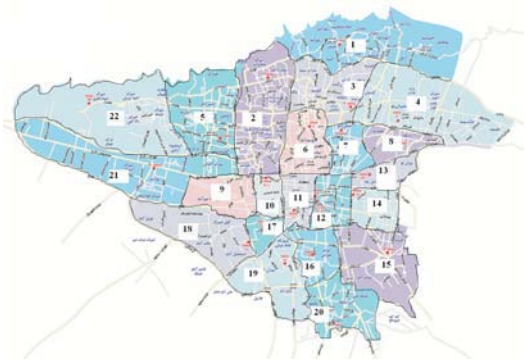


Figure 3. example of geographical separation of Tehran districts

V. EXPERIMENTAL RESULTS

Simulator software Aimsun V6.1 and its AAPI environment for C++ programming are the tools have been used for implementing this method. This software is one of the good ones can be used to simulate the traffic networks. It has a lot of tools for designing, adjusting, simulating and controlling the traffic networks in different scales.

In this paper, the performance of this method is shown on a small traffic network consisting of 17 main two-way streets and 6 junctions. In this network junctions have 4 phases and the number of the signal lights; it is the number of phases. In this simulator, the entry traffic volume can be determined with two following methods:

- 1) origin and destination matrices
- 2) using “traffic State”

In this research, the second method is used for determining the traffic volume entering to the network. The simulation time is 10 hours and the interval of checking the traffic volume of the streets according to the streets length and network size is 30 minutes. In this network sodium 250 W lamps are used for the streets equipped with double-side poles installed in the middle of streets. Fig. 6 shows a view of the designed network in the software and some of the “low” and “high” consumption holons formed in the simulation time. In Fig. 6 the yellow districts with the red lines represent the high consumption holons and the blue districts represent the low consumption holons. Based on the results of the simulation, many streets can be in low consumption mode instead of high consumption mode in the simulation time and this can contribute to save the energy consumption significantly. The energy is consumed by a street lamp is computed in Watt Hour (Wh) as follows:

$$E = P \times t \quad (1)$$

E is the consumed energy, P is the power in Watt and t is the time of energy consumption in hour. The energy consumed by the network in the simulation time while using the proposed method, is calculated as follows:

The consumed energy of street #1 is:

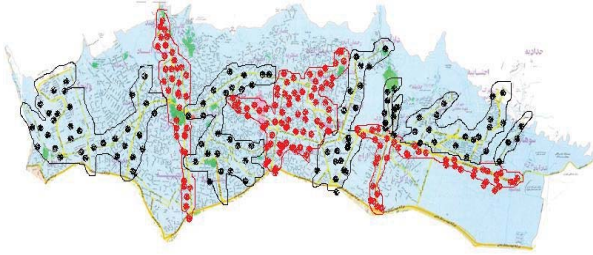


Figure 4. a sample of forming high and low consumption holons

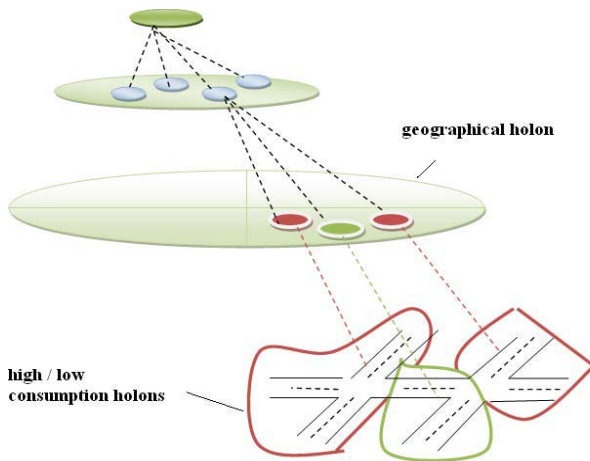


Figure 5. a general view of the proposed structure

Number of poles in this street: 20
 Band #1: $20 * [10 * 125] = 25 \text{ kWh}$
 Band #2: $20 * [10 * 250] = 50 \text{ kWh}$
 $25 + 50 = 75 \text{ kWh}$

The consumed energy of other streets also computed the same as the above example. The energy consumed by lighting system of the streets with the proposed method in 10-hour simulation is in table 2. The Fig. 7 shows a comparison of energy consumption of the streets in normal mode and when using the proposed structure. The diagram of Fig. 7 shows that using this method can cause a significant energy saving in the street lighting system.

VI. CONCLUSION AND FUTURE WORKS

Providing a solution for economizing the electrical energy in the street lighting systems can help significantly the energy saving in the country. The results of the implementing the proposed approach shows that performing this approach on the street lighting system and simultaneously controlling traffic will result in saving energy properly.

Since the proposed structure for smart lighting control is on the traffic volume of the streets, using a smart traffic control will result in saving more electrical energy. Some of

the holonic structure characteristics are flexibility, dynamism and capability of hierarchically controlling and these characteristics make proposed approach more efficient.

Table 2. energy consumed by street lighting system

Street	Number of poles	Energy (kWh)
1	20	75
2	20	75
3	30	114.25
4	20	75
5	30	118.125
6	20	75
7	20	75
8	25	104.6875
9	25	87.5
10	25	85.9375
11	20	75
12	20	75
13	30	75
14	20	75
15	30	75
16	20	75
17	20	75

Another advantage of this approach is giving a complete and a comprehensive design for creating a great flexible, interactive smart system. Some of the future projects are generalizing this smart control structure to all of thoroughfares such as alloys, parks and other places.

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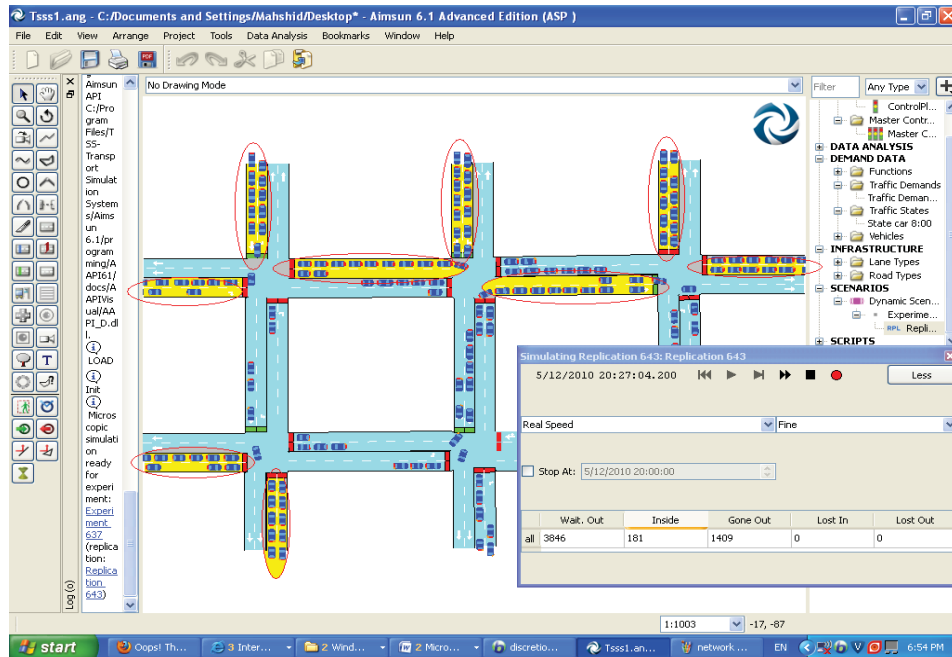


Figure 6. Simulated network in Aimsun 6.1 and forming high/ low consumption holon

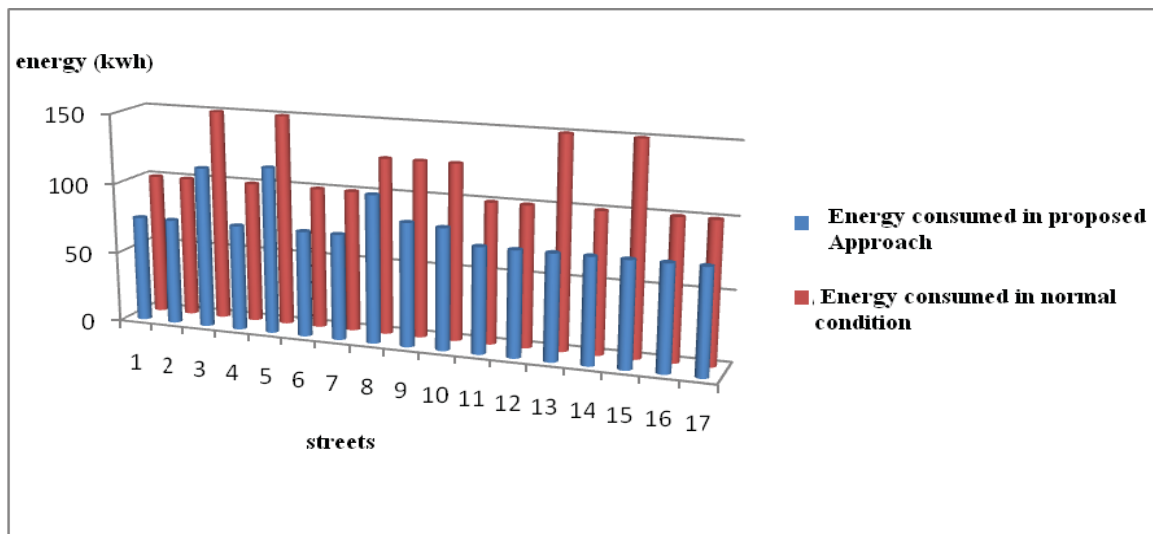


Figure 7. Energy consumed in proposed approach and in normal condition

Automatic Street Light System Using Arduino & LDR Sensor

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Abstract— The massive quantity of electricity gets used in lights through roadways in India. Still, roadways remain empty for nearly all of the day after at night, with a handful of cars passing during the entire time. In India, the typical roadway and bridge illumination system isn't automated. The number of vehicles and visitors is particularly high in the streets at night, and the lights are on all night. A greater proportion of light is squandered on roads ineffectively. This paper shows a street light system is utilized with different innovations. Power LEDs are utilized due to its excellent electrical efficiency, extra duration, cost-effectiveness, and capacity to lower pollution levels. Using a smart street light system, we will save our electrical costs. In addition, smart street lights will control power more effectively by reducing the risk of the automated street light system warming as well as the risk of incidents. Wasting energy by keeping the street lights on all night, we can build an affordable, high-performance smart street light system with the Arduino, LDR sensors. In this article, we'll discover how to create an automatic street light which turns itself on or off while a car or other object is passing by.

Keywords—Sensors, street light monitoring system, Arduino, LDR.

I. INTRODUCTION

For safety reasons and to prevent incidents in the dark, street lights are a crucial component of modern vehicles. Even this, hardly one thinks to turn off or on the lights whenever not in use in modern life. I will provide you with a way to cut street-level power usage and remove labour in this paper. This is among the greatest methods currently in usage of the automatic street light system, which offers a smart lighting technology. The major goals of employing this kind of street lighting include to preserve power to future generations to come, since it uses less electricity. Automated devices today do a little work manually & are very flexible and efficient. The use of (LDR) in an automatic street light controller removes the need for manual labour.[1]

It has previously been suggested to manage lighting systems with an Arduino controller & a Light Dependent Resistor

(LDR). Many investigations have been inspired by the important role of smart lighting systems, and many studies are being conducted so street light is controlled by solar energy [2].

II. LITRATURE REVIEW

Wireless detector network is based on street lamps control systems. Streetlamps are typically controlled by the network in automated modes using minimum power and the Morning and Evening algorithm [3]. This mechanism is also capable of being controlled. We have control over the street lamps in this phase. The technique might be significantly applied everywhere which demands quick command, involving streets, stops, colleges, and the electrical sector. It also features a high-power sensor supply [4].

The Automatic Street Lighting System uses a Light Dependent Resistor to recognize and switch on street lights. A microcontroller is used to regulate outcomes whenever feedback comes from a source of resistance. This system is suitable for various areas, including streetlights, bus stations and outdoor lights. The report aims to introduce an efficient road lighting system with reduced power usage, using LEDs as more efficient than other.[5]. The in built system in laptop designed to show specific tasks with ongoing data requirements, often installed as a part of a complete device. These systems are adaptable and meet various end-user needs [6].

The application of LDR within automatic street light controllers removes the need for manual labor. Whenever light falls down the zone of the human body, street lights turn on automatically. It turns on simply in the evening & turns on when its sun rises. applying the LDR, users may automate a streetlight functioning [7].

Specialists have widely researched the incorporation of Light Ward Resistors (LDRs) and microcontrollers, like Arduino, to make mechanized road lighting frameworks. These frameworks use LDRs to identify changes in

surrounding light levels and trigger the programmed actuation or deactivation of streetlamps.[8] Through trial and error and examination, researchers have inspected the viability of various edge values set for exchanging among constantly modes, endeavoring to find some kind of harmony between giving sufficient brightening and limiting energy utilization.[9].

III. SYSTEM COMPONENTS

Materials	Standards
LDR	Voltage: DC 3-5V, 5mm, 1.8gm.
Arduino Uno	22pins, operating voltage 6-20V
LED	5mm, operating voltage 5V
IR sensor	Voltage: DC 3-5V, Range 2-30cm, Angle 35
Resistor	10k ohm, 220ohm

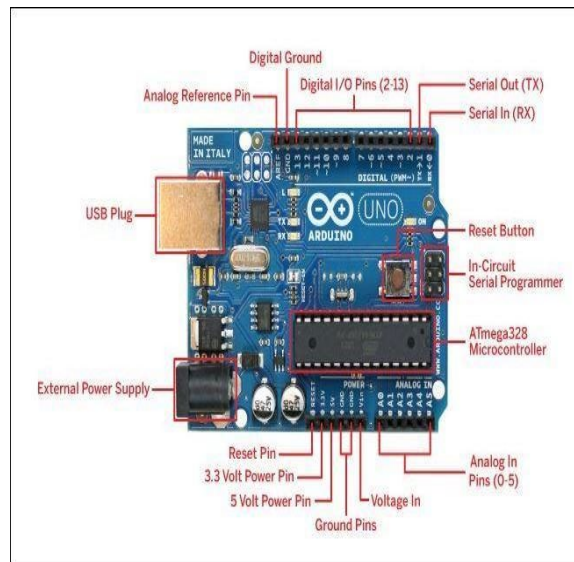
A. Arduino Uno:

It is an open-source microcontroller board with an ATmega32 series controller that can be programmed using an IDE (Integrated Development Environment). The Arduino's input and operating voltages range from 7 to 20 volts. The USB port on it allows for power and data transfer from a personal computer. The ATmega32 processor uses the sensor's input signal to execute the code that has been uploaded. C++ and other programming languages are supported by the Arduino IDE[9].

The microcontroller contains a port for an (Universal Serial Bus) connector, a 9-volt electrical outlet for a second power source, six analogue input/output pins from A0 to A5, and an In-Circuit Serial Programmer (ICSP) module of the 14-digital input/output pins, six of these pins maybe utilized to output. Additionally, it features Flash storage capacity of 48 kilobytes, of which the boot loader uses 0.5 kb. The microprocessor receives data from the PIR motion sensor and LDR sensor, processes it, and outputs it to LEDs [10].

B. LDR:

A light-dependent resistor (LDR), also known as a photo resistor or photocell, is a variable capacitor that decreases with increased light intensity and increases with decreased intensity [11]. It is primarily used for detecting day and night, turning on/off street lights based on changes in light intensity or day/night time.



C. Resistor:

Resistors are Lamps or detectors, among other electrical devices, are utilized with the remaining various part to interrupt circulation. They resemble cylinders with colored bands and wires. They come in two, four, and five-band varieties. Resistors may prove utilized in automated traffic light controllers such as a 220ohm resistor with LED and a 10k ohm resistor with an LDR sensor. Careful selection is essential to avoid incorrect values[12].

D. Light Emitting Diode: Whenever tiny current is put across the (LED), it generates lights. It has two legs, the longer for the positive connection and the shorter for the negative connection. In our project, we use an LED as an output, which is turned on/off based on sensor data. LEDs have a two-terminal device called Cathode and Anode, with the longer terminal being the Anode and the shorter terminal being the Cathode. They cannot be connected directly to a battery or voltage source.[13].

E. Breadboards: A breadboard is basically an item used in circuit design or construction. It enables users to create designs avoiding welding by enabling you to set up parts and connectivity to the board.

F. IR sensors:(IR) monitors identify and gauge the amount of energy that is given off or absorbed through items. They are made up of an ultraviolet generator and an antenna. A device that emits a wavelength of light that isn't observable to the naked normal eye, but the device receiving it recognizes when interacting with a particular object. The quantity of infrared

light received by the receiver varies as an item enters its field of vision. Movement identification, monitoring of temperatures, location detecting, as well as remotely management tools all employ infrared sensors [14].

*G. Jumper Wire:*An electrical wire, or collection of electrical wires in a cable, with a connection or pin at either end is known as a jump wire and is typically used to link parts of a bread board, or other trial or test circuit, directly or with other machinery or parts, without connecting.

H. USB B Cable:

It is used for linking your project's Arduino Uno, Arduino Mega 2560, Arduino 101, or any additional board to your laptop's USB connection female A connector.

Component price:

TABLE I. Cost Table

Arduino Uno	750
Light Dependent Resistor (LDR)	12
Resistor	20
Light Emitting Diode	18
Breadboards	90
IR sensors	120
Jumper wire	50
USB B Cable	70
Total cost	1130

IV. METHODOLOGY

When the sun goes down at night, the LDR detects the amount of light and signals a microcontroller to switch on the street lighting system. When there are no vehicles on the road, the infrared sensor installed on the road divider will detect an absence of the vehicle and send a signal to the microcontroller to reduce the intensity of the street light. The electrical power usage rate will be decreased by up to 60% through to this technique. The sensing device provides a Communication of signals to the microprocessor & the microprocessor orders the LED to get activate at 100% efficiency when it senses the existence of any motor[14].

Two various kinds of sensors, each with a particular function, are primarily employed in this system. The first component is the Light Dependent Resistor (LDR), which is used to turn on or off the LEDs through the day and at night by sensing the brightness of the light. Infrared sensors are also included to detect the existence when the IR sensor detects no vehicles, it instructs the micro controller to decrease the street light's intensity [14].

A. Block Diagram:

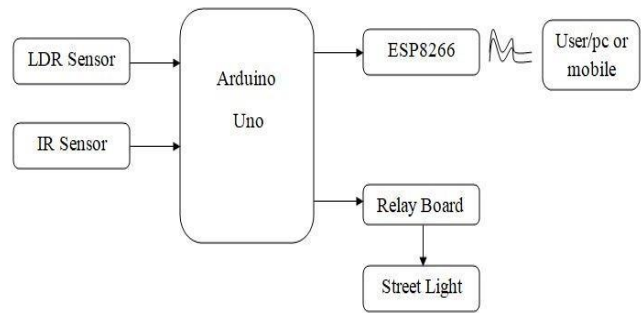


Fig.1 Circuit diagram

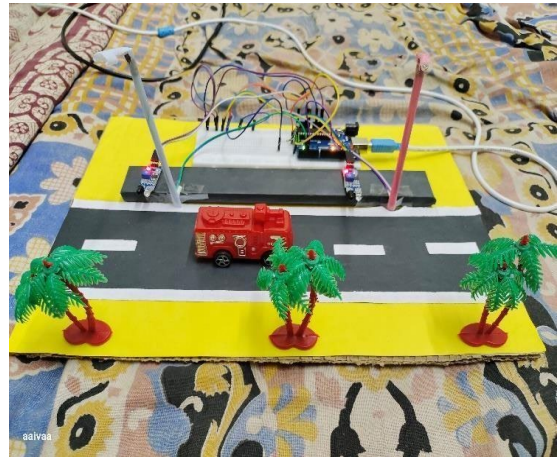


Fig. 1 Proposed Model of Automatic Street Light System

B. Software: The language for Arduino programming is used to create the code. The software known as the Arduino IDE is used to program the Arduino Uno Board1.The ATmega328P processor on the Arduino Uno board includes a set up boot loading which enables you to transfer fresh software without the use of a third-party hardware programmer.

- Connecting the Arduino board with the laptop and choose the appropriate COM connection in the Arduino IDE.
- Download the software in the Arduino Software IDE. Choose the appropriate board from the Tools Board option and submit the code.
- The outline is at the center of the system, performing all important tasks.
- The Arduino IDE 2.04 is used for compilation and download it.

*C. CODE:*The code reads the LDR data and enables the relay according to the threshold value.

```

intldrpin = A0;int led = 7;
int threshold = 40;
void setup()
  
```

```

{
serial.begin(9600);
pinMode(led, OUTPUT);
}
Void loop()
{
int data = analogRead(ldrpin);
serial.println("");
serial.print("Light Sensor");
serial.print("Value = ");
serial.print(data);
if(data <= threshold)
{
digitalWrite(led, HIGH);
}
else
{
digitalWrite(led, LOW);
}
}

```

TABLE II. Outside lighting Vs LED Sensitivity

Outside lighting	LED Sensitivity
63	515
56	547
52	541
209	118
212	116

D. EQUATION: The automatic street light system using an LDR and Arduino can be designed using the following equation:

$$R = (VCC * LDR) / (VCC - LDR * V)$$

where:

R = Resistance of the LDR

VCC = Supply voltage of the circuit (in volts)

LDR = Resistance of the LDR in ambient light conditions (in ohms)

V = Voltage loss across the LDR (in volts)

This system's basic concept is that the LDR adjusts its resistance in accordance with the quantity of light entering it. The Arduino determines when to turn on or off the street

light depending on the voltage across the LDR.

V. RESULT & DISCUSSION

A) Daytime situation with maximum brightness

No LEDs are ON throughout the day because of the ambient light, and the LDR value stays fairly stable. Brightness has an impact on intensity.

B) Situation at Night in Total Darkness

Since there is no sunlight throughout the night, all LEDs are on at maximum intensity, and the LDR value also stays almost constant because there is no ambient light to detect. Table displays the LDR value, also known as external brightness, as well as the intensity value. Table provided us with a graph, which was used as According to the graph, as the brightness of the outside environment grows, the intensity of the LEDs reduces, and as the brightness of the outside environment lowers, the value of the LED rising. The intensity increases while it is completely it's night out there.

C. Output:



Fig. 2 Diagrams demonstrating the automatic streetlight control system's results

VI. FUTURE IMPROVEMENTS

The automated street light technology offers an opportunity for additional improvements that will make it more effective, smart, and flexible. Below are a few suggestions for future enhancements.

- Systems for Intelligent Automation
- Bluetooth Networking

- ML Connectivity
- The Effectiveness of Solar Panels
- New Technology sensors
- The Tracking and Reporting of Power
- Estimation of Maintaining

VII. CONCLUSION:

The most secure cost-effective and environmentally beneficial method of energy conservation in this article of “automatic street lights”. Statistics show that we can reduce the amount of electricity now used by motorways by more than 40%. The project's disadvantages may include initial expense and ongoing upkeep. The LEDs may be utilized for quick changing generate cool light, don't contain any harmful materials, and have an extended lifespan. Due to these factors, our project offers far more benefits than it does drawbacks. Since the investment return period is so short, balancing the initial expense with the long-term rewards will never be an issue. The project has potential for a number of other purposes, such as providing illumination for campuses, industry, and large shopping parking spaces. This may also be employed for monitoring on the grounds and in the workplace. With this technique, it will help preserve electricity and protect the natural world that will help our nation grow organically. The present system can be easily and inexpensively replaced by this one. The design depends entirely on an Arduino, which in this case functions as a microcontroller by receiving data using an LDR. Modern Indian sidewalks, especially require this kind of system. Automated devices today do a little work manually & are very flexible and efficient. The use of a light dependent resistor (LDR) in an automatic street light controller removes the need for manual labour. In this system bulb get dim because of more usage but electric bulb are in condition of condition usage. So there is still need of Automatic street light design to controlled dim light and connect power on-off with object motion detector.

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Brightness Controlled Solar Powered Intelligent Street Light

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Abstract— Smart city is a term that can be described as technology that promotes sustainable development practices. Due to inefficient design, most street lighting nowadays wastes a lot of electricity. A smart city's key aims in streetlight management are energy economy and simplicity of maintenance with little manual labour. This research has designed an intelligent street lighting system that incorporates an automatic control mechanism for energy savings. The light intensity, the Wi-Fi network and the status of the streetlights were designed and monitored. An ESP32 microcontroller was used to process the sensor values and controls the output devices from the motion sensors and Light-Dependent-Resistor (LDR) sensors. The controller and sensors are powered by their Photovoltaic (PV) system. The solar panel is connected to TP4056 which acts as a charge controller for the Li-ion battery. The result has identified that the low-dropout (LDO) regulator power achieved consumption is 4mA. The clock speed of the ESP32 is reduced from 240Mhz to 80Mhz in reducing the power consumption. There is no performance issue was observed with the reduced clock speed. The use of light-emitting diode (LED) as the light bulb and controlled brightness provides lower power consumption of about 63.5% for 11 hours of usage. The use of solar power shows that street lighting is no longer needed to connect to the grid, and it is significant compared to conventional street lighting which consumes a lot of energy and requires a high cost of maintenance.

Keywords—Solar Power, Street Light, Artificial Intelligence

I. INTRODUCTION

A smart city requires technology that can enhance the quality of life and the performance of public services[1]. This includes the reduction of resource consumption, waste and all other costs. More intelligent technological services and infrastructures must be developed and integrated to optimize public resources[2]. Street lighting is an essential service that lights up a city from the darkness, reducing crimes. gives a sense of security at night, and sufficient visibility to ensure safety[3]. Smart street lighting has the potential to change the way cities are managed while also saving a lot of money. Local governments and utility providers are looking for new ways to minimize energy usage and expenses, with street lighting accounting for roughly 40% of many cities' overall

energy expenses. Switching from halogen to LED luminaires can assist in achieving this aim by generating immediate energy savings of 50% to 80% and it lasts longer[4]. Furthermore, by adapting output to ambient light levels and dimming or brightening as needed, smart LEDs can save an extra 10 to 20%[5]. They can also be programmed to turn on just when motion is detected, then dim or turn off after a defined period.

Furthermore, cities receive expanded capabilities and functionalities in addition to saving money. Cities and utility companies can add a wide range of equipment and sensors at a low cost by repurposing existing brackets and poles. With the capabilities such as motion sensing, monitoring, and others, cities can improve operational efficiency, improve people satisfaction, and decrease costs.

The study aims to develop an integrated intelligent street lighting system based on intensity control that achieves energy efficiency, with the use of ultrasonic sensors to detect the object, LDR sensors for measuring light intensity to reduce power consumption, and Wi-Fi to provide store the data for monitoring. By using solar power and LED that provide a longer lifespan and are less harmful to the environment, it is hoped that this research could contribute to the energy-saving goal without jeopardizing road safety. The study goes into further detail on the devices and their use, as well as the outcomes.

II. LITERATURE REVIEW

Conventional streetlight that must be manually controlled and cannot be monitored is still used by a lot of cities. This type of street lighting consumes a lot of energy at 35%-50% of a city's energy consumption[6]. Furthermore, a conventional streetlight that still uses bulbs such as incandescent light, CFL, and metal halide/HPS, contributed 30%-60% of overall carbon emissions which causes the greenhouse effect. Replacing an HPS lamp with one solar streetlight for 20 years can save 18.47% of CO₂ emissions[7].

There are various designs proposed by researchers to reduce the energy consumption of street lighting. Some proposed the integration of renewable energy into street lighting[8]. With the use of renewable energy such as

sunlight, wind, rain, and waves, the use of non-renewable energy, such as fossil fuels, which produce harmful effects on the earth can be reduced[9]. As conventional street lighting consumes a lot of energy, utilization of renewable energy is one of the most important factors in reducing energy consumption and ensuring a nation's energy security.

Conventional street lighting uses must be manually controlled or uses a sensor to turn on at night and turn off in the morning. This is a huge waste of energy consumption and is not efficient. However, some researchers proposed the integration of an intensity controller, that provides a way to control the intensity based on the situations, such as movements, weather, vehicle speed, and amount of traffic. A proposed system has been presented that uses Arduino as the microprocessor, a Light Dependent Resistor (LDR), that act as the light sensor, an infrared (IR) sensor, that detects movements and another IR sensor, that when use simultaneously, can be used to the calculated speed of moving object[10]. Figure 1 shows the different settings or intensities for different moving objects. Another proposed solution has presented an energy-efficient streetlight using a few sensors, communications technologies and various other IoT enabling technologies[11]. Although the costs of integrating IoT into streetlighting may be costly and not very clear, they may reduce the cost of maintenance as communication devices allow remote troubleshooting. The use of wireless-based communication device based on Zig-Bee protocol using 2.4GHz band, communicating with other streetlight posts. The device communicates with other light posts to adjust the intensity beforehand when the sensor detects movements[12].



Figure 1: Different settings for movement

Figure 2 shows the streetlights' communication when detecting an object. The system is powered by a Zig-Bee network that establishes the remote and automatic on and off control, adjustment of power usage and issues monitoring[12]. When the sensor detects an object, the streetlight intensity is increased and will send a command wirelessly to other streetlights to increase the brightness. This leads to huge energy savings as the roads are mostly empty during nighttime.

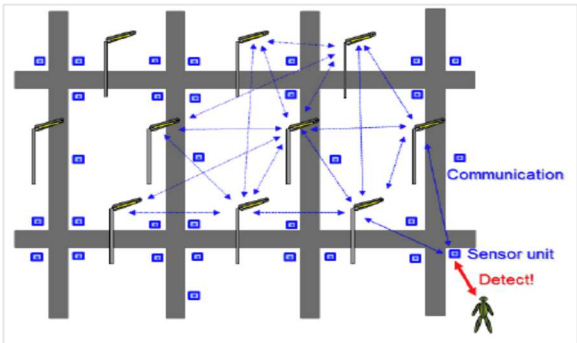


Figure 2: Streetlights' communication with other streetlights

Figure 3 shows the coverage of the object sensor, which can be improved with a better sensor or placement of the sensor. With the current technological advances and increase in accessibility to these technologies, more projects related to improving street lighting become publicly available[13]. Some researchers proposed streetlights using an intensity controller. The researchers from IIUM Gombak proposed the use of LED lamps, to reduce energy consumption, in conjunction with controlled sensors, such as an IR sensor to detect an object, and brightness control based on the speed of an object and the surrounding light[10]. This existing project is quite similar to the proposed research regarding the features provided, with the key differences being, that sensors to detect movement is Time-of-Flight sensors, which is a better alternative to IR sensor in terms of distance measurement and object detection, with a minor price difference.

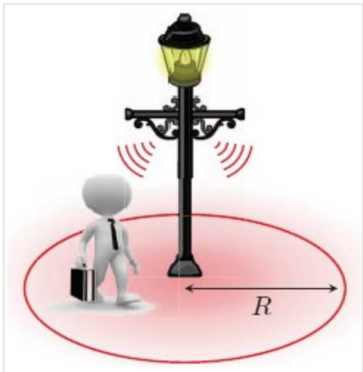


Figure 3: Coverage of sensing unit

Figure 4 shows the flowchart of an existing project from Bangalore, India entitled Automatic Street Light Control by Detecting Vehicle Movement. The paper describes a project that detects vehicle movement on highways and turns on just a block of road lights in front of it while turning off the trailing lights to save energy[14]. This existing project is similar in a sense of detection of movement such as vehicles, but as some city requires that streetlights cannot be turned off, this project only reduces the brightness of the streetlights. To conclude, the key difference between most existing projects and this project, is the components related to object detection, the ability to remote monitoring, so that maintenance can be planned faster and the integration of solar energy into streetlights.

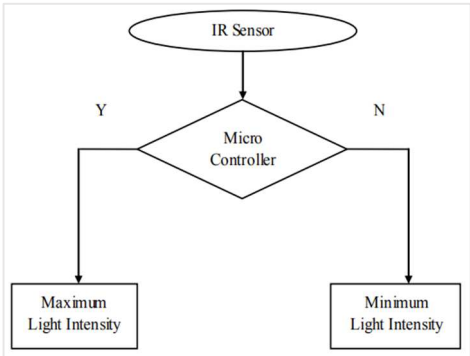


Figure 4: Flowchart of an existing project

III. METHODOLOGY

The project makes use of sensors to measure light intensity so that the streetlights will not be turned on during the day but will be switched on at night and during overcast weather when the light intensity is low, assuring good visibility. The intensity will be regulated by the microprocessor using the object detection sensor. When the sensor detects an object such as a person walking or a vehicle, the light intensity will be changed, and the lights will remain on until no motion is detected. Furthermore, the system employs LED lighting, which is very energy-efficient and has a long lifespan, as well as Pulse Width Modulation (PWM), which entails turning LEDs on and off at a pace that is invisible to humans, assuring energy efficiency. The project's microcontroller, the ESP32, processes the sensor values and controls the output devices appropriately. Motion sensors and Light Dependent Resistor (LDR) sensors are both employed in this project.

A. Research Flowchart

Figure 5 presents the research flowchart that starts with checking the set time for acquiring the LDR value, which is set to every 5 minutes, and when the value is obtained, the system checks if the lights are turned on, and decides to turn on or off the lights based on the LDR value. If the lights are turned on, the system starts the ultrasonic sensors to check the presence of an object. When an object is detected from one or two of the sensors, the system will increase the intensity of the LED lights for a few minutes. If another object is detected within this time frame, the LED will continue to be turned on at high intensity, until no object is detected. The other set time, which is set at every hour, is to turn on the Wi-Fi and send the sensors' data to the server, with the acquired values from all the sensors. Wi-Fi will be turned off when the process completes.

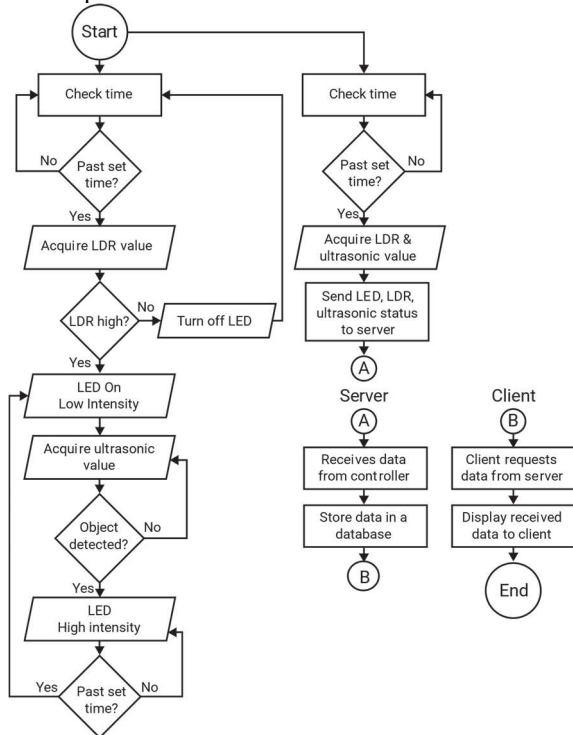


Figure 5: Research flowchart

B. Project Schematic

Figure 6 illustrated the schematic diagram that used an ESP32 as the controller, HC-SR04 as distance or motion sensor, LDR as the light sensor, and is powered by PV systems.

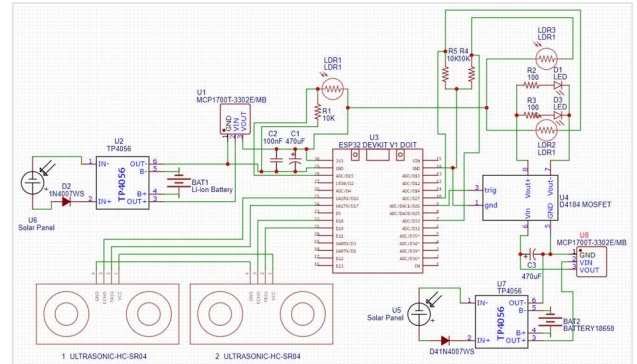


Figure 6: Project schematic diagram

C. Project Working Principle

Figure 7 shows the working principle of the research project. The controller will receive the data from sensors and control the intensity of the streetlights, and the controller sends the sensors' data to a server, where a client can monitor the streetlights through the cloud.

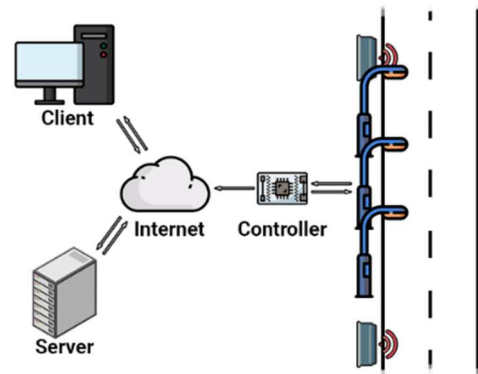


Figure 7: Project working principle.

D. ESP32 Microcontroller

Figure 8 presents the use of the ESP32 microcontroller that controls the system tasks as programmed. The ESP32 microcontroller contains 34 programmable GPIOs and is a low-cost, low-power microcontroller with integrated Wi-Fi and Bluetooth. It operates at 240 MHz but is underclocked to 80 MHz to save energy. The operating voltage is 3.3V, which is provided by a lithium-ion battery through the 3.3V pin of an LDO regulator. The Vin pin is not utilized in this project since it is attached to an AMS1117 regulator, which has a high quiescent current and is not ideal for a battery-powered device.



Figure 8: ESP32 microcontroller

E. LDR Sensor

Figure 9 presents a Light-Dependent Resistor (LDR) sensor to measure light intensity using the acquired varying resistance. When the intensity of the light is high, for example, the resistance of LDR increases, and vice versa. An LDR sensor module is utilized in this project to detect the level of light intensity in the surroundings. It is attached to the controller's digital pin.



Figure 9: LDR sensor

F. Ultrasonic Sensor

Figure 10 presents the employment of the ultrasonic sensor model HC-SR04 to detect the movements of passing vehicles and pedestrians. Ultrasonic sensors are devices that generate and detect ultrasonic waves. Transmitters convert electrical impulses to ultrasound, receivers convert ultrasound back to electrical signals, and transceivers both send and receive ultrasound. The sensor's data is utilized to automatically alter the streetlight level.



Figure 10: Ultrasonic sensor

G. Monitoring of Lights and Sensors

To monitor the lights and sensors, the project employs the use of Wi-Fi to send the data. The server uses Node.js as the backend framework and MongoDB as the database. The data are sent from the controller in JSON format to the server and are saved to the database. The frontend uses Vue.js, where when a client accesses the page, it will request the data from the server and display them to the client. The stack employed in the project is shown in Figure 11.



Figure 11: Monitoring stack

IV. RESULT AND DISCUSSION

A. Use of Ultrasonic sensor and LDR

Figure 12 illustrates the developed prototype of the system. The proposed design system is to reduce the power consumption of street lighting. LDR sensor to detect light levels to turn on and off the LED lights were used. When the LED is turned on, the system will increase the LED light's brightness when the ultrasonic detects an object.

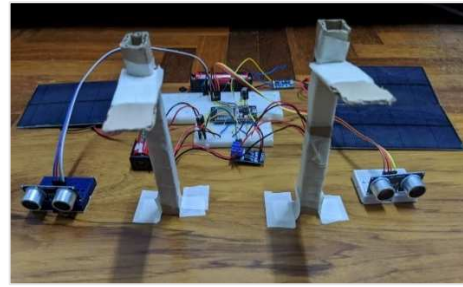


Figure 12: Project Prototype

B. Solar System

Figure 13 presents the developed PV solar system. A solar system is used to reduce the power consumption of the system[15]. With the use of solar, the streetlights do not have to connect to the grid, hence reducing the negative effect of using fossil-based fuels. The controller and sensors are powered by their PV system. The solar panel is connected to TP4056 which acts as a charge controller for the Li-ion battery. The LED lights are powered by another solar system, and the battery is connected to the D4184 MOSFET.

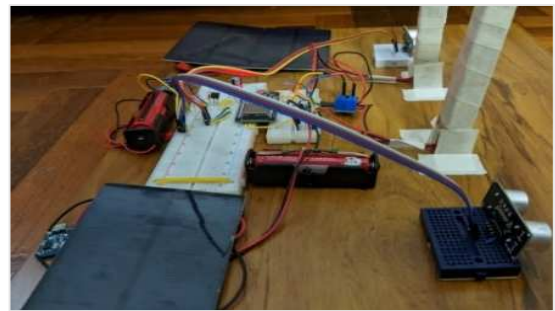


Figure 13: PV System

C. Power Consumption of ESP32

The project uses ESP32 as the microcontroller. Table I shows the power consumption of ESP32. The controller is powered by a single Li-ion battery that is charged with a solar panel. However, the ESP32 power consumption is around 160mA to 260mA, which is quite a lot. And with the use of Wi-Fi to send data, the power consumption can peak at up to 300mA for a few seconds. Furthermore, when the ESP32 is powered through the Vin pin, an increase of around 20mA is observed as the Vin pin will use the AMS1117 regulator to step down the voltage to 3.3V. To reduce the power consumption of the controller, a low-dropout (LDO) regulator is used to step down the supply voltage to 3.3V, which is connected directly to the 3.3V pin. This bypasses the use of AMS1117 in the ESP32. The LDO regulator power consumption is 4mA. And to further reduce the power consumption, the clock speed of the ESP32 is reduced from 240Mhz to 80Mhz. With the clock speed reduced, no performance issue was observed. Furthermore, Wi-Fi is turned off when not in use. This further reduces the power consumption to the data shown in Table III. And the peak current usage is around 112mA when connecting and sending data through Wi-Fi.

TABLE I. ESP32 POWER CONSUMPTION

	LED OFF	LED LOW	LED HIGH
Current (mA)	44	45	46

D. Effect of LED on power consumption

PWM control was used to adjust the intensity of the LED based on the presence of an object to reduce the power consumption of an LED. The project uses a 1W-rated LED that is limited to 40mA. Hence, when the LED is turned on without intensity control, the LED will use about 116mW of power. Table III shows the analysis of power consumption at 20% and 100% of intensity.

TABLE II. LED INTENSITY AND POWER CONSUMPTION

Intensity	Current (mA)	Power (mW)
20%	8.6	24.94
100%	39.4	114.26

TableIII presents that when the common streetlights are turned on for 12 hours, 6.00 p.m. to 6 a.m., at 100% intensity, the energy consumption is 1.276 Wh, while if the streetlights are turned on at 100% for 6 hours, and at 20% for 5 hours when less street activity occurs, the total energy consumption is only 0.8103 Wh. This result in 0.4657 Wh daily savings as shown in equation 1 and 13.971 monthly savings as shown in equation 2 or 63.5% in energy savings.

TABLE III. STREETLIGHTS POWER CONSUMPTION ON FOR 11 HOURS

	Hours	Intensity (%)	Energy (Wh)
No intensity control	11	100%	1.276
Total			1.276
Intensity control	6	100%	0.6856
	5	20%	0.1247
Total			0.8103

$$\text{Energy savings} = 1.276 - 0.8103 = 0.4657 \text{ Wh} \quad (1)$$

$$\text{Monthly energy savings} = 0.4657 \times 30 = 13.971 \text{ Wh} \quad (2)$$

E. Lights and Sensors Monitoring

The controller will acquire the values of the sensors and send and save the data to the server to monitor the status of the lights and sensors. Figure 14 shows the data that is saved to the database when the sensor detects an object, and the streetlights are at high intensity.

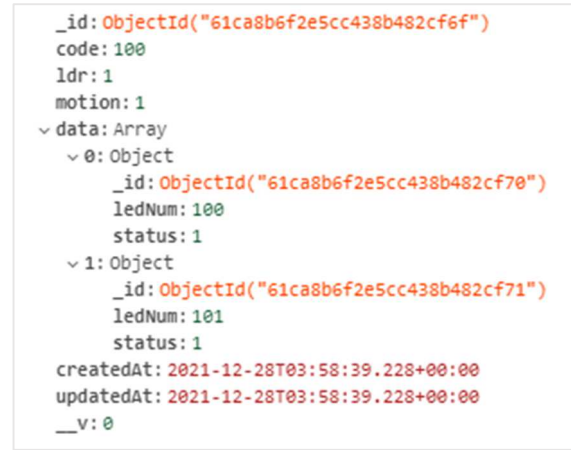


Figure 14: Data saved in a database

Figure 15 illustrated the system displays the requests for the latest data from the server and displays the status of the streetlights when a client accesses the webpage.



Figure 15: Client-side monitoring

VI. CONCLUSION

Conventional street lighting consumes a lot of energy and requires a high cost of maintenance. This also can affect the environment as most countries still utilize fossil-based fuels to power the streetlights. To summarize, the project goal is to design and develop an intelligent street lighting system based on intensity control and online monitoring that will save energy and be easier to maintain. The project uses LDR sensors to turn on and off the streetlights and a controller with ultrasonic sensors to adjust the intensity of the lights based on the presence of an object. Furthermore, it uses LED light bulbs that consume lower energy, are brighter and last longer when compared to other types of light bulbs, to further reduce energy consumption. Other than that, the project is powered by a PV system, which completely disconnects streetlights from the electrical grid. According to the findings, when compared to conventional LED street lighting without any control mechanism, the suggested system offers potential for energy savings of up to 63.5%. This suggests that the suggested system may provide environmental benefits in terms of Greenhouse Gas (GHG) emissions, as well as cost savings. Furthermore, with the identification of broken bulbs, this method makes streetlight maintenance much easier. The data from the controller and sensors is transferred to a database and preserved, laying the groundwork for the construction of a complete internet of things (IoT) system. In a nutshell, to further improve the system, a time-based control

can be added to control when the streetlight will be turned on. And a current sensor can be added to each streetlight instead of an LDR sensor, to monitor the power consumption.

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CHANGE IN LIGHT INTENSITY AND AUTOMATION OF STREET LAMP THROUGH MACHINE LEARNING ALGORITHM IMAGE PROCESSING

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Abstract—Street lighting provides a number of important benefits. It can be used to promote security in urban areas and also improve safety for drivers, riders, and pedestrians. Over the past decade, there has been an increase in energy demand. As a result of this, halogen-based street lights were replaced by LED technology, which significantly reduced energy consumption. Since the LED light system was introduced in street lights, quite less improvement has been implemented. There are a lot of improvements that can be made to overcome manual operating, difficult-to-address faults, and to make the street light more energy efficient. The proposed model deploys Raspberry Pi and a microcontroller in street lights to make them automated and to conserve energy by adjusting the street light intensity to low when no movement is detected. Movement detection is done using a camera through image processing. Using OpenCV, Image processing is done. As this system uses a Raspberry Pi, every light can be connected to a network which helps in maintenance and also remote access can be taken to control the street lights, and faults can be addressed easily.

Keywords—Automated, Camera, Fault Detection, Image processing, Light Intensity, openCV, Raspberrypi, Street Light.

I. INTRODUCTION

The advent of led lighting had a significant effect. Since LEDs are more energy-efficient and have a longer lifespan than other light sources, they have replaced tungsten and halogen lights. Eventually halogen street lights are also replaced by LED lights. Even though LED lights use less energy, there is still room for development to improve energy consumption, and automation. Street lights are usually turned on and off manually. A timer was used to turn the lights on and off automatically, however this method is ineffective since it depends on time. The timer-based system does not function as intended if there is a significant change in climate. There are some types of faults that might occur in street lights, like LED array faults, core supply failure, and LED driver failure. It seems challenging to manually troubleshoot these errors.

The development of a model that remotely detects a street light malfunction and intelligently regulates the street light is essential. This proposed idea uses image processing and machine learning methods to control the light's intensity based on movement sensed by the camera on the road. OpenCV is used to read live footage from the camera, identify objects from the input as either vehicles or people, and adjust the intensity in response to movement observed. The Raspberry Pi functions as a processing unit in this concept, with one unit being connected to another via network. Each unit can be remotely accessed over the network to make changes to the system. It can also be used to accurately identify faults and take appropriate action.

II. LITERATURE SURVEY

A S Mohammed Shariff et al work suggests a method for detecting the vehicle's number plate. The suggested approach uses python and open cv. The solution suggested analyzes images using PyTesseract and the OpenCV library, converts input to grayscale and removes unnecessary characters, and subsequently performs optical character recognition using the edge location technique[1].

Apeksha P Kulkarni Vishwanath P Baligar work is about real time vehicle detection and counting. The proposed methodology calculates the number of cars within a monitored area by identifying, tracking, and gathering them using a Raspberry Pi. Furthermore, it computes data, records automobile video streams, and formats it in OpenCV format [2].

Ahamed Yasar Z et al suggest an approach using OpenCV for broader Surveillance. The concept suggests employing LoRa modules to create a computer vision system for sound notifications and protest site monitoring. It is divided into two primary sections: a receiver end equipped with an Arduino and a warning flag marker, and location detection and reaction[3].

Murugan V et al suggested a system for surveillance of vehicles and traffic using deep learning RCNN. In the vehicle discovery process which entails outline extraction, foundation estimation, and classification using the RCNN lowers computing complexity and guarantees precise, affectable, specific, and correct vehicle identification[4].

Shivraj Dhonde et al have proposed a method to detect over-speed and license plate of the vehicle. This research project explores the subject of video-based vehicle locating technology, focusing on practical and affordable data collection techniques. It proposes to provide a coordinate framework for permit plate location and speed using optical character recognition, YOLOv5 computing, and vehicle speed [5].

Modabbi,Arshad Mohammad proposed a method for The Street Lighting System's Energy and Economic Analysis of Smart Technologies. Smart PIR sensors and LED street lights can work together to lower operating expenses and power consumption in metropolitan areas. For six months, a traffic-aware smart PIR street lighting system in Aligarh that was installed on 36 street lights demonstrated a 96–97% reduction in electricity usage over traditional lighting systems. These sophisticated lighting systems are capable of controlling power and expense usage [6].

Khin Pa Pa Aung et al proposed an automatic system for the detection of license plates in Myanmar. The suggested License Plate Detection is an automation system that performs automatic license plate recognition (ALPR) using image/video data. By using image processing technology, the system is able to identify license plates by looking for white borders on them. Utilizing bounding box technology, morphological operation, and edge recognition, the system precisely extracts the license plate region. Lighting, backdrop scenery, and a variety of plate kinds are ALPR challenges. In open spaces, the system is essential for precise license plate recognition [7].

Ranjitha L et al work represents a smart light system for the streets and also a density based system traffic. The suggested method uses IoT. In the suggested approach the traffic signals are produced according to the density of the traffic. For monitoring the density of the traffic Infrared Sensors are used which also alerts timing and output. Using GPS modems, the device is able to locate hospitals and ambulance services nearby. Red signals are turned off and green ones are put in place during critical exchanges. Using timers and LEDs, an automated system senses the presence of cars and pedestrians and turns off the lights when they leave [8].

J.V. Anchita Alagammai et al work suggests a solution for detecting the faults and reporting it along with automatic control of street light. This solution uses IoT which enables remote monitoring ,good maintenance along with reduced downtime. These solutions help save money and provide an atmosphere that is more effective and environmentally friendly [9].

Muhammad Umair Khan et al suggested a methodology for implementing smart street lighting based on embedded

systems using Raspberry pi and Computer vision. The project features a Raspberry Pi 4 camera module and OpenCV-powered smart street lighting system. The system achieves great accuracy in a variety of conditions by detecting approaching cars, estimating how fast they will be traveling, and determining how many lights need to be turned on [10].

Priyansh Maheshwari et al have made a comparative analysis on a surveillance system for road traffic density which uses Infrared sensor, Raspberry pi and image processing. An increasing number of cars, which causes traffic jams and noise pollution, is covered in the study. It recommends de-clogging techniques including effective traffic signals and round-the-clock IR sensor and Raspberry Pi monitoring. The findings provide affordable and effective alternatives for places with moderate traffic volume [11].

III. PROPOSED SYSTEM

The proposed idea aims to automate street lighting as well as conserve energy as much as possible and to avoid manual fault detection. Raspberry Pi 4 serves as the central processing unit for this proposed model, which features a quad-core ARM Cortex-A72 processor with 4GB LPDDR4 RAM, providing significant improvements in processing power. This increased processing capability and Having more memory enables the model to handle larger datasets and run more memory-intensive applications. It offers extensive connectivity options, including Gigabit Ethernet and dual-band 802.11ac Wi-Fi, which facilitate fast data transfer, reliable network connectivity, and seamless integration with a wide range of peripherals and devices. This is very useful for reliable remote access for trouble shooting in the street light and also modify as required Despite its increased processing power and feature set, the Raspberry Pi 4 remains energy-efficient, making it suitable for battery-powered and energy-conscious applications.

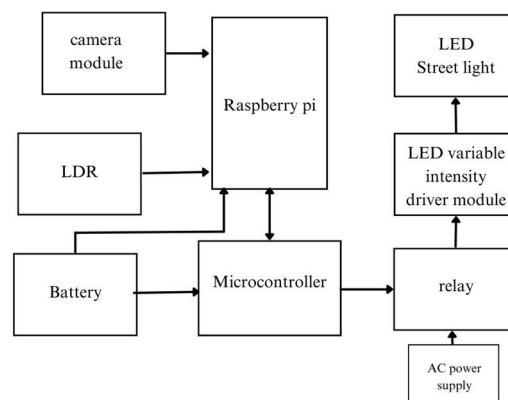


Fig 3.1 Hardware block diagram of proposed model

The proposed solution's hardware block diagram is shown in Fig. 3.1. With the Raspberry Pi, the camera module and LDR sensor are interfaced. The ambient light level is detected by an LDR sensor. With the use of this data, street lights are able to be turned on and off automatically according to the ambient circumstances. A microcontroller and Raspberry Pi are linked to operate the street lamp.

Microcontroller is interfaced with street lights through an LED driver and relay. Raspberry pi and microcontroller are powered by batteries and each model can control a series of Street lights.

The OpenCV framework is used to identify objects and movement from the camera feed. The script continuously analyzes video frames for objects (such people or cars), classifies them according to their attributes, then examines every object in detail, frame by frame to identify movement using methods like contour detection and background subtraction. Based on the detection results, the street light's intensity is adjusted. If a human or vehicle is detected, and movement of the detected object is observed, the light intensity is set to high. Otherwise, it remains at low intensity. This control is achieved through GPIO pins on the Raspberry Pi, which interface with the street light's control system through Microcontroller.

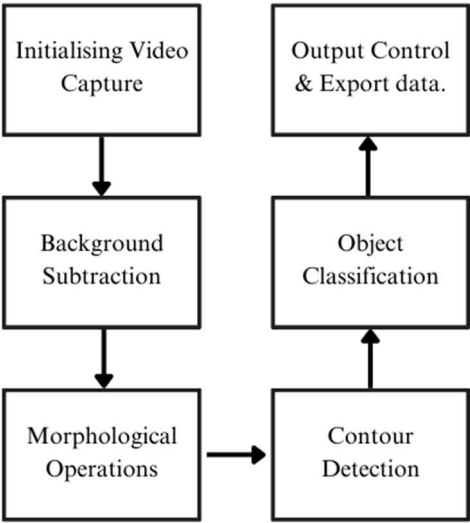


Fig 3.2 Software block diagram of the proposed model

Computer vision employs the Mixture of Gaussians algorithm for background subtraction, separating moving objects from a stationary background in video sequences. This algorithm updates the background model parameters based on pixel values observed in video frames. The 'cv2.createBackgroundSubtractorMOG2()' function in OpenCV creates a background subtractor object using the Mixture of Gaussians algorithm, specifying parameters like history length, foreground pixels detection threshold, and shadow detection.

Morphological operations, performed using the cv2.morphologyEx() function in OpenCV, are pivotal in refining the foreground mask derived from background subtraction for accurate object detection. These operations, including erosion, dilation, opening, and closing, manipulate binary images to eliminate noise, fill gaps, and enhance object shapes.morphological operations serve as a crucial preprocessing step, enhancing the reliability and accuracy of object detection algorithms by preserving meaningful objects while suppressing noise and artifacts within the foreground mask.

Contour detection is a crucial process in computer vision,

identifying continuous curves outlining objects in an image. It uses OpenCV's findContours() function to classify contours as "Person" or "Vehicle" based on their aspect ratio. This process provides visual representation of detected objects, providing insight into their location and size within the image, making it essential for various computer vision applications. Depending on whether movement is detected or not, you're updating the output state of GPIO pin 8 and displaying a corresponding message on the video frame.

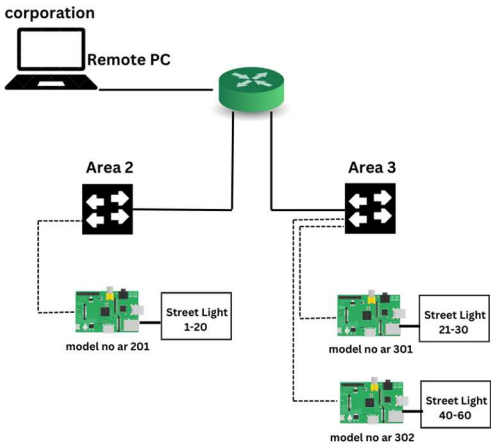


Fig 3.3 Network diagram of proposed model

fig 3.3 depicts the network diagram of the proposed system. The suggested method can control a series of street lights. All the raspberry pi controllers will be connected to a network.The raspberry pi can be remotely accessed using VNC viewer and putty via SSH protocol as they are connected in a network. The proposed model significantly reduces manual fault addressing, instead using remote access faults that can be detected. Fault will be detected with the help of a series of LDR sensors placed near the light array. if all the series of LDR did not get the light input then there might be a core supply failure or a driver fault . if only some LDR did not get the light input, there might be damage in a certain section of the LED strip.

IV. RESULTS.

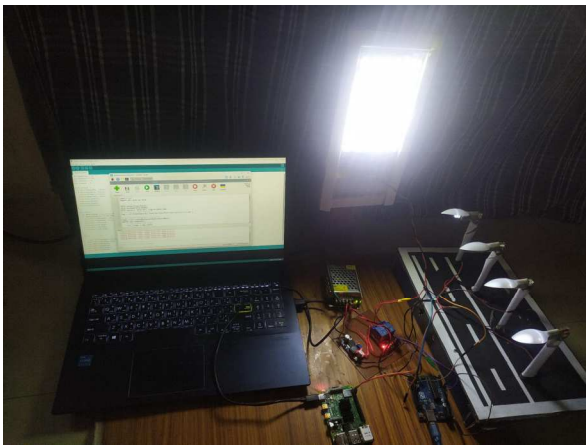


Fig 4.1 Hardware block diagram of proposed model

Fig 4.1 represents the working model of the suggested system.The figure helps to understand how the components

are integrated together. The raspberry pi is interfaced with the microcontroller and the following results are obtained upon execution of the program.

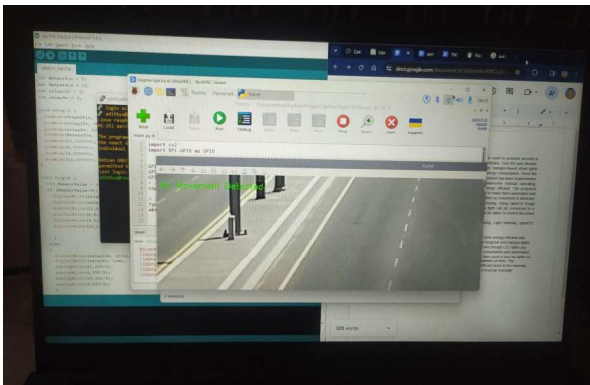


Fig 4.2 Street light at low intensity (no movement detected)

The results of this suggested model are seen in Fig. 4.2, when the vehicles are not present. The light's intensity is rather low because no movement has been noticed.

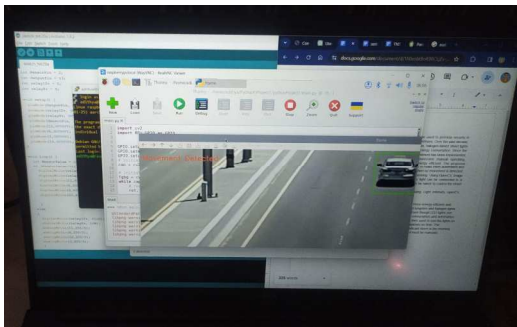


Fig 4.3 Street light at high intensity (movement detected)

When automobiles are present, the study's findings are displayed in Fig. 4.3. When the vehicle is identified to be moving, the light's intensity increases. Along with that in the frame there will be an indication whether the movement is detected or not.



Fig 4.4 lux value of light at high and low intensity

The fig 4.4 depicts the luminance value of the street light when the intensity is high and low. When the intensity of the

light is high the lumens value varies between 2000 and 3200 for a maximum energy consumption of 24 watts whereas when the intensity of the light is low the lumens value varies between 600 to 800 for a maximum energy consumption of 16 watts.

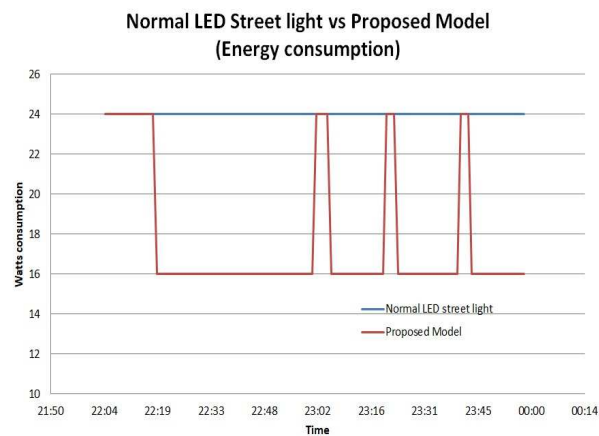


Fig 4.5 Power consumption of Existing and the proposed model

Fig 4.5 shows the comparison between normal LED street light and the proposed solution in terms of the energy consumption. This data has been taken for a period of 1 hour. The blue color depicts the normal led light which consumes 24 watts for an hour, whereas the red color depicts the proposed model which consumes 24 watts at maximum intensity and 16 watts at low intensity. In this time period four vehicle movement has been detected. When the vehicle is encountered light will glow with high intensity otherwise it will be maintained in low intensity. In the given input feed vehicle movement is detected for 15 mins in 1 hour of time. That is for 18 mins 24 watts of energy is consumed, the remaining 42 mins street light only consumes 16 watts. The average energy consumption of the proposed system is 18 watts in this test example. That is 25 percent of the energy is conserved.

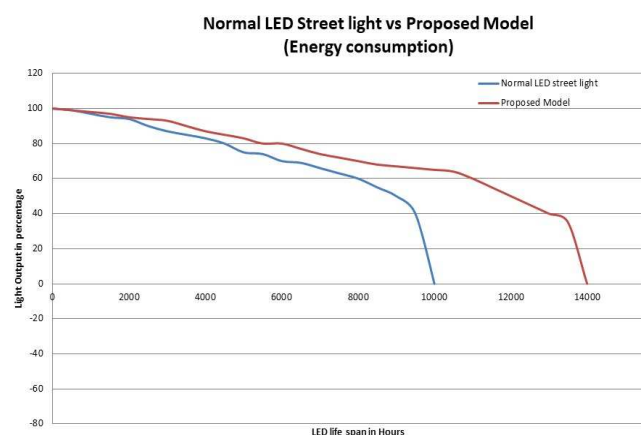


Fig 4.6 Life span of Existing and the proposed model

fig4.6 shows the comparison between a normal street lamp and the proposed system. As the power consumed by the light is less in the proposed model it eventually increases the lifespan of the light. Average lifespan of street lights is 10000 hours. Through implementing this proposed model, streetlights only consume 75 percent of maximum power.

Hence the lifespan of the street lamp will be improved 20 to 30 percent above the average lifespan.

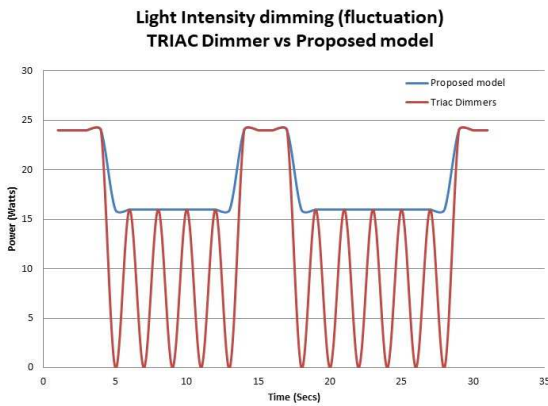


Fig 4.7 power fluctuation in TRIAC and Proposed model.

There are many methods to control the intensity of the light. TRIAC dimming technique is used to control the intensity of the light through AC supply. dimming the light using TRIAC will result in flickering of light because of the fluctuated AC supply. fig 4.7 shows the fluctuation effect of TRIAC dimmers and Proposed model.

There are a few difficulties that arise throughout the research and implementation phase. The initial implementation requires more time and can only be carried out in batches. If this system has to be implemented in highway roads, a high fps camera has to be used whose initial investment is high but energy conservation will be more effective in this type of roads.

V. FUTURE WORKS.

The project will modernize the city's street lights and infrastructure as part of the Indian government's plan for smart cities. This project may be amended to meet future needs. power consumption can be still reduced by implementing solar powered light in this proposed model. This project may include functionality for detecting accidents and fire occurrences in future versions. Future automotive tracking systems will benefit from this research.

VI. CONCLUSION

The development of artificial intelligence has the potential to solve issues and raise living conditions. The proposed model achieved its main objectives, that are minimizing the power consumption, automating the street lamp and easing the conventional fault detection method. Regular training of object and movement detection is also required. The importance of street lighting in our daily lives cannot be overstated. This discussed model works with all street lights and can be practically implemented. The proposed model will be more helpful in upgrading the infrastructure of the cities. This would greatly impact infrastructure development for smart cities in India. In comparison to an equivalent model, this one has more features and is more adaptable to changing demands. With this lighting system, we can reduce the amount of electricity consumed for street lights even while the demand for power is always rising. Through the application of modern technology, this approach might be

used to improve traffic laws even more. Future updates to this model could add features like fire and accident detection. In the future, this proposed model will be useful for tracking automobiles.

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Energy Efficient Street Light Controller for Smart Cities

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Abstract—Smart City-based electronic product applications are gaining importance nowadays. It contains efficient urban mobility, efficient public transportation, e-governance, safety and security, smart lighting system, etc. The main focus of this work is to present a design of street light controller to provide a reduction in power consumption and wireless control which eventually will provide a reduction in the required budget of electricity for street lights. Reducing power consumption leads to a reduction in brightness of lamps. This kind of street lighting states to open street lighting that adapts to movement by cars, walkers or cyclists. The system will work as adaptive street lighting. Lights will be obscured when no action is detected and will light up when some movement is detected. For this situation, the lighting framework is not quite the same as old-fashioned, static enlightenment, or dimmable road lighting that diminishes at pre-decided circumstances. Excessive lighting can be banned by targeted dimming of zones of the city, roads or individual luminaires. According to traffic and crowd, the light intensity will be controlled. This paper also incorporates light sensors which will be useful in a rainy environment. Number of street lights will be controlled by the user at a remote place. The complete system will work on RTC where lights will be running at 100% intensity in peak traffic time and with reduced intensity after peak traffic time.

Keywords—Energy Efficient, LED Street Lights, Dimming System, Real-time Control, Pulse width modulation (PWM), ZigBee network, Centralized user interface;

I. INTRODUCTION

A. Background and motivation

The street light controller is a smarter version of the mechanical or electronic equipment previously used for the street light ON-OFF procedure. Energy effectiveness is the main issue when arranging enclosed or open air lighting frameworks. Lighting can represent 10–38% of the aggregate vitality charge in a run of the mill urban areas around the world [1]. Delivering street lighting is considered the major and costly task of a city. Street lighting is mostly a vital issue for civic consultants in developing countries and the reason behind this is tactical standing for communal as well as financial steadiness.

Ineffective lighting system trashes substantial economic assets each year. On the other hand, the poor lighting system can produce risky circumstances. Energy effective utensils and policy can change street lighting budgets radically (frequently by 25-60%) [1]. These funds can cut the requisite for new power plants. It will deliver the resources for another energy way out for the people who live in distant regions. Those saved funds can also empower cities for enlarging street lighting to extra regions, growing entree to lighting in underserved regions. Likewise, advancement in lighting feature and advancement in facilities may lead to progress security conditions for walkers and automobile traffic.

A perfectly designed, energy efficient street light controller has to allow people traveling at night with better visibility with security and ease when decreasing energy usage and charges. This also should enhance the presence of the vicinity. Quite often, street lighting practices outdated lighting technology, ill-designed and defectively sustained and, hence consuming enormous amounts of energy and monetary assets, and its effects to fail to afford superior lighting.

The broadest explanations behind wasteful road lighting frameworks in cities are:

- Poor design and installation
- Selection of inefficient luminaires
- Deprived power value
- Deprived setup and maintenance performs

There are enormous latent for progressing lighting features while dropping energy usage, charges, greenhouse gas emissions, over energy effective retrofits for street lighting, better-quality operation as well as conservation performs.

On the old-fashioned lighting criterions and disorganized appliances, the lighting schemes do not cause energy losses and regular replacement of parts. These obsolete techniques experience the ill effects of the deficiency of inescapable and viable communications, observing, computerization, and fault diagnostics issues. To address these difficulties, a few innovations have been

utilized in the writing to spare energy, for example, the use of the light emitting diode (LED) rather than metal halide (MH) lights [1][2]. Be that as it may, the system in light of these innovations need additionally progress to build the energy profitability Effective energy-efficient street lighting strategy integrates minimum energy and meeting necessities for visibility and proper light levels, resourceful lamp technologies, finest pole height and assignment, effective light distribution.

India, US, Canada, and many other countries have started introducing street light control system to their road lighting for street light management, energy conservation and maintenance purpose. It helps to reduce carbon footprint and saves the environment. In a commercial point of view, it also helps to earn carbon credit for the monetary benefit [1][3]. Objectives to present are

1. To show a smart lighting system which concentrates on the self-sufficient operation and the vitality saving money on temperate moderate for the avenues.
2. Manufacture an energy saving smart lighting framework with coordinated sensors and controllers. Plot a smart lighting system for which makes the structure flexibility and expandability.
3. Plan a smart lighting structure with uniformity and versatility with the different business thing and motorization system, which may join more than lighting systems.

B. Present Scenario

In the greater part of the urban areas, the road lights are settled and maintained by municipalities. Most urban and semi urban communities are as yet utilizing a blend of fluorescent, CFL, high-pressure sodium lights or metal halide bulbs, that are not proposed to meet area-wise lighting requirements. Little study or methodology has gone into the illuminance needs in various zones of roads, to report the fundamentals of vehicular activity and people on foot. For example, the lighting necessities of vehicle activity in rapid locales are unique from low-speed high movement districts. So also, lighting needs in street intersections are not the same as auxiliary streets. Of course, the lighting necessities of an area with vehicular movement will contrast from that of a locale with high person on foot activity.

Lighting can represent 10–38% of the aggregate energy charge in a run of the mill urban communities around the world. In this, Gas based units – 22971.25 MW, Coal based units generate 154170.89 MW, Diesel – 1199.75 MW, , Hydro – 40867.43 MW, Renewable Energy - 31692.14 MW and Nuclear – 4780 MW. The part savvy offer of introduced limit is State-claimed units – 37%, Private units - 36% and Central units – 27% [4].

The road light plan is about the enlightenment of light as well as the tallness of the lighting post, and this thus varies in view of the prerequisites of that particular zone in the city. Because of an absence of district insightful review, typical recommendations are allotted on a city-wise premise, prompting the advanced spending plan experienced in the city lighting framework. Regularly, one notification that the road lights remain ON in the great sunshine. These things happen in view of the lights are turned off in light of a pre-chosen time instead of lighting needs, and this changes in view of season and area of the city. Thus there is a need for detailing a well thoroughly considered approach to keep

the use of electrical energy. May be, the organization can consider applying Street Light Control System by Light Dependent Resistor, that can actually kill bulb light while the daylight drops on it.

Denied support of road lights is extra issue gone up against by general residents. The cities are hard constrained for capitals, and the general population needs to confront the effect. These days, sun power era has turned out to be more pragmatic and have most recent improvements in viable Photovoltaic Units. Outcomes of these are in the utilization of free road lights that have an insignificant to-no reliance on matrix based electric supply. At the point when road light is utilized with greatly low-power devouring however high on glow LED globules, the potential power spared from the grid based supply is critical. With all this volume generation, the each capital cost of LED lamps and the solar panel will decrease, and will make the reception significantly more feasible.

II. LITERATURE REVIEW

A. Related Work

Automatic Street Lights is all about to control the power consumption at the streets and to eliminate workforce. This includes controlling a street lights circuit with some specific sensors, Micro-controllers and LDR during day and night. During daytime street lights are not required to glow, so the LDR keeps the street light off until the daylight level is low and glows in night time while the resistance of LDR is high. [5].

The energy effective ZigBee-based open air light checking and control framework reflects the client's requirement and framework energy exhaustion. The anticipated ZigBee framework likewise connected the standard routing calculation that impacts in enhanced network introduction when contrasted with the customary frameworks. This can carefully modify the brightness of the LED lights as per the daylight conditions. Likewise, the planned framework can distantly screen the lights status, control utilizations, and the charge of the individual lights and also of the aggregate framework [6].

To additionally slice the energy utilizations and to condense the wiring assembly, various lighting controller frameworks have been recommended to determine that issue, for example, inhabitation detecting technique [7], light level change [8], and Power line communication. In spite of dropping the wiring assembly in PLC centered plans as given in [9], periodic falls may happen in PLC systems working on low voltage transmission lines.

Various designs are termed in [10] offers a summary of numerous studies commerce with smooth and smart street lighting systems. From design examination, it can be seen that efficient street LED lighting systems, with greater charges, do have the peak energy savings as well as the speediest rate of profitability time.

According to the investigation in [11], transforming the original brightness into "brilliant state" and the "dim state" acknowledged energy sparing lamps, as well as guarantees the typical passerby walk. For this reason they utilize induction coils. That energy sparing control framework's energy utilization has a positive association with the energy of bulbs in the dimmest state. In this manner, under the enlightenment standard to limit the energy in the dimmest state can viably enhance the energy

productivity.

The structure is focused on a WSN arrangement of the broad scale sort that empowers the remote switch of street lighting bulbs [12]. The framework likewise permits reserve funds in the electric power and support costs. The engineering utilizes incorporated Doppler sensors that allow for vehicle recognition and helps finish the power productivity objective. The road lighting checking framework empowers the show of periodical reports concerning the energy utilization and giving points of interest on the upkeep work accomplished or modified for a future date. Aside from the server presentation, the control focuses on Apache 2.44 web server considers the distant contact of the manager by means of a website page.

In [13] author suggests the Lighting-Enabled Smart City Applications and Ecosystems outline. It comprises a) controllers, to incite physical procedures, for example, lighting, b) broad remote Open-air Lighting Set-ups, to interface the streetlights using either work, or cell systems, and c) sensors, to gather heterogeneous spatiotemporal information about the city, d) cloud-based applications, that will prepare the data gathered and also spread by the all-inclusive remote sensors organize.

B. Lamp technology

A discrete programmed energy sparing gadget bound to the light darkening in general lighting system commanded by High-Intensity Discharge lamps and Magnetic ballast [12][14]. The circuit input voltage inside three levels 170V, 190V and 230V. The determination of the level is done naturally and relies on upon the hour time. That dimming method was first mimicked and afterward tentatively validated utilizing two power levels. The fundamental downside is the low power factor as well as consonant waste, that must be enhanced via a superior plan.

The most urgent component of the illumination framework is the light source. These have the energy efficiency, visual element and spending parts of the light framework. A light source is an element that interprets electrical power, into electromagnetic radiation known as light. The rate at which electrical energy changes over into obvious light is called luminance productivity. It is measured in lumens per watt.

Light-emanating diode (LED) innovation is an advancing innovation with vitality sparing point of view. Operating for an average of 10 hours per day, LEDs have a lifespan of up to 13 years, and provide a nice spectrum of light. The lifespan and performance depend on system design, the superiority of the LED, working environment, and lumen devaluation factor over a period.

Even though the direct cost of the LED is more than the cost of HID lamps, the energy depleted by the LED is either half of the lamp's energy or less than that. In addition to that LED lamps are last longer than conventional lamps and these causes in vital savings. The LED lamp installation does not require a capacitor or ballast. It changes to low voltage direct current from the source voltage using a small power supply.

In Akola Municipal Corporation, Urban Local Body situated in Maharashtra state, lights were distributed with more than 11,500 customary street lights with capable T5 fluorescent tube lights. The venture that was using an energy speculation reserves execution contracting approach,

has realized imperativeness stores of 2.1 million kWh consistently a 56% diminishment in the ULB's essentialness usage for street lighting. Those electrical energy sparing assets have realized cost stores of about INR 6.4 million consistently, and the venture funded for it in only 11 months. The project's achievement headed related projects in other parts of Maharashtra [1].

Los Angeles city in the USA has permitted a \$57 million retrofit advancement, including the substitution of 140,000 city road light stuffs with LED apparatuses and the establishment of a remote watching framework to gather status and report constant execution information for every installation furnished with the tools [1].

III. PROPOSED SYSTEM MODEL

A. Single Unit

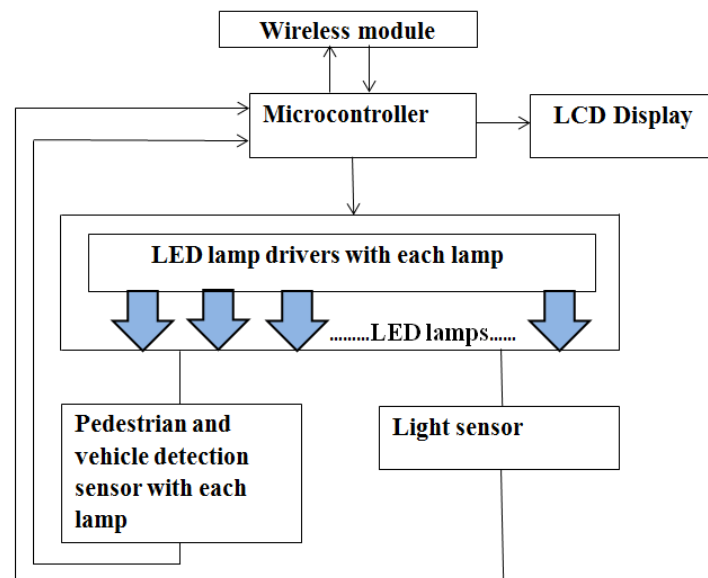


Fig. 1. Single Unit

The single unit consists of microcontroller, LCD display for microcontroller, LED lamp circuit driver, Light sensor and pedestrian/vehicle detection sensor as shown in Fig. 1. To reduce power consumption and to achieve security sensor's role is important. Hence to detect whether pedestrian or vehicle is going on the street, the related system is required to use. In rainy season or bad-light condition, sufficient light is required for visibility even in regular daytime. So the corresponding light sensor is used for that purpose.

Public activity stream around evening time are not the similar each time. Urban street public stream and movement stream in the 18 o'clock to 24 o'clock is at the peak in mean number however following 12 in the night the activity stream will be essentially diminished. Nonetheless, urban street lighting has stayed at a high shine for 8 hours, expending tremendous measures of power, so that the energy continues depleting endlessly. To address this issue, the system should work on the real-time clock, so that after 12 in the night, light intensity should be reduced.

B. Whole System

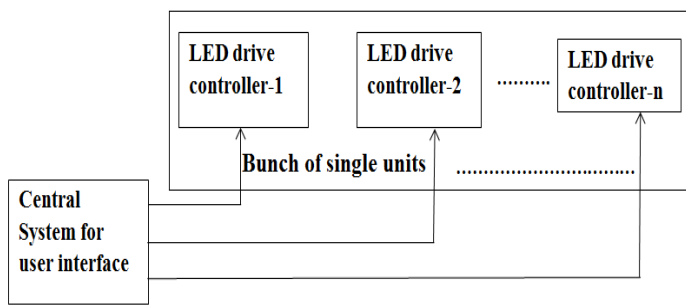


Fig. 2. Whole System

Limited number of lamps will be controlled by one microcontroller. To expand the system over a large area, number of controllers will be required to control each lamp. Hence more than one module is designed to work under such circumstances as shown in Fig. 2. It is also important to know the status of LED lamps that is at which brightness they are. In addition to that system should be operated manually so that manual ON/ OFF or brightness of lamp can be changed. For that purpose base station or central PC is used on which manipulation can be done by GUI.

IV. SYSTEM DESCRIPTION

The concept of adjusting the intensity of street light is mainly interrupted by LDR, Real Time Clock (RTC) and an infrared sensor as shown in Fig. 3. The ARM controller is interfaced with LDR and Infrared vehicle detection circuitry before inbuilt RTC. The controller output is given to LED driver circuit which will have a connection with actual LED street lights. Changing the duty cycle of PWM is responsible for brightness and dimness of street lights.

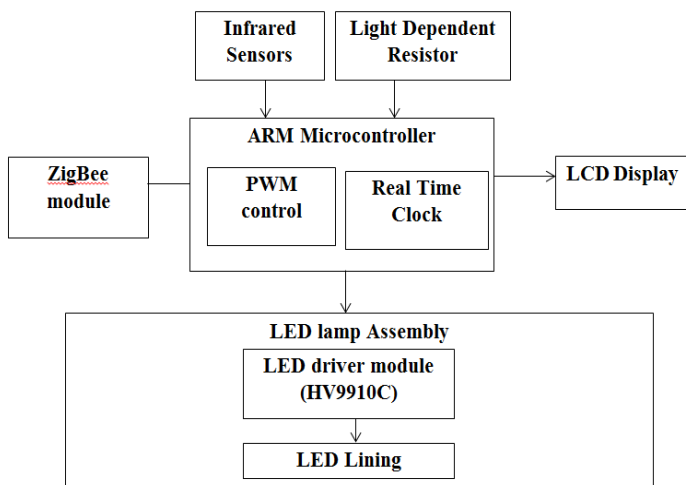


Fig. 3. System Description block diagram

The brightness control helps in sparing energy in late evenings while movement mass on the streets is little. A programmable ARM processor is locked in to give distinctive brightness at the diverse circumstances of night utilizing PWM method, for energy sparing. LDR interfaced with ARM controller follow the switching operation

precisely. The infrared sensor will keep track on intensity depending on the vehicle detection. Street lights are driven through LED driver card (FL7733).

Again we want low-power, low-cost wireless network. This is achieved with ZigBee network. A ZigBee module is used with every controller. This ZigBee module will transmit and receive data and controls between base station and controller. Its low power utilization limits correspondence separations to 10 to 100 meters observable pathway, subject to power yield and eco-accommodating components. ZigBee gadgets can impact data over long separations by going data through a working system of in the middle of gadgets to touch more inaccessible ones. In proposed system ZigBee network is used. For implementation purpose two nodes are used and those will communicate with central PC.

The framework was intended to work in stand – alone, by the energy provided from the AC mains. The benefits from this kind of energy are important thus dodging the tedious and expensive wiring and association with outside power systems, supporting significant assets and effortlessness of utilization. The plan is anticipated to be a low-control, diminishing the battery limit and furthermore the drive accomplished from the AC mains. The program that controls the framework is outlined essentially to abstain from wasteful energy. Fundamentally, as a result that the system works exceptionally in the night, evading overabundance of energy during sunshine hours. Moreover, the sensors enable the system to drive particularly when required. Once more, the framework works to a great degree prudent LEDs to ensure precise brightness and guarantee energy investment funds. In conclusion, when the framework is halted, all gadgets are in the rest mode, that led to minor power utilization. The wake-up is activated by the variety of situations. The assurance of the battery depends on upon the conditions where the structure is established.

This offered framework might be broke down as being overpriced; anyway we should consider its advantages. Insignificantly higher costs of the smaller scale controller are repaid by the inadequacy of costly outer wiring and the accessibility of energy system and perceptibly bring down rates of upkeep. Energy reserve funds are of most extreme significance today. The objective is the sparing of working estimations of road lighting framework portrayed by the straight forward establishment and low power utilization.

V. COMPLEXITIES INVOLVED

A. Utilization of port pins

As a solitary port of controller is associated with LEDs yet the calculation which we outlined will give a change of forces at various port pins. Thus, so as to have force varieties at the single port we may require mux-demux.

B. Befuddle in measurements of LPC header mount

As the PCB format is planned in ARES experts the genuine measurements of header mount of LPC 2148 contrasts from the accessibility of header mount the market.

C. Two-Way Streets and Adjacent Roads

Two stresses are emphasized for the proposed structure. The main is, the thing that to fulfill for the bidirectional roads or one-way street. In proposed models, the infrared sensor can choose the status of the control signal in perspective of the speed, position and the heading of each vehicle. The second issue is for the two adjacent roads, in which the Light Nodes is set in the midst of the two lanes. For this circumstance, Light Node has two lights and two stabilizers. Each offset can be controlled independently. In like manner, the infrared sensor can send control signs to each road independently. The lights all over the place that have vehicles will be lit and on the other adjoining street lights will be turned off or dimmed if no vehicles exist on it. This subsequently can reduce the energy usage.

D. Dimmable Light Knobs

In the event that the street or road lighting framework is dimmable, encourage energy sparing can be acquired. With the quantity of vehicles and the climate conditions the light power can be controlled. For instance, toward the start of the night, the light force can sit to 60% for the zones having vehicles. Furthermore, if the zone has just a single vehicle, the light force can sit to 40%, however in the event that there are numerous vehicles in a zone, the light power can sit to 100%. There exist numerous advances for the dimmable road lighting frameworks. Reducing light intensity decreases the perceivability of the drivers, which will hamper the security of drivers and individuals by walking. Look into about this theme has been made, the aftereffect of which demonstrates that the measure of diminishing does not appear to impact the perceivability of eyewitnesses until half Existence of Pedestrians. In the interstates or on the streets where there are no walkers turning the light off is allowed. That can enhance to energy sparing without risk. Be that as it may, in the avenues or in the downtown areas where there exist walkers, finish switching off of the light is not permitted all the night regardless of the possibility that there are no vehicles in the city. This is for the wellbeing of individuals and to take out the violations and so forth.

E. Light Node Maintenance and Monitoring

A fundamental as well as repetitive undertaking finished by the development experts is the support and checking of the street lighting structure. The Lamps have specific life-time and can be isolated at whatever point. Subsequently, it must be replaced rapidly to give the required level of detectable quality and from this time forward the prosperity. In any case, recognizing the prosperity status of the impressive number of Lamps in colossal urban regions containing countless scattered on an immense number of long roads and lanes is a tough and impractical task

F. Instance of Emergency

Concerning emergency cases, the offered scheme

gives the required level of detectable quality to the drivers. On the interstates, just vehicles are passing. The structure is proposed to cut the parts of the paths that have no vehicles at whatever point. Nevertheless, in the downtown territories, where people may exist in the city, the system does not cut the light, yet rather dimming it to the most lessened allowed keep running in a couple of periods which is half from the full power. It can be executed so that the modified control of the light can be evaded.

VI. RESULTS

The table given bellow gives an idea about power consumption of five LED lamps of 9W for one month. LED Street lamps are of about 36W and higher. We used 9W LED lamp as a prototype.

As LED lamp is of 9W, energy consumption for 1 hour will be 9Watt-hour. But electricity measures in kilowatt- hour, hence it is 0.009kWh. We used 5 LED lamps, hence total energy consumption for an hour will be 0.045kWh, and for a month (30 days) it will be 1.35kWh. So if we use LED lamp with full intensity, it will have an energy consumption of 1.35kWh for an hour using daily for a month. But with RTC we are decreasing intensity after peak hours so as to save energy. It is as given in the Table I.

Table I. Hour wise Power Consumption

Sr. No.	Scheduled RTC duration	Intensity (in %)	Theoretical Energy consumption (KWh) With reduced intensity	Energy Consumption At 100 % Intensity(KWh)	Actual Avg. Energy Consumption	Estimated Energy Saving (KWh)
1.	6:00pm to 7:00pm	80	1.08	1.35	1.0258	0.27
2.	7:00pm to 8:00pm	90	1.215	1.35	1.1540	0.135
3.	8:00pm to 9:00pm	100	1.35	1.35	1.28225	0
4.	9:00pm to 10:00pm	100	1.35	1.35	1.28225	0
5.	10:00pm to 11:00pm	100	1.35	1.35	1.28225	0
6.	11:00pm to 12:00am	100	1.35	1.35	1.28225	0
7.	12:00am to 1:00am	70	0.945	1.35	0.8975	0.405
8.	1:00am to 2:00am	50	0.675	1.35	0.6411	0.675
9.	2:00am to 3:00am	50	0.675	1.35	0.6411	0.675
10.	3:00am to 4:00am	50	0.675	1.35	0.6411	0.675
11.	4:00am to 5:00am	50	0.675	1.35	0.6411	0.675
12.	5:00am to 6:00am	30	0.405	1.35	0.3846	0.945
13.	6:00am to 7:00am	30	0.405	1.35	0.3846	0.945
14.	7:00am to 8:00am	10	0.135	1.35	0.128	1.215
	Total			18.9	11.6679	6.615

$$\text{Estimated Energy saving} = \frac{\text{Estimated energy saving}}{\text{Energy consumption full brightness}}$$

$$\text{Estimated Energy saving} = \frac{6.615}{18.9}$$

$$\text{Estimated Energy saving} = 35\%$$

$$\text{Actual Energy saving} = 1 - \frac{\text{actual energy saving}}{\text{full actual Energy consumed}}$$

$$\text{Actual Energy saving} = 1 - \frac{11.6679}{14 * 1.28225}$$

$$\text{Actual Energy saving} = 1 - \frac{11.6679}{17.95}$$

$$\text{Actual Energy saving} = 35\%$$

In the table it is shown actual energy consumption and theoretical energy consumption. Hence total percentage saving using RTC only for 14hrs is 35 % which is estimated. But it may save less than 35% energy when we take in consideration of interrupts from IR, consumption of other hardware components.

As we are using LDR and IR, the light intensity will vary according to that. Other peripherals like microcontroller, sensors and communication nodes also consume some energy. So this system can save energy up to 30 to 35% for a month optimally.

VII. CONCLUSION

In this paper a plan of an advanced street lighting system is described that adds different know-hows, proposing comfort of energy savings and maintenance. This is gained by using the innovative controlling of the street light.

The projected scheme is particularly applicable for street lighting in distant urban and rural regions where the traffic flow is low at eras. The individuality of the power network licenses to implement it in distant regions where the traditional schemes are excessively expensive. The system is flexible, extendable and entirely modifiable to user requirements.

This project is an economical, practical, ecological and the harmless way to save energy. It noticeably challenges the two difficulties that world are facing today, saving energy and also the removal of glowing lamps very powerfully. According to statistical information we can save more than 35% of the electrical energy that is now consumed by the highways. Initial cost and maintenance can be the drawbacks of this project. With the improvements in technology and good resource design the budget of the project can be cut down and also with the usage of good equipment the maintenance can also be reduced regarding periodic checks. For these causes our paper presents far more benefits which can over shadow the present boundaries.

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IoT based Smart Street Light Automation using Thermoelectric Transducers

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Abstract -The non-renewable energy sources are getting depleted rapidly and as it is also causing global warming, it's high time to act and take actions such as looking for innovative and renewable energy sources. Thermoelectric energy is ideal for stable energy generation. The generation and distribution of thermoelectric energy are cost-efficient and time saving. In major cities and rural areas, energy management plays a vital role in controlling global warming. The central part of public energy consumption comes from street lights, clearly, street light energy management is very poor and here the loss of energy is taking place, IoT-based Smart Street light automation using Thermoelectric Transducers comes into the picture, where the switching ON/OFF of the street light is done by the sunrise and the sunset timings which are different for different locations and can be obtained in real-time from statistically calculated data of the latitude and longitude. The method of switching saves a lot of energy and helps in the proper functioning of streetlights. This method is cost-effective and needs low maintenance compared to the conventional LDR sensing method.

Keywords—*Thermoelectric Transducers, Street Light Automation, Internet of Things-based Relay, ESP8266 Node Micro Controller Unit.*

I. INTRODUCTION

Considering the degradation of the Environment and recent climate changes it's high time to adapt to sustainable sources of energy from conventional sources such as Coal, Thermal, and Nuclear Energy. Energy sources such as Thermoelectric and hydroelectric are suitable alternatives for traditional fossil fuels. With the help of this transition from nonrenewable to renewable sources of energy, the harm created to the environment can be mitigated. Conservation of energy also plays a major role in the path of transition towards adopting sustainable energy. The conventional mechanism of automating streetlights is in a place where LDR (Light Dependent Resistor) is directly placed on top of streetlights where the resistance of the LDR changes with the change in the intensity of the light directed towards it. It senses the light and sends a signal to the controller so that the energy consumption can be reduced by turning off the Street Lights when the resistance increases, and the energy can be saved.

The major issue with street light automation using LDR is that the accuracy and the sensitivity of the LDR are usually not high as many influential factors should be taken into

consideration such as dust, humidity, and temperature. Sudden changes in weather conditions may also cause the LDR to malfunction, and the response time of the LDR is high which leads to irritability of the mechanism of automation of streetlights using LDR [5]. On the other hand, IoT-based street light automation using a thermoelectric transducer satisfies both the generation and conservation of renewable sources of energy, as a thermoelectric transducer generates electricity by converting heat. Thermoelectric transducers include both Solar and Peltier modules for energy generation which is a renewable source of energy generation [1].

The ESP8266 Node MCU is connected to the internet where reliable data is collected and processed by the device to control the relay. Using a relay connected to the Node MCU, switching streetlights is done efficiently based on the data of sunrise and sunset of that specific Latitude and Longitude. This mechanism is cost-efficient and requires less maintenance compared to conventional models that are currently in place for energy generation and management [10] [7].

The proposed model harvests the energy during the daytime, the harvested energy is used to power up the streetlights during the night time automatically based on the real-time data. Implementation of the proposed work helps in reducing the carbon footprint. In rural areas where access to the electric grid is difficult, a microgrid integrated with thermal harvesting and automation of streetlights enhances sustainability and overall quality of life in rural areas.

The paper consists of eight sections in total. The first section deals with the introduction of the proposed mechanism and the drawbacks of the previous models, the second section deals with the block diagram and the working of the proposed model, the third section deals with its components, the fourth section deals with the hardware model of the proposed mechanism, the fifth section deals with the result and data collection of the voltage and current values, the sixth section describes and concludes about the statistics of the observations that are made, the seventh section deals with the precautions that are to be taken while constructing the model, the eight and the final section consist conclusion and future scope.

II. WORKING OF THE PROPOSED MODEL

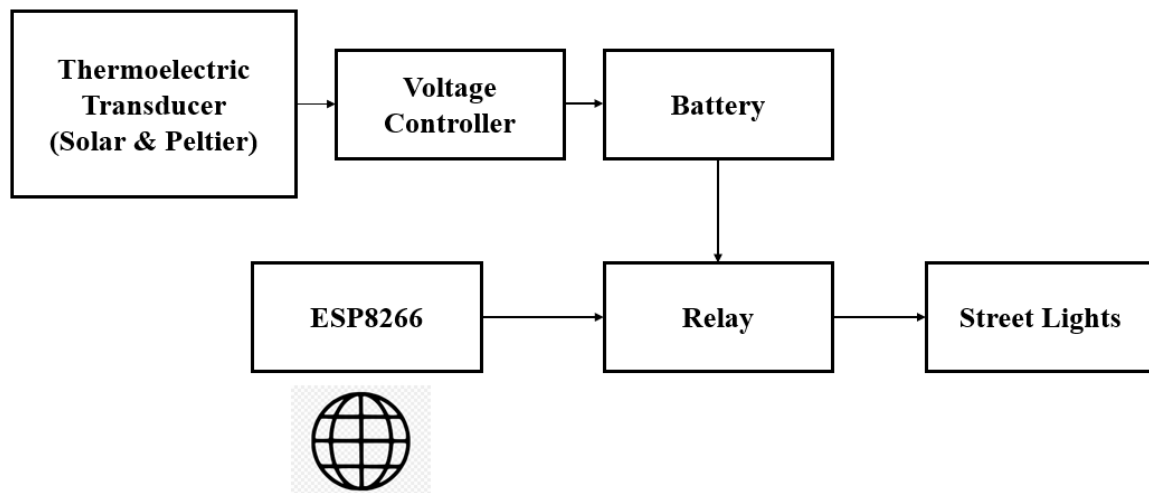


Fig.1. Block Diagram of the proposed model

In the proposed model shown in Fig.1, for the generation of electricity, a set of thermoelectric transducers which is a combination of both Solar and Peltier modules is deployed. The solar panel is a compact version of the thermoelectric transducer, the electricity is generated with the help of sunlight, when the light rays fall upon the solar panel the light that is the heat energy is then converted into electricity, and the solar panel is deemed as very efficient and can be used for various energy generation purposes. The photovoltaic cells that are present inside the module absorb the sunlight and convert it into electricity, a protective layer that is present on top of the solar panels will make sure that the panel will last longer.

The Solar panels are connected in series so that the output from the solar panel can add up to 12V and a maximum output of 100mA can be achieved by generating electricity from solar panels. The panel can generate a maximum power of 1.2W. The output is based on the sunlight and temperature to which the solar panel is subjected. The Peltier module is a thermoelectric transducer, the Peltier module is a dual channel energy flow module where one face of the module consists of a cool face and the other consists of a hot face. The internal structure of a Peltier module consists of coils that are wound together in a sequential manner which are used to produce or absorb heat.

The Peltier module has a rated current and a rated voltage on which the energy can be produced. When the Peltier module is subjected to heat or radiation, the heat inside the module flows from the hotter direction to the cooler direction. The coils which are present inside the Peltier module are subjected to the flow of electrons which are caused by the heat, which is flowing from the hotter side to the cooler side. The current flows through the Peltier module as the heat transfer takes place across the module. This method of energy generation is considered an effective method as a

particular set of heat sinks are being used for fast energy absorption and energy dissipation so that the efficiency of this method can be increased by generating more electricity. A voltage of 12V can be generated from the Peltier module when the modules are connected in series and a maximum power of 0.96 Watts can be achieved. The heat sinks are placed above and beneath the Peltier modules where a 12V DC fan is allocated for the job of dissipating the heat faster from the cool side of the module where the heat is subjected towards the hotter side of the module and the current is generated opposite to the flow of electrons i.e., opposite to the flow of heat.

The energy is generated from the Solar panel and the Peltier modules are combined by connecting both the thermoelectric transducers in parallel so that the voltage of the generated energy is maintained, and the current is added up together. The output terminals of the generated energy are connected to a Voltage Regulator commonly known as a Buck Converter, where the Voltage controller module maintains the voltage constantly by increasing or decreasing the flow of current through the output terminals of the Voltage Regulator. The output terminals of the Voltage Regulator are monitored constantly as the output voltage can be adjusted using an inbuilt potentiometer present in the module. The output terminals of the Voltage Regulator are then connected to the battery for storing the energy that has been generated. The developments in using the LDR to detect the sunlight and switching ON/OFF streetlights are not very promising as the system is failing in commercial and crowded areas as the light of nearby surroundings is affecting the sensing capability of the LDR, this is where the proposed model comes into picture for managing and conserving the energy.

A. Working of the IoT-based relay using ESP8266

The data that is collected is processed and updated at specific intervals of time. The microcontroller is connected to the internet with the help of Wi-Fi in which the SSID and the password can be configured accordingly. A relay is connected to the microcontroller and a digital pin is defined so that the relay can act when it is required. Based on the sunset timings at a particular latitude and longitude which is procured from the internet, the relay is switched on. Similarly based on the Sunrise timings the relay is turned off at that specific latitude and longitude. One end of the relay is connected to the battery and the other end is connected to the street light where the signal port of the relay is connected to the microcontroller. This method of automation of streetlights is very efficient and cost-effective, unlike the conventional LDR method where the efficiency is less and maintenance is high.

III. COMPONENTS

This section consists of the data regarding the components that are used in the construction of the proposed model and the specifications of the components.

A. Solar Panel:

A 1.2Watt solar panel as in Fig.2, can produce a voltage of 12V and a current of 100mA when it is subjected to sunlight. The amount of sunlight that falls on the solar panel, which is nothing, but the sunlight intensity is responsible for the amount of power generation. The solar panel is nothing, but a collection of solar cells connected in both series and parallel to obtain the maximum efficiency which is required to obtain maximum power. These are generally made with the type of material which will exhibit photovoltaic effects, in most cases, silicon is used to make solar cells. Voltage Regulator is used to regulate the output of the solar panel which helps to charge the battery. The solar cells are delicate and can be damaged easily so the solar panels are covered with an extra protection layer to avoid damage from wind, and rain so that they can last longer and can avoid damage.



Fig. 2. 12V Solar Panel

B. Peltier Module:

A 12V Peltier module shown in Fig.3, when it is heated up produces a current of up to 80mA, and the total output power will be 0.96 watts. This module works on the principle that heat is generally transferred from a hotter body to a colder body. The Peltier module operates based on

the principle of heat transfer from a hotter body to a colder body. A desirable amount of current is generated when a differential amount of heat is applied. The module contains a combinational array of thermoelectric transducer elements which in total contribute to a thermoelectric transducer. The efficiency of the module depends on the temperature differential and heat dissipation.



Fig. 3. Peltier Module

C. ESP8266 Node MCU:

The ESP8266 Node MCU in Fig.4, is the most popularly used communication device, runs on an open-source Lua-based firmware that can be connected to the internet. The device is highly efficient and user-friendly and can be easily programmed using Arduino IDE Software. The ESP8266 has a RAM of 128 KB and an internal storage of 4 MB of flash memory to store data and programs. It runs on I2C (Inter-Integrated Circuit) protocol and can be operated using both Wi-Fi and Bluetooth which makes the developer board ideal for IoT-based projects. The API (Application Programming Interface) keys can be obtained and can be programmed based on specific applications. This device is programmed in such a way that the algorithm enables the microcontroller to collect the sunrise and sunset data of a particular latitude and longitude, in real-time with the help of a designated API key when the microcontroller is connected to the internet.



Fig. 4 ESP8266 Node MCU

D. Voltage Controller:

A voltage controller popularly known as a buck converter as in Fig.5, can provide a stable output by adjusting the

potentiometer. The voltage can be maintained constant by adjusting the current output in various load conditions. A feedback loop is also used to continuously monitor the output voltage and compare it to the adjusted value. This type of voltage controller is used to enhance the stability of the system.



Fig.5 Voltage Controller

E. TP4056 charging module:

A rechargeable 12V lead acid battery shown in Fig.6, is mostly used in energy generation, and storing mechanisms. The battery is made up of lead dioxide and sponge lead which has a capacity of 1300mAh. The power rating of the battery is 15.6 watts, and the generated energy from the solar and Peltier modules is stored in the 12V lead acid battery. The battery comprises long-lasting cells, as the life cycle of a battery depends upon the temperature and the discharge rate of the battery. The Lead acid battery is highly efficient and cost-effective and is a compact option for such applications.



Fig. 6. Lead Acid Battery

F. 5V Relay:

A 5V relay in Fig.7, is a switch that is controlled using a microcontroller with the help of a digital or analog input connected to the relay. The relay consists of a coils and a set of switches which are in contact. A 5V relay signal is attached to the relay so that it can be controlled. It is a switching mechanism where the microcontroller won't be in contact with the external power supply which is connected to the relay. The relay is capable of handling up to 250V AC and 30V DC and can handle up to 10A of current.



Fig.7. 5V Relay

G. LED:

Light emitting diode also popularly known as LED is a solid-state device that is a diode that converts electrical energy directly into light energy. The light is emitted across a specific range of wavelengths.



Fig.8 . LED

H. Heat Sink and Heat Dissipator:

With the help of Heat sinks and heat dissipators as in Fig.9, the management and regulation of heat are done, which it makes ideal for managing the thermoelectric devices so that the thermoelectric devices can work efficiently. This passive cooling component is designed to dissipate heat energy. These materials are made with materials of high thermal conductivity. The fins on the heat sinks increase the area of dissipation of heat and hence faster generation of electricity, proper insulation is provided to attain high efficiency.



Fig.9 . Heat Sink and Dissipator

IV. HARDWARE IMPLEMENTATION



Fig.10. Hardware model

The hardware circuit is shown in Fig.10. The Peltier module is connected to a solar panel of 12V in parallel, and a pair of heat sinks are attached to the Peltier module for faster heat dissipation. The output terminals of the generated energy are connected to the buck converter and the output of the buck converter is connected to the battery so that the street light can be turned on. A relay is connected to the A0 pin of the Node MCU for switching the power supply of the street light which is connected to the battery. The generated energy is measured and analyzed with the help of a multimeter.

V. RESULT – DATA COLLECTION AND ANALYSIS

The energy generated from the Thermoelectric transducer, (Peltier) and Solar Panel is connected to the buck converter and the output is connected to the battery. The harvested energy is stored in the battery for future use. The stored energy is being used to power the streetlights. The switching ON/OFF of the street lights is done with the help of an IoT-based device ESP8266. This device enables the ability of the grid to connect with the Internet to procure real-time values of the sunrise and sunset of the desired location.

Date and Time	Voltage (Volts)	Current (Milli-Amps)
05/06/2023 & 9:00	11.4	136
05/06/2023 & 9:15	11.3	124
05/06/2023 & 9:30	11.9	176
05/06/2023 & 9:45	11.7	164
05/06/2023 & 10:00	11.3	122
05/06/2023 & 10:15	11.1	134
05/06/2023 & 10:30	11.2	173
05/06/2023 & 10:45	12	180
05/06/2023 & 11:00	11.9	178
05/06/2023 & 11:15	11.8	165
05/06/2023 & 11:30	12	172
05/06/2023 & 11:45	11.9	168
05/06/2023 & 12:00	11.8	174
05/06/2023 & 12:15	11.7	177
05/06/2023 & 12:30	11.9	175

Fig.11. Data Collection of Voltage and Current

The voltage and current readings are taken and are being recorded as shown in Fig.10, for the analysis and monitoring of the energy generation from the thermoelectric transducers.

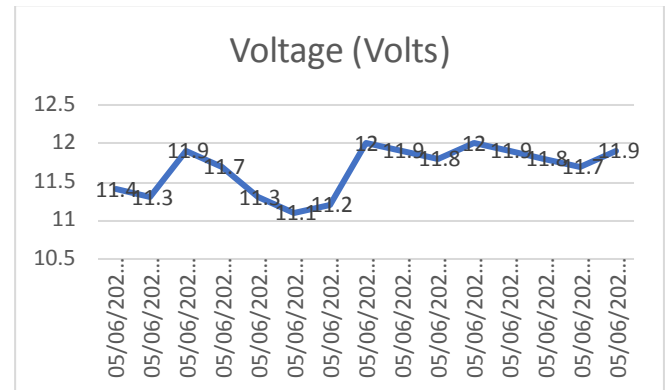


Fig.11. Graph of Voltage Generated

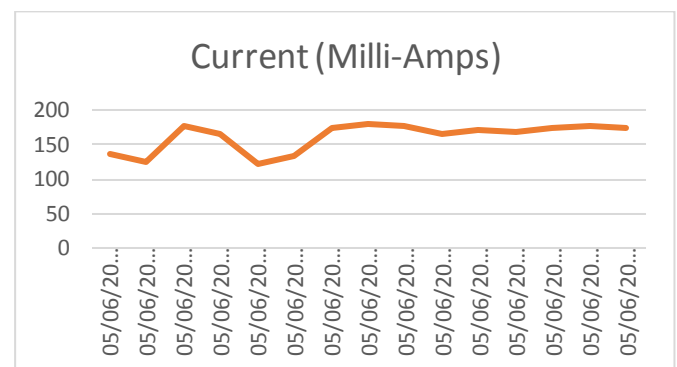


Fig.12. Graph of Current Generated

The voltage, current readings are measured and analyzed from the solar panel and the thermoelectric transducers. The thermoelectric transducers are connected in parallel so that maximum efficiency can be achieved. The constant power output is ensured in such a way that a voltage regulator is connected between the terminals of the battery and the voltage output of the thermoelectric transducer. The above-collected data is obtained from tapping the voltage output of the thermoelectric transducers for future prediction and is shown in Fig.11 and Fig.12 respectively.

The average voltage fluctuates from 11.1 to 12V depending upon the intensity of sunlight and energy generation by heat dissipation from Peltier. The maximum current is generated when the system is working under optimal conditions. Hence the maximum power efficiency of the model can be determined based on the maximum voltage and current outputs respectively. The performance efficiency of the model can be improved by regular maintenance of the components and by selecting the components of the desired power rating that are suitable for obtaining maximum efficiency.

VI. STATISTICS

- Maximum voltage generated by Thermoelectric Transducers **12V**.
- Maximum current generated by Thermoelectric Transducers **180mA**.

- Average voltage generated by Thermoelectric Transducers **11.66V**.
- Average voltage generated by Thermoelectric Transducers **161.2mA**.

VII. PRECAUTIONS

1. The terminals of the thermoelectric transducers are to be connected by checking the polarity.
2. The ESP8266 needs to be configured and the libraries are to be installed for proper output.
3. The terminals of the battery need to be connected according to the polarity of the Voltage Regulator.
4. The battery must be handled safely under specific temperature conditions.
5. Proper insulation and Heat Sinks are to be applied for proper heat transfer.

VIII. CONCLUSION AND FUTURE SCOPE

The energy generation using thermoelectric transducers and the switching of streetlights using a relay that is connected to ESP8266 node MCU ensures that the adaptation of sustainable and renewable energy from the conventional source of energy. The energy is generated through a Solar panel and Peltier module. The combination of harvesting the thermoelectric energy gives a high yield compared to other types of energy generation. The energy that is generated can be stored and can be used for later purposes as the storing of energy is made simple by using a lead acid battery which is durable and has high life cycles compared to other options. The maximum power generation depends on various factors such as the intensity of the sunlight that is directed toward the Solar Panels and the amount of heat supplied to the Peltier Module. Generation also depends on the power rating and the quality of the modules being used. The stability of the system can be improved by constantly monitoring the data and predicting any abnormalities in the generation of electricity.

The switching ON/OFF of street light is done using the Node MCU as the data of the sunrise and sunset of a particular location can be procured. This method ensures saving energy as the sunrise and sunset data gets updated frequently and the switching is made simple by simply connecting the NODE MCU to the internet. A lot of energy is saved by adopting this method which contributes to provide a safer and more sustainable environment for future generations.

The proposed model can be developed and can be integrated with major cities, microgrids, and renewable energy generation stations. This proposed model can also be implemented using a cloud-based platform where the grid and the energy-generated data can be constantly analyzed and controlled based on the requirements of the specific area. The generation and utilization of energy can be regulated using smart meters which help in real-time monitoring of the energy which is being generated. The model can be scaled up so that the energy losses can be minimized. Energy management and switching can be done directly through satellites in real-time.

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IoT Based Street Light Controller and Monitoring System

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Abstract— *Street lighting has become an essential commodity for human life as it provides lighting in cities, towns and most especially on roads to reduce risk of accidents at night. Though necessary, streetlight systems not properly managed can result in risk of accidents, excessive loss of energy, financial losses and dissatisfaction of customers. This paper proposes a smart system to manage streetlight operation, relying on sensors and internet of things technologies. The proposed smart streetlight system consisted of hardware and software design. While the hardware makes use of sensors, Arduino board and GPS devices to build an integrated system capable of reporting the state of the street light through available wireless network, the software aspect consisted of an online application, capable of remote monitoring and control. The smart streetlight was conceived, built and tested. Results showed an effective operation, leading to reduced energy consumption of the streetlight, subsequent reduction of carbon footprint, effective detection of fault, effective location of fault owing to GPS devices and reduced downtime. The system is further expandable to cover a whole town/nation streetlight system.*

Keywords— streetlight, automatic system, remote control, microcomputing, energy efficiency

Introduction (Heading 1)

Street lights are one of the essential services required for smooth transportation and security at night. Street lights have always been a great concern for public authorities because of their significant roles in providing reliable lighting and security to communities [1]. The amount of electrical power consumed by public street lights keeps rising, as well as the importance of street lights in our daily lives to protect and keep the citizenry safe. Millions of dollars are spent each day on street lights to provide the required electrical energy [2].

In Ghana street lights are present on all major and minor roads. They are normally powered from sunset to sunrise at full capacity every day. According to The Guardian online news portal “Ghana’s total road network length is almost 117,000km.” In a 2011 street light policy document, the

Government stated that a “lack of adequate developmental, regulatory and operational framework on street lights has rendered their continuous expansion untenable.” The policy stipulates basic requirements for operation and maintenance existing and new street lighting units. The basic requirements are: the maintenance of an up-to-date electronic database of all units to ensure the satisfactory management of a maintenance process that meets legal obligations and provides information for the calculation and payment for electricity consumed, controlling, monitoring, and maintenance and repair procedures that ensure expeditious responses to identified defects.

The policy goes on to encourage energy conservation and remote monitoring and indicated that the most efficient street lighting technology (including low energy consuming and remote monitoring technologies) shall be used as far as practicable. The Ministry of Energy (MoEn) in collaboration with the Ministry of Local Government is to promote new technologies such as Smart-grid and Automation in new and existing street lights leading to optimized operation, energy efficiency and reduced CO₂ and other greenhouse gas emissions. [3] reports that inadequate visibility is one of the key risk factors for pedestrian road traffic injuries (RTI). Reports from studies carried out in Australia reported a 59% reduction in pedestrian crushes following improvement in roadway lighting. “In Ghana, the rate of pedestrian RTIs is proportionately high, making up more than 40% of all road-user fatalities.”

Additionally, the increasing number of faulty street lights along our roads is very alarming. The major problems are the capacity of the responsible authorities to attend to these faulty lights and the mode of reporting and identifying the street lights with faults. The current systems employed have several problems like maintenance issues, timing problems, connectivity problems and inability to control display levels.

Evidently seen on most streets of the country are either no street lights or street lights that are not functioning properly – completely damaged by vehicles and left unattended to, bulbs are dead, or too dim to light the environment properly. With the current system, when a street light does not turn on during the dark hours or does not turn off during day time due to a fault, it is very difficult for the appropriate authorities to identify and solve the problem.

Another problem is the increasing amount of electricity consumed by street lights. In Ghana and Accra specifically, currently, it is reported that the capital city spends about GHC200 million annually on street lighting alone. As reported by Gunasundari et al. (2017), millions of dollars are spent each day on street lights to provide the required electrical energy on a global scale.

Another pressing problem is the inability to control the brightness intensity of the street light bulbs to help conserve energy. There are times when the highest levels of light intensities are not required because there is little or no human movement that requires too much light for visibility.

In view of the above challenges, this paper proposes an innovative design and implementation of a smart street light monitoring and controlling system to ensure low power consumption, consumption monitoring, and fault detection of street lights in Ghana. The paper contributes significantly in deriving sustainable solutions to automation, reduced power consumption and cost effectiveness of street light systems. The rest of the paper is structured as follows: section two deals with the literature review; section three presents the methodology; section four summarizes the results and analyses them and section five presents the conclusion.

I. LITERATURE REVIEW

A survey of literature shows that a myriad of technologies has been deployed in the past to remotely control traffic lights and attend to their challenges timely. Most of these technologies however are still in the trial period and come with a number of limitations. This section provides a summary of the most important ones. **Internet of Things**
The Internet of Things (IoT) refers to the network of physical objects accessible through the Internet [2], [4]. These objects are able to communicate with each other by connecting them together using either wired or wireless connections. Evans (2011) predicted that the IoT represents the next evolution of the Internet considering its ability to automate, gather, analyze, and distribute data that we can turn into information and knowledge. IoT enables the interaction of objects by interconnecting objects and assigning private or public Internet Protocol (IP) addresses (Patel & Patel, 2016). IoT finds extreme use in controlling street light remotely however its reliability depends on the network infrastructure which in itself has a number of vulnerabilities and has proved unreliable over the time in many developing countries [1], [5], [6].

Zigbee Technology

ZigBee is created on the IEEE's 802.15.4 personal-area network standard, and is considered as an alternative to Wi-Fi and Bluetooth for some applications including low-powered devices that don't require a lot of bandwidth [7], [8]. These Zigbee's WPANs operate at 868 MHz, 902-928MHz and 2.4 GHz frequencies. This communication system is less expensive and simpler than the other proprietary short-range wireless sensor networks as Bluetooth and Wi-Fi. Zigbee networks are extendable with the use of routers and allow many nodes to interconnect with each other for building a wider area network. The Zigbee protocol is very secure, as it uses a variety of security mechanisms such as AES-128 encryption, network and device keys as well as frame counters [9] Cost and expandability are the major issues limiting the full deployment of Zigbee [10], [11].

Power Line Carrier (PLC)

PLC technology carries the signal over power lines that are already connected to most street lights. For that reason, outside of the United States and the United Kingdom, PLC has traditionally been the leading method of communicating with outdoor lights in a networked system. Although this is changing, as RF mesh technology is now being deployed in various European smart city trials, PLC is still a significant network technology for lighting controls. Large numbers of lights run on dedicated power lines without the interruption of frequent substations. As a result, these power lines can be used as the backbone for networked communications. PLC communication is reliable as illustrated by [12], [13]. PLC are sometimes combined with RF Mesh network to form hybrid PLC/Mesh technology which allows communication to hop from PLC to wireless and back seamlessly while always staying connected. PLC deployment demands change in technology and imposes huge costs on the government. Currently PLC are not yet deployed in Ghana. Again, using PLC will mean that communication can be deployed only in areas where power lines are accessible which becomes another limitation.

Smart Street Lighting on a Global Scale

According to [8], [14]-[16], large projects on street lighting involving complex intelligences are being implemented across the globe. The choice of communication medium indicated in their paper as the preferred medium for implementing the projects are usually RF mesh, PLC, and UNB communications. Recently, there has been a continued rise of new lower power wide area network (LPWAN) communication technologies such as LoRa, NB-IoT, and LTE-Cat-M1." Tseng (2016) in an attempt to provide a solution for intelligent street lamp monitoring and energy management, stated that with the requirements of the public, management units, safety, and energy saving, the combination of street lamps and intelligent techniques is an inevitable trend, as well as a requirement of developing or developed countries. However, Woods (2017) in a white paper for Navigant Research denotes that communication options such as the power line carrier (PLC) and the RF systems for connecting street lights provide only limited options of control and no opportunity to accommodate future updates.

Street Lighting in Ghana

The provision of street lighting facilities in Ghana had been the responsibility of the Metropolitan, Municipal and District Assemblies (MMDAs) and Communities until the late 1980's. The weak financial base as well as the inadequate and, in a lot of cases, unavailable technical capacity of the local authorities led to the deterioration of street lighting facilities over the years.

In 1988, the Government appointed the Electricity Company of Ghana (ECG) to enter into a maintenance Contract Agreement with a local street lights contractor to help remedy the deteriorating situation. A 1% street lighting levy on all domestic electricity sales was introduced to provide funding for the operation and maintenance of the street lighting in the country. The street lighting levy was revised to ₵0.20 per kWh of electricity sold in 1991, which was further increased to ₵0.50 per kWh of electricity sold in June 1994. From 1994 to date, however, there has been no review of the street lighting levy despite the expansion of the installation of streetlights in nearly every corner of the country, and an increase in electricity rates over the years. This has contributed massively to the increasing deteriorating rate of most street lights in the country.

II. METHODOLOGY

This paper adopted the Rapid Application Development approach which according to Breu et al., (2008) is a software development methodology that focuses on rapid prototyping and application development to ensure faster product delivery. The phases of RAD include: requirement planning, user or system design, development or construction and cutover. The proposed system has software and hardware components. The hardware components used are listed below: Arduino Uno, Relay SPDT, GPS Receiver, SIM 800L.

As illustrated in Figure one, the system block diagram consists of a central control unit connected to a photocell unit, a GPRS module and a GPS module. The system outputs consist of the street light setup, the LEDs, the dashboard for display etc.

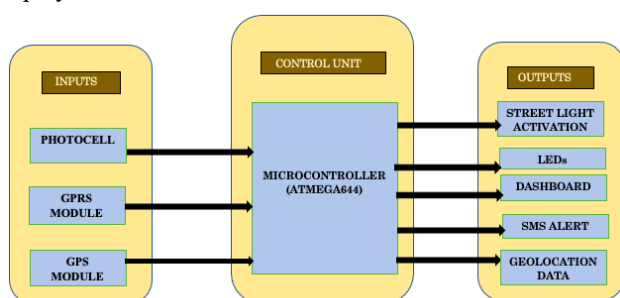


Figure 1. Block diagram of the proposed system

Figure two presents the circuit diagram of the proposed system. The circuit was designed using KiCAD software.

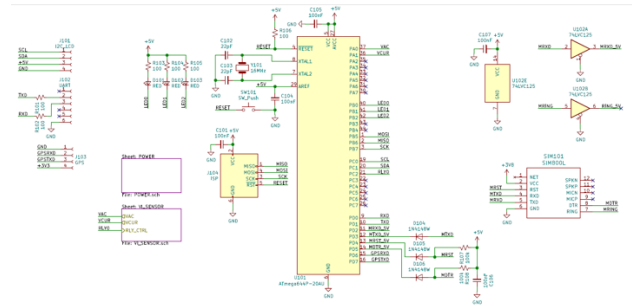


Figure 2. Circuit diagram of the street light controller

To operate the system successfully, a data sim shall first be inserted in the GPRS module for connectivity. The system is then connected to a power source. When done properly, the LED blinks every 1 second to indicate the module is running but has not made connection to the cellular network yet. When it blinks every 2 seconds it indicates that the GPRS data connection has become active. When it blinks every 3 seconds, the module has made contact with the cellular network and can send/receive voice and SMS. When it blinks very fast, it means it has full connectivity and is transmitting data. The photocell senses whether or not there is enough lighting. When the lighting is adequate, the street lights go off and when it is not adequate, the lights come on. The dashboard provides an interface to visualize and monitor the system.

When the system is powered and the photocell senses light, the streetlight goes off. The data is transmitted via GPRS and then made visible on the dashboard. In case the streetlight does not go off when the photocell is exposed to light, the streetlight will be able to be turned off from the dashboard. When the photocell senses no lights, the streetlight goes on. The data is transmitted via GPRS and made visible on the dashboard as well. In case the streetlight does not come on when the photocell is not exposed to light, the streetlight will be able to be turned on from the dashboard.

The operation of the system is also illustrated in flowchart of Figure three. Figure four and five show the implementation of the proposed circuit diagram on a PCB through simulation with Multisim software.

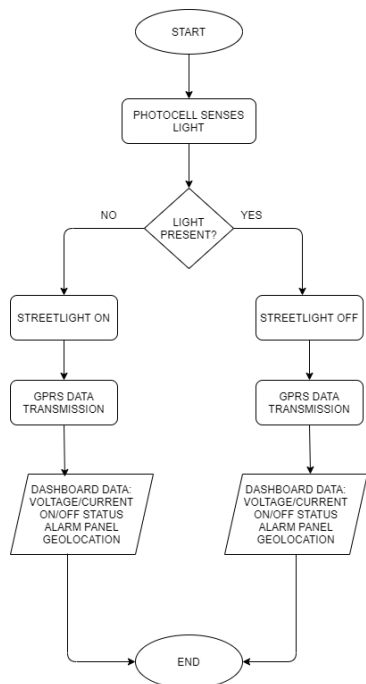


Figure 3. Flowchart diagram of the proposed street light controller

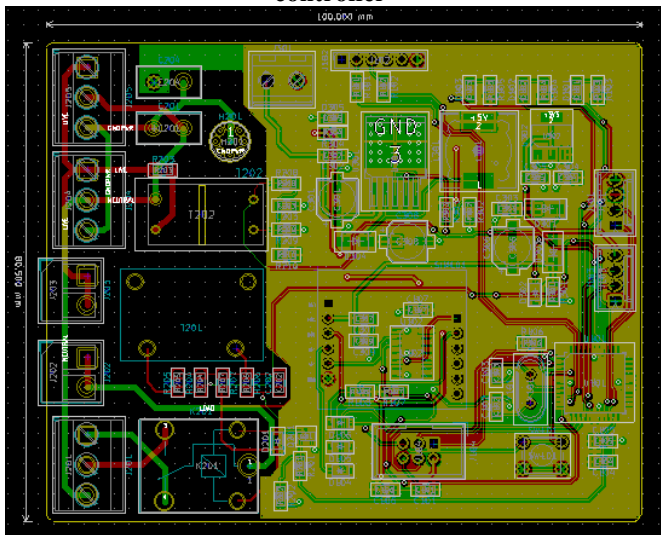


Figure 4. PCB diagram showing routing line of the street light controller

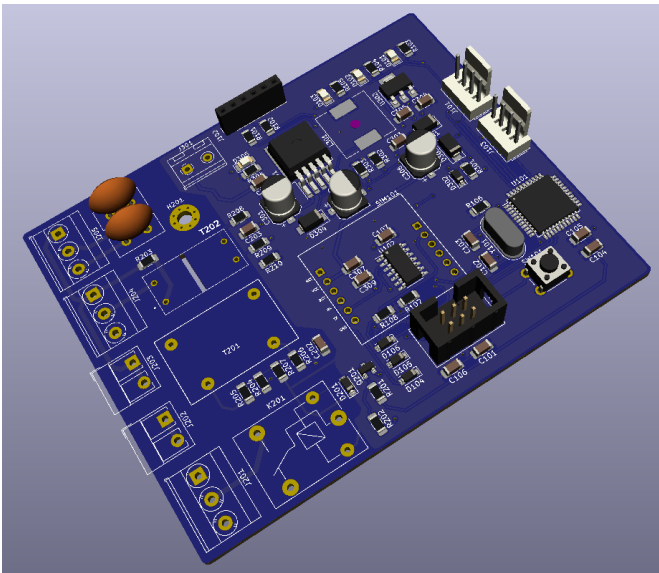


Figure 5. 3D view of the designed street light controller

III. RESULTS AND DISCUSSION

Figure six below shows the assembled circuit in different modes of operations

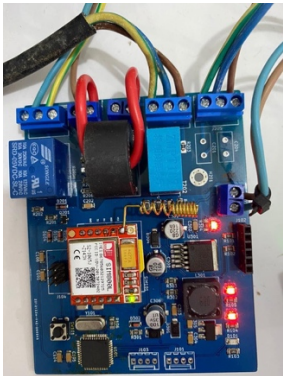


Figure 6a. Assembled circuit illustrating a connection for transmitting of data



Figure 6b. Lights come on when the photocell sensor is exposed to darkness

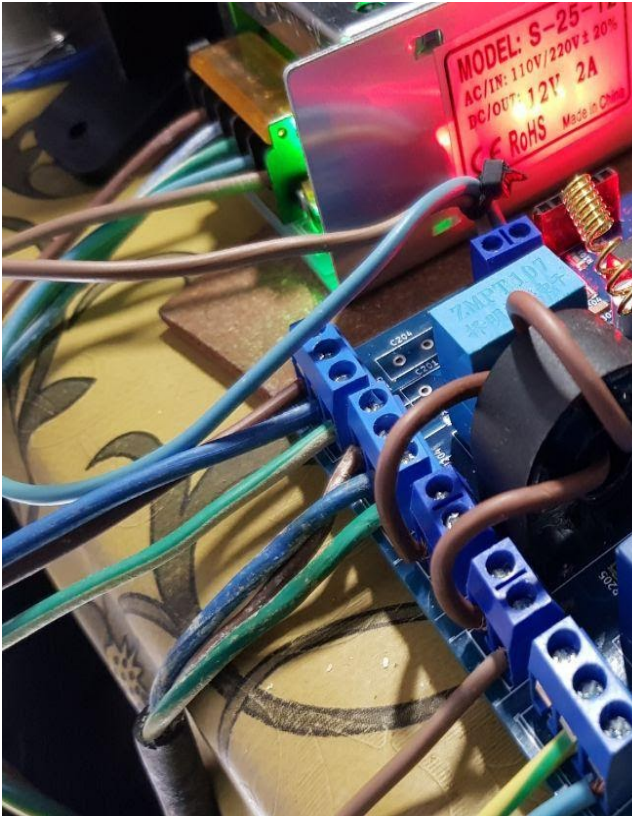


Figure 6c. Closer view of the circuit board when the lights are on

Figure 6. Illustrations of assembled circuits

Table one present the summary of the simulation outcome for different use case scenarios

Table 1. Simulation Results

TEST	ACTION	OUTPUT
1	Show light on the photocell sensor.	LED light goes off
2	Stop showing light on the photocell sensor.	LED light comes on
3	Toggle lamp switch on the dashboard off while light is not on the photocell.	LED light goes off
4	Toggle lamp switch on the	LED light comes on

	dashboard while light is on the photocell.	
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Additionally, some few screenshots of the software interface are presented in the subsequent paragraphs. Figure six illustrates the Login page of the proposed software

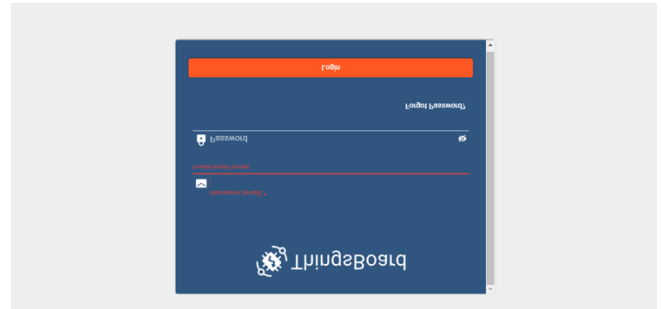


Figure 6. Login page of the software interface.

Figure seven shows the landing page of the dashboard showing the status of the streetlight. In this case, it is in the active state and the indicator (light) is on. Figure 7 also displays the connect time and date, the disconnect time and date and location.

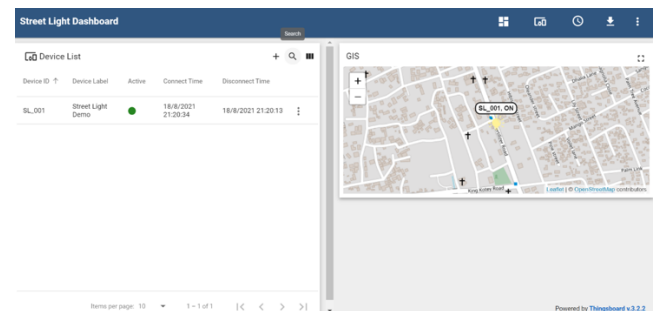


Figure 7. Landing page of the dashboard illustrating an active state of the system

Figure 8 illustrates the landing page of the dashboard showing the inactive status of the streetlight (the light is off). The Figure also displays the connect time and date, the disconnect time and date and the location.

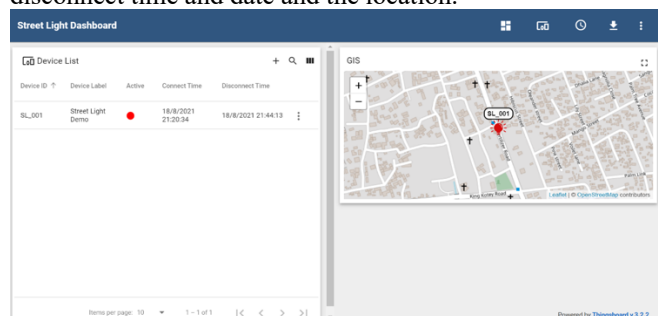


Figure 8. Landing page of the dashboard illustrating the inactive state of the system

Figure 9 illustrates the state of the lamp which switches off and on the streetlight in case the photocell is damaged. Current and Voltage data transmitted by the streetlight is also visualized here with their respective graphs. The figure also shows the alarms and their status. Currently, the lamp is turned on.

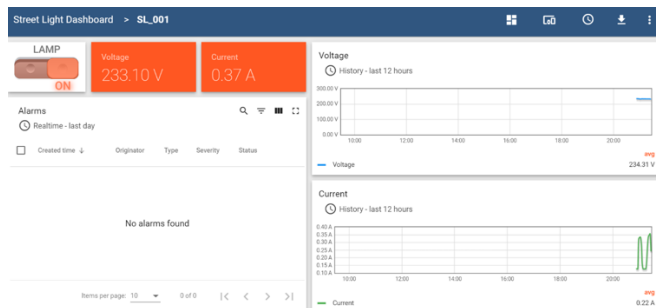


Figure 9. State of lamp, current and voltage visualization

IV. RESULTS AND DISCUSSION

This paper focused on the design of an improved streetlight management system intended to resolve most of the issues encountered with the existing or conventional ones. The advantages of the developed systems are listed and discussed below.

- ***reduced energy consumption due to the integration of photosensors***

Conventional streetlights have proven to be less energy efficient and very expensive to maintain. It is common to observe street lights being ON during daytime usually because their operation is manual and the operator for one or other reasons could not turn it OFF or ON at the appropriate time [17]. Thus, the energy wasted for not controlling the streetlight operation timely or automatically is considerable.

The introduction of the photosensor helps solve this problem by automatically turning the light ON at night and off at dusk. This results in a more efficient energy utilization

On the other hand, “not turning the streetlight ON” can reduce visibility at night and lead to road accidents. It has been reported that the majority of road accidents were due to lack of visibility especially in situations like Ghana where the road conditions are not too good and too narrow. Lighting on the road is therefore primordial and should be provided timely as rendered by the use of photosensors [17], [18].

- ***Timely maintenance and reduced downtime***

An important challenge with the management of streetlight has to do with the responsiveness in terms of maintenance. A number of issues can make the light not turn ON at night. Usually, it takes quite a long time to get the maintenance team aware of the challenge and dealing with it. The downtime may lead to further

negative impacts for the road use as this results in lack of visibility on the roads.

However, with the advancement in technology, and the internet of things (IOT) in our proposed system, the streetlight can be controlled and monitored remotely in real-time [19]– [21]. A set of sensors helps to detect in real time, the state of the streetlight and also direct to the probable causes of its breakdown. Additionally, the proposed system has an inbuilt GPS that can predict the exact location of the light to help reduce the troubleshooting time. This reduces downtime, increases the reliability of the system and increases the security of road users.

- ***Innovative interface for remote monitoring and control***

The study also proposed a software interface that can be used for remote monitoring and control of the operation of the streetlights. The operator can take a few control actions like turning ON or OFF a specific light through the online interface or monitor the state of the light to detect whether they are in good working conditions or otherwise. Also, the interface provides relevant information on voltages and currents, GPS location in an area map. Subsequently this interface helps in augmenting the resilience and effectiveness of the streetlight management system and further lessens the burden on authorities that were handling the same tasks manually in the past.

V. CONCLUSION

This paper proposes an innovative design and construction of a streetlight management system. With the help of the IoT, sensors have been installed to remotely monitor the state of the streetlight. A software with user-friendly interface has also been designed to monitor and control the operation of the streetlight remotely. The proposed smart streetlight offers an effective way to measure and save energy by preventing unnecessary wastage of electricity associated with poor management of the existing streetlight. The smart street lighting system is eco-friendly and has interesting features including remote switching capabilities over the internet and real-time monitoring of the state of the light. Furthermore, the smart streetlight is durable, expandable, improves energy efficiency and contributes to cost saving.

ACKNOWLEDGMENT

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SMART LIGHT ILLUMINATION

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Abstract--Street light ensures nocturnal safety for drivers, pedestrians, and cyclists. It plays a pivotal role in fostering a secure and comfortable environment. The malfunctioning of these lights poses a significant risk of road accidents. In response to this, Smart Light Illumination System is proposed, representing a technologically advanced framework for automatic control mechanisms and to implement fault detection protocols for street lamps. Leveraging Light-Dependent Resistor(LDR) sensor, the smart street light system monitors ambient lighting conditions, dynamically regulating street lamps' ON/OFF status based on real-time weather assessments. In residential areas, a motion sensor is employed to continuously monitor the surroundings for motion and activates street lights when motion is detected, optimizing energy efficiency. On highways, Infrared (IR) sensors detect vehicle presence through infrared radiation monitoring. The sensors interface with the Pico W to dynamically adjust street light brightness and activate lights based on real-time movement detection. The system optimizes energy efficiency by dimming lights during periods of inactivity. A pivotal feature is advanced fault detection capabilities. In the event of a malfunction, the system identifies the fault and generates a notification message. Users will get access to a dedicated website to monitor real-time functionalities and identify any malfunctions promptly.

Keywords:LDR,Motion sensor, IR sensor, Pico W

I INTRODUCTION

Effective public lighting is a key element of a nation's infrastructure, essential for ensuring safety and enhancing the quality of life. Traditional lighting systems, which depend on manual switching and basic timer controls, often

lead to significant inefficiencies. The systems frequently waste energy by operating at high intensity even when it's unnecessary, such as during low-traffic periods. Additionally, the lack of ability to quickly identify and repair defective lights, resulting in extended downtime and compromised reliability.

The proposed system overcomes the issues by introducing an advanced public lighting system that employs automated controls and real-time monitoring. The system adjusts lighting intensity based on actual needs, resulting in lower energy consumption and enhanced efficiency. It also includes fault detection capabilities, enabling rapid identification and repair of defective lights, thus reducing downtime and ensuring consistent, reliable public lighting.

The system prioritizes:

•Safety and Security:

Enhanced Visibility: Ensure well-lit streets at all times, improving visibility for pedestrians, cyclists, and drivers, thus reducing accidents and enhancing personal safety.

Fault Detection: Expediently detects and resolves lighting issues such as outages normal functions, ensuring consistent lighting levels and minimizes areas prone to crime.

•Cost Efficiency and Environmental Impact:

Energy Savings: Adjusts lighting levels based on real-time conditions, reducing energy consumption and operational costs.

II. LITERATURE REVIEW

Yusaku Fujii et. al discuss the development of a smart street light system that uses LED, brightness sensors, motion sensors, and a short-distance communication network to conserve energy. The system developed, make it difficult for pedestrians and vehicles to distinguish between smart street lamps and conventional ones. The author explores the current status and future prospects of this smart street light system [1].

Yu-Sheng Yang et. al. proposes an efficient system for configuring, deploying, and managing smart street lights, utilizing container-based system management for scalability and quick deployment. It uses NoSQL and in-memory databases for flexible data management, and uses an SSH encrypted tunnel for data delivery and token validation [2].

Amjad Omar et. al compares and contrasts various smart street lighting systems, highlighting their shortcomings and highlighting future developments. It highlights the need for energy savings, remote lamp failure detection, and sensor adjustments to mitigate global warming effects [3].

Karthikeyan et. al. proposes an automated system for street illumination that detects motion and automatically dims lights for areas where it's needed, reducing energy usage. The system also allows for adjustable light intensity based on the user's needs [4].

Dr A. Senthil Kumar, in collaboration with other researchers, states that the Internet of Things (IoT) is revolutionizing various areas like traffic control, street lights, and health monitoring, reducing energy waste and improving efficiency. The survey explores how IoT is being used to create smart street lights, addressing the global energy crisis and developing sustainable street lighting solutions [5].

III METHODOLOGY

The system focuses on integrating advanced sensor technologies to create a robust network of smart street lights capable of proactive fault detection and efficient management. The system strategically deploys sensors, including infrared (IR) sensors, motion detectors, and light-dependent resistors (LDRs), to continuously monitor lighting conditions and environmental parameters. The sensors interface with the Raspberry Pi Pico W via connections to the ground (GND), power (VCC), and data pins. Real-time data collected by the sensors is analyzed to optimize street light operation and promptly detect any malfunctions. Upon identifying a fault, the system generates a notification message, allowing users to access a dedicated website for real-time monitoring and quick identification of any issues, as illustrated in the block diagram in Fig. 1.

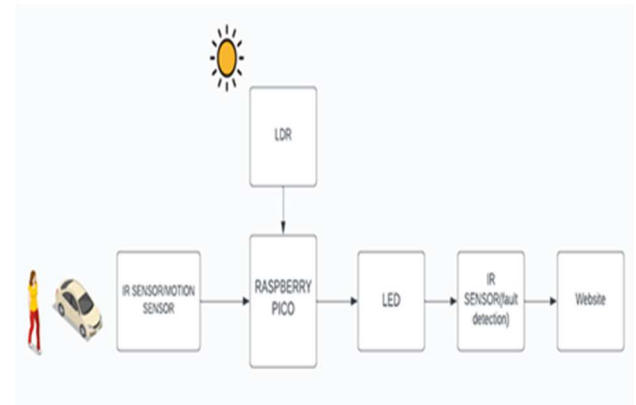


Fig 1. Block diagram of proposed system.

A. Photoelectric control:

The Light Dependent Resistor (LDR) serves as a pivotal ambient light sensor, strategically positioned, it interfaces with a GPIO pin on the Raspberry Pi Pico W, enabling continuous monitoring of its digital output by the microcontroller.

Employing a predetermined threshold, the microcontroller distinguishes day from night based on LDR readings. Upon detecting darkness (ambient light below the threshold), the Raspberry Pi Pico W activates street lights. Conversely, in daylight, when the LDR detects sufficient light, the microcontroller ensures lights remain off for energy efficiency.

B. Pervasive lighting system:

- High ways:

Infrared (IR) sensors detect vehicle presence through infrared radiation monitoring and it is most often used in highways as depicted in Fig 2. The sensors interface with the Pico W, to dynamically adjust street light brightness and activate lights based on real-time movement detection. The system optimizes energy efficiency by intelligently dimming lights during periods of inactivity.



Fig 2 Street light in High ways

- Residential areas:

The Motion sensor is connected to GPIO pins and continuously monitors the surroundings for movement, particularly in Residential areas. The Raspberry Pi Pico W processes the data and activates street lights while detecting motion, optimizing energy efficiency. The system dynamically adjusts illumination levels based on activity,

contributing to both energy conservation and enhanced security.

C. Integration of sensors:

The integration of a Light Dependent Resistor (LDR) sensor and a motion sensor enables a lighting system that activates only when both sensors concurrently detect the respective condition. The LDR, responsible for monitoring ambient light levels, works in conjunction with the motion sensor, which detects dynamic changes. The LDR sensor is connected to GPIO pin 2 and the IR sensor is connected to GPIO pin 3, as shown in Fig 3. Their respective values are measured and displayed using the Arduino IDE, as illustrated in Fig. 4. The lighting system is activated solely when both sensors signal that predetermined thresholds have been met, thereby optimizing energy efficiency.

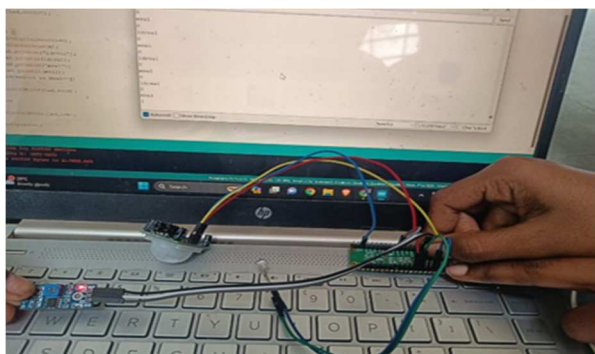


Fig 3. Integration of sensors

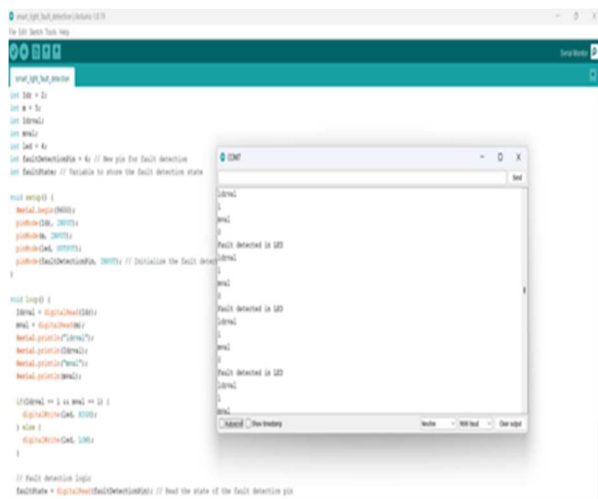


Fig 4.(Serial monitor of Arduino IDE)

D. Website Development:

A fault detection mechanism has been developed in tandem with infrared (IR) sensors. The system integrates the output of an LED with an IR sensor, enabling the sensor to detect faults conveyed through the LED output.

A website has been created to aid in identifying and reporting faults in street lights. Users have the convenience of checking the status of nearby lights and submitting

complaints if necessary, thereby facilitating efficient maintenance and fostering community engagement in the upkeep of public infrastructure. A custom logo has been created as shown in Fig. 5.



LUMINA- SPOT THE GLITCH, ILLUMINATE THE SWITCH

Fig 5. Logo of Website

The website offers a comprehensive platform designed for efficient street light management. Users can easily log in to access a personalized dashboard that provides real-time monitoring of nearby street lights, allowing them to quickly identify any faults. The site's fault detection feature enables users to report issues directly through the platform as shown in Fig 6 and Fig 7 . It keeps users informed with timely alerts and notifications about the status of the reports and any scheduled maintenance.

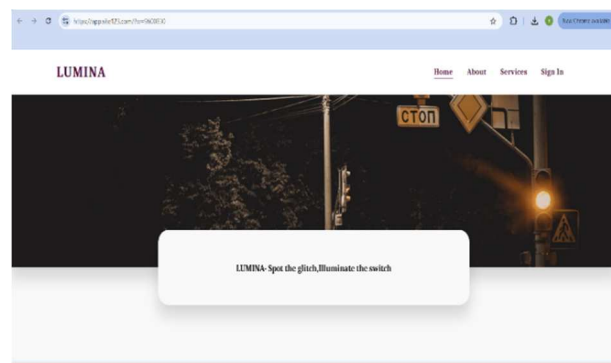


Fig 6. Home page of website

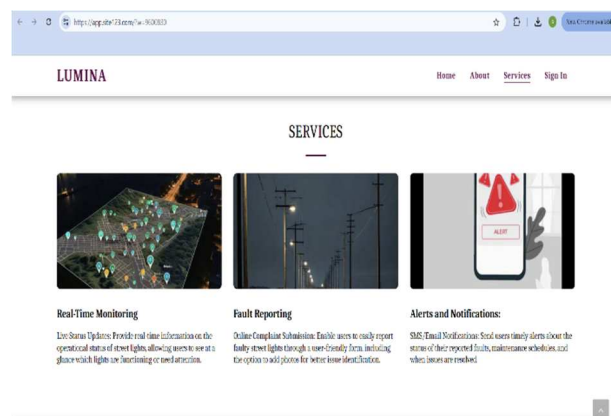


Fig 7.Services

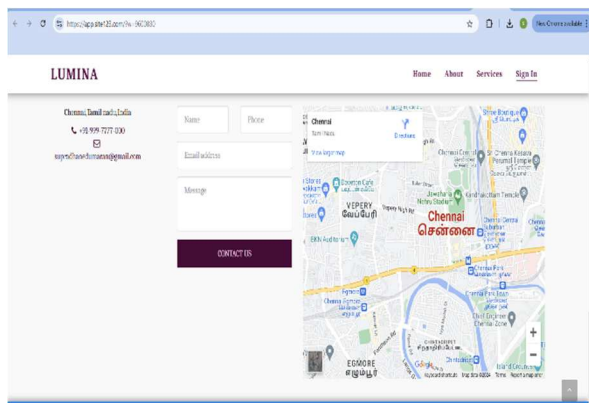


Fig 8. Log in page of website

IV RESULTS AND CONCLUSION

The sensors are successfully integrated with the light and it significantly improved energy efficiency, safety, operational cost reduction, and environmental impact. The system autonomously adjust lighting levels in response to ambient light conditions and detects motion, achieving energy savings, when compared to traditional lighting systems. Faults are accurately detected and automatically reported through the website, triggering swift notifications and enabling prompt resolution, thereby ensuring optimal street light maintenance and enhancing public safety.

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Smart Street Light Lamps

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Abstract- In an era of smart living, people strive to innovate existing technologies to achieve the concept of smart city. Street lighting systems is one of them. Most urban, towns and semi-urban cities still use conventional lighting systems. Current lighting systems use compact fluorescent lamps (CFL) with combination of high pressure sodium lamps to provide illumination. But the conventional use leads to huge demand and wastage of electricity between 20 to 40% of the total energy produced. Although a single unit of lighting consumes less power, their enormous quantity and long hours of operation requires more energy. So, energy saving is an essential need for smart lighting systems. Energy efficiency is achieved by this smart lighting system. It will provide light that varies intensity as per illumination requirement. There are water level sensors that will alert and send data of the increasing water levels in an area to authorities which can prove lifesaving during rains. There are emergency lights that would light up to warn of dangers or emergencies along the road. The LCD screens can be used to display statistics of available information. It will display weather forecast and emergency notifications. The smart street light that is deployed on a pole, also has provision to send alert for broken lights that requires maintenance. So, the proposed system will overcome all the drawbacks of conventional system and provide energy saving, smart alerting solutions.

Keywords: Compact Fluorescent Lamps; smart street lights; light illumination; Light Dependent Resistor (LDR); flood detection.

I. INTRODUCTION

Modernization calls for innovative automation. It considerably reduces the amount human efforts to perform tasks. Environmental concerns, limited exploration in renewable alternative energy sources makes it paramount to use energy efficiently. Energy is an important parameter when considering impacts of technologies on environment. Smart street lighting is in an increasing demand to convert cities into smart cities with developed roads and highways. Street lighting system is the most noticeable feature depicting the urbanization of a city. Public lighting systems are meant to provide safer traffic and pedestrian movement besides adding to a city's ambience. Conventional lighting systems use a combination of fluorescent, CFL, metal halide or high-pressure sodium lamps bulbs. A single unit of lighting lamp consumes less power, but their enormous quantity and long hours of operation requires more energy from the total energy produced which can be further reduced. The lamp life of

conventional sources is less. It requires maintenance more often.

The paper proposes an energy efficient street lighting system with added benefits of intensity variant lighting, flood monitoring, emergency lighting, and maintenance alerting system. Using light efficient diodes is a promising solution which is replacing the conventional lights. It is more energy efficient and provides more brightness, even the lamp life is more than the conventional sources. The intensity of the light emitting diodes can be controlled as per requirement using light dependent resistors. When there is no or less light incident on the pole the light emitting diodes will shine with more intensity. Flood Sensor uses an ultrasonic module to sense the increase in water levels. It starts to send warnings when a minimum level is reached by the water and continues to send data till it crosses the highest mark provided. It also a Wi-Fi module to alert and provide data of increasing water levels in an area as recorded by the sensor module to authorities. Such data can provide location of the pole, with its recorded rain information to help the authorities plan their actions before the area gets submerged in water. This is very beneficial and can help save a lot of lives during rains. The smart street pole also has emergency lights that would light up to warn of dangers or emergencies along the road. Liquid Crystal Display (LCD) screens to be used to display information of weather or emergencies. It also has provision to send alerts for broken lights needing maintenance, resulting in smarter street lights that provide more than just illumination while saving on energy.

II. LITERATURE REVIEW

Keeping in mind the long-established and ordinary approaches of smart street lighting system as a sense following articles are considered for review:

Sindhu A. M et al. worked on smart street lights using infrared sensors (IR) and by introducing Light Emitting Diode (LED) lamps instead of High Intensity Diode (HID) lamps since HID lamps are not cost effective and reliable. The system was implemented using IR sensors, using which the movement of vehicles and pedestrians was monitored and the system was programmed to get activated once there was an input to the sensor. But the range of IR sensors is less. IR sensors make lamps light up on detection of movement which will lessen lamp life in the long run due to being on/off again

and again if the vehicular traffic is discontinuous. So LDR can be used instead to vary the intensity of light than shutting it down completely. [1]

Shagun Malhotra et al. worked on a smart street lighting system that suggested an intelligent use of street lights that is energy efficient and offered effective way for the maintenance of street light by using a Global System for Mobile (GSM) module that sends the relevant information to the centrally located control station. But using a GSM module is not much reliable because of connectivity complexity, so instead an ESP8266 Wi-Fi module can be used to send the required data to the control station. [2]

K. Y. Rajput et al. in their work on intelligent street light using GSM, also suggested the concept of smart lamp posts in which there will be a 24-hour monitoring of street lamps where necessary actions will be taken and maintenance will be done to check the condition of lamps using GSM module. But surveillance system of lamps requires frequent maintenance increasing human dependency. [3]

Chetna Badgaiyan et al. proposed a smart street lighting system using ZigBee and PIR sensors for higher efficiency and maximum throughput. ZigBee is a wireless communication module which is generally very complex and its implementation is much more difficult. [4]

B. K. Subramanyam et al. proposed a wireless street lighting system, which uses a combination of current technologies and research made in this field to provide efficient and convenient use of electricity. By replacing the conventional lamps with solar lamps considerably large amount of electricity was saved. But using LDR in place of solar lamps it costs less per unit of street lamps in comparison and works well in all weather conditions. Also, this system works in any weather condition very effectively. [5]

P. Nithya et al. suggested and implemented an intelligent monitoring and maintenance of street lamps, which used a wireless zig-bee module to send collected data to a central station. [6]

Anila Devi et al. Implemented a street lamp which was completely based on a GSM module along with a zig-bee module implemented with some additional sensors. The system was implemented on the concept of wireless technology for further monitoring and repairing of lamps. Led lamps were installed which are more reliable and cost effective and requires much less maintenance. [7]

R. Kavitha et al. proposed street lights based on traffic. The street lights pole detects traffic in vision direction, necessary actions are taken. The system also gives facilities to pass emergency vehicles efficiently. [8]

S. Suganya et al. worked on street light automated system which detected the movement of vehicles and pedestrians on the street. The maintenance of the lamp post and provide various statistics a zig-bee module was implemented to maintain a wireless communication. LED, zig-bee module, power transistors and many other sensors were collectively used to implement the smart street light. [9]

K. Santha et al. have proposed a system that will make use of movements of vehicles and the street lights are controlled accordingly. The system was programmed to work automatically according to the road conditions. The intensity

of the light was managed on the basis of vehicle movements. The system was implemented using a pic microcontroller, which was programmed to suit the necessary situations. [10]

Srikanth et al. proposed an automatic lighting system based on remote control that uses ZigBee. The implemented system is completely automatic and the functions of turning the lights on and off or simply dim the light were done efficiently depending on the movements taking place on roads. Using PIR sensors the motion of all the objects were detected. [11]

III. PROPOSED SYSTEM

Since the conventional street light system is not energy effective, smart street light system has overcome by replacing traditional CFL lamps with LED. To achieve energy efficiency, it is necessary for a light to glow with intensity as per illumination required. So, these LEDs will be controlled by Light Dependent Resistor. Using LDR, depending on the weather conditions the intensity of light can be controlled. This will help with automatic switching of light in varying intensity, thus achieving energy efficiency. A street light is not only meant for lighting purpose, but also it can be utilized for other purposes. This system will be used for detection of water level during rainy seasons, by just adding an ultrasonic sensor above a certain height on the pole. These sensors will detect the level of water continuously and if the value reaches a particular level, an alert notification will be sent to the control center using the ESP8266 Wi-Fi module and then required measures will be taken. This smart street light poles will be further used for displaying weather and accident alerts on an LCD display screen which will be installed on poles. When an accident alert is notified to control room, the LED can be turned red for emergencies. This system also has an action of providing alerts of blown out street lights.

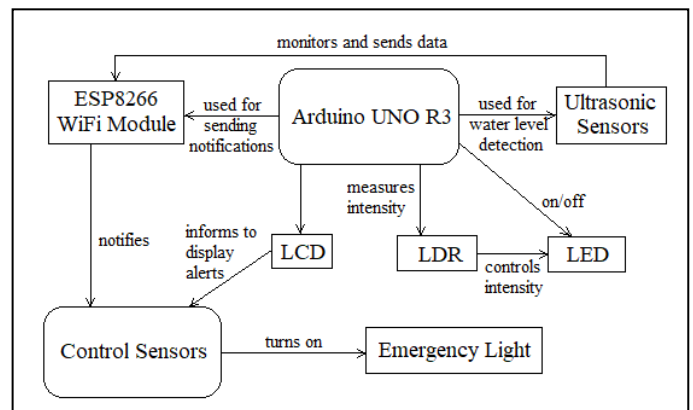


Fig. 1. System Architecture.

Following are the modules of this system:

A. Flood Sensing Module

This module works mainly with ultrasonic sensors, which are used to measure increasing level of rainwater. The transmitter in an ultrasonic sensor sends a sound wave and the receiver waits till that sound wave is reflected back. The length is calculated by taking time into consideration, for that the distance between object and barrier is measured. If the distance is less than a particular limit, then an alert notification will be sent to the concerned authority using an espresso chipset ESP8266 which is a Wi-Fi module and then necessary measures can be taken. The acquired data will be now useful to analyze if a particular region vulnerable to floods. If so,

required action can be taken by Government to develop the height and water effluent of such regions. [12]

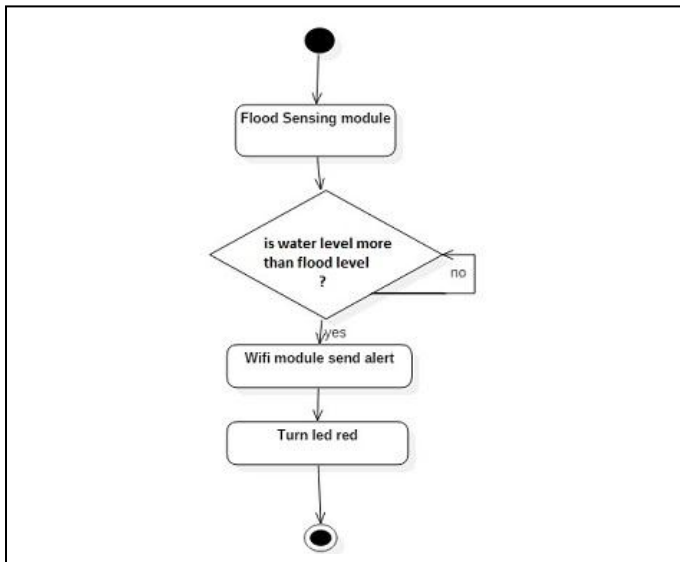


Fig. 2. Flood Sensing and Alert Notification Module.

B. Light Module

This proposed system uses both LDR and micro-controller for lighting using intensity through sensitivity. This detected intensity will be now used for illuminating street lights, using LDR. When the intensity of the light falling on LDR decreases the LDR resistance increases, while on the other hand, the resistance will be decreased if the intensity of the light falling on LDR is increased. So, depending on the environmental lighting conditions the resistance of LDR will vary between hundred ohms and mega ohms. The opposition measured of LDR will be some mega ohms. This means there is complete absence of light. When the resistance is reduced by some hundred ohms then there is glaze. This entire circuit is connected to Arduino UNO R3 which is a micro-controller board. [13]

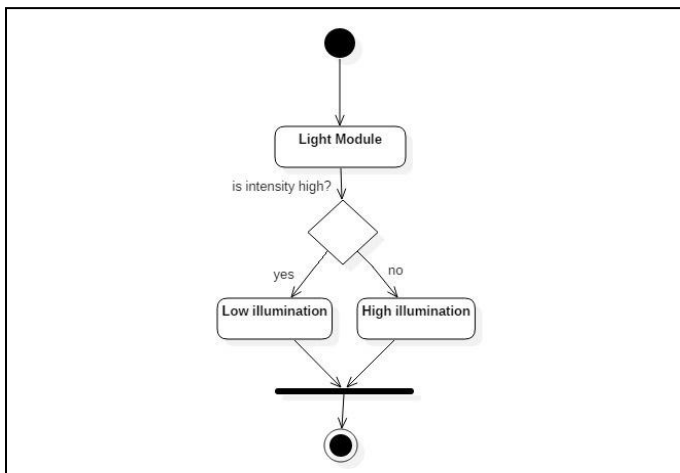


Fig. 3. Light Module.

C. Emergency Light and LCD Display Screens

In this module whenever an alert is issued such as flood alerts or accident alerts the white LEDs in the lamps would be turned off and red LEDs would be powered on to make people

aware about the emergency. This will provide the commuters and people present in the area a chance to tread carefully and think about their safety. If the danger is higher, then it warns them, in order to start looking for alternative paths. To exhibit alert announcements LCD Screens will be used to denote crisis like situations. During normal conditions the same screens will be used for flashing weather details. The LCD can also be used to carry advertisements in order to generate some revenue. This entire module functions with the help of Arduino UNO R3. [14]

IV. RESULTS AND DISCUSSION

This section depicts the entire setup of flood sensing circuit. All the components are connected and screenshots depicts the entire circuit connection.

Following are the screenshots in an orderly manner:

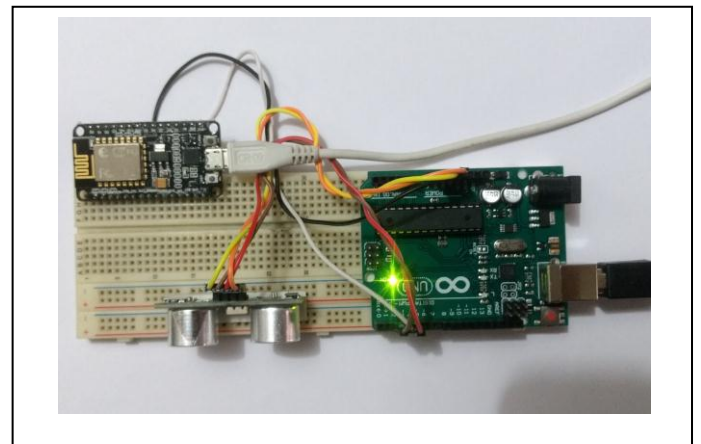
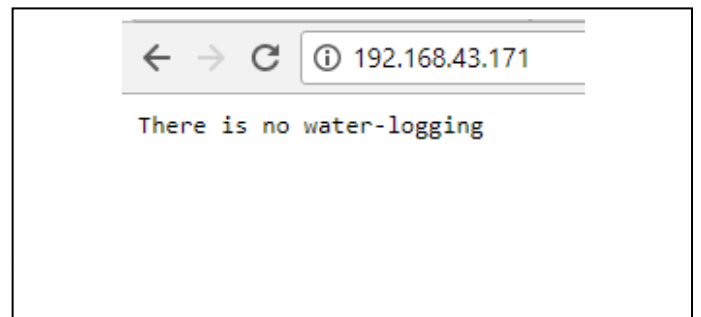
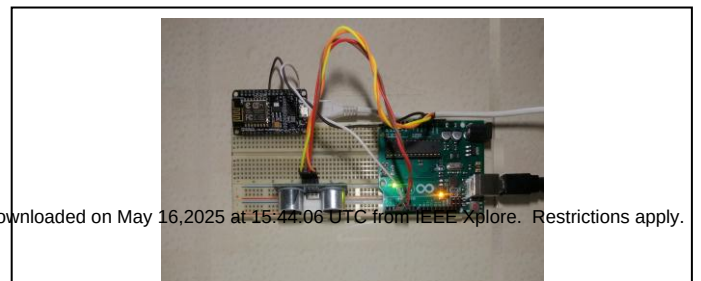


Fig. 4. Initial setup of hardware.

Figure 4 depicts the initial setup of the hardware. The Arduino UNO R3 board and ESP8266 are connected to the external power supply for the flow of current. The ultrasonic sensor and ESP8266 are connected to the Arduino board. The LED on Arduino board blinks normally. This means that the ultrasonic sensor is transmitting waves continuously to detect level of rainwater.



Since the level of water has not reached a desired limit, the message displayed on sever will be "There is no water-logging".



According to test results, it can be concluded that the flood sensing module operates well to detect the increasing level of rain-water and sends alerts accordingly.

V. EFFICIENCY OF THE PROPOSED SYSTEM

A. Power consumption and savings

The following table shows evaluation of power consumption between CFL and LED lamps for 12 hours. The table depicts that the LED lamps consume less power than CFL and are more efficient.

The most effective solution is to use LED lamps. They are also efficient by maintaining intensity of light. The proposed solution saves up to 47% of total energy.

TABLE I. POWER CONSUMPTION COMPARISON BETWEEN PROPOSED SYSTEM AND EXISTING TECHNOLOGY.

Installation area	Power Consumption using CFL (Watt)	Power Consumption using LED (Watt)
City	4755.3	970.56
Highways	8400.0	1592.44

B. Cost Analysis

The table below shows operating and purchasing cost of different types of street lamps. It is clearly seen that the initial cost for LED is higher when compared to CFL, but as the life of LED Lamps is more it requires less maintenance and thus is beneficial than other types of lamps.

TABLE II. COST FOR STREET LIGHT LAMPS.

Type of Lamp	Cost of Lamps (₹)	Cost of Lamps required for 10 years (₹)
CFL	103.30	4023.05
LED	322.18	2029.28

C. Efficiency of emergency alerts

The emergency alerts are very quick. The ultrasonic sensor takes approximately 10 μ s to notify ESP8266 module. The time taken to display it on web server depends on internet speed. Similarly, the time taken to turn on red LED also depends on various factors like internet speed, route used to send message and protocol used by client and server.

VI. CONCLUSION

The system proposed in this paper can efficiently save a large amount of energy by managing power consumption effectively. Usual conventional lamps are replaced with LED lamps to minimize the wastage of power. This system is completely sensor based because of which the complete system is automated. The system is developed to minimize the energy consumption by automatically controlling the switching of lights ON and OFF. Additionally, the system provides a provision to detect rainwater level and send alerts at the time of floods. Also, an LCD screen is installed to display warnings or alert messages. Smart Street Light proves to be more beneficial than the current street lights in variety of aspects.

Fig. 6. Actual implementation of circuit.

Figure 6 depicts the second phase where the orange colored LED on Arduino board blinks continuously, which means that the rain-water level is increasing and has reached a desired limit to occur a flood.

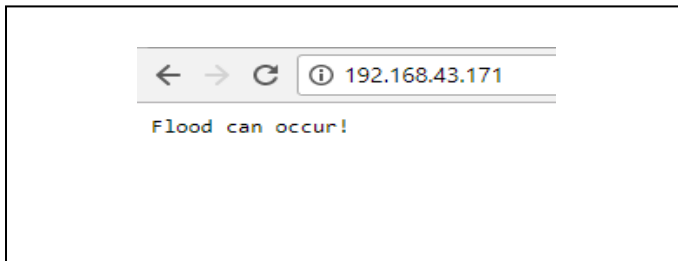


Fig. 7. Message displayed when water level is high.

Once LED starts blinking continuously, the server will display a message that flood can occur.

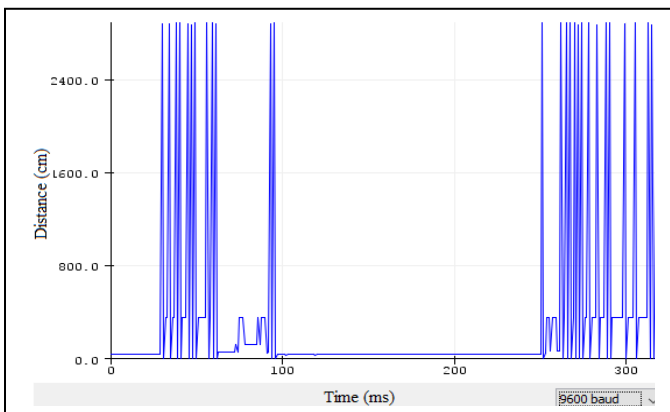


Fig. 8. Graph of distance against time for water level detection.

Figure 8 shows the accuracy of flood sensor module. As the level of water increases the distance decreases. At time 50ms - 100ms the level of water is increasing. After 100ms - 250ms the output is a straight line because the water level has reached its limit. When the level of water is not increasing i.e. after 250ms the distance remains constant.

Smart Street Lights has been implemented using latest technologies and advanced sensors. Further with multiple additions, the current system can be used to serve multiple purposes. LED lamps can be replaced with solar panels or solar lamps. Also, the system can be modified to provide charging stations and a public Wi-Fi can also be provided. The same pole could be used to install roadside CCTV cameras with advance features of accident detection based on Artificial Intelligence. Pollution detection sensors could be added on poles to measure the amount CO₂ emitted in a particular city. This system is adaptable to further additional implementations.

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Smart Street Light System using Wi-SUN and oneM2M

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Abstract—As urbanization continues to accelerate, the demand for energy-efficient and sustainable urban infrastructure has become paramount. Street lights are an elemental part of any city because they facilitate better night visibility, secured roads, and luminescence in public areas. Also, because street lights are present homogeneously on city roads, if they are used for wireless communication, other smart nodes can make use of their backhaul. In this paper, we present a smart street lighting system based on Wi-SUN (Wireless Smart Ubiquitous Network), which offers excellent redundancy owing to its use of a mesh topology. Wi-SUN is based on the IEEE 802.15.4g standard and uses license-free sub-GHz bands. We have used oneM2M as the middleware platform to enable possible interoperability with other smart city data verticals such as weather stations, environmental monitoring, crowd monitoring and so on. The proposed system provides a robust and low-cost method for controlling street lighting in any area, providing IP-addressable lighting. The proposed system additionally aims to provide consistent Wi-SUN coverage for future nodes in the area of deployment.

Keywords—oneM2M, Smart street lighting, Smart city, Urban deployment, Wi-SUN

I. INTRODUCTION

As the concept of smart cities has gained considerable momentum in recent years. The urban areas face the challenges of rapid population growth, resource constraints, and environmental concerns. The rapid urbanization and increasing energy demands have led to the development of innovative solutions for sustainable and smart urban infrastructure as well as smart homes [1], [2]. In recent years, smart street lighting systems (SSLs) have gained significant attention as a promising solution to address the challenges faced by traditional street lighting. Smart street lights apart from providing remote monitoring and control, adaptive lighting, energy efficiency and data analytics also provide a networking capability for a dense network of IOT sensors like pollution monitoring [3]–[5], environmental parameters [6]–[8], traffic management [9], and so on. This paper aims to provide a smart street lighting solution with Wi-SUN based communication and oneM2M as the backhaul server. We explore the technologies and components involved in smart street lighting systems and discuss their potential to revolutionize urban lighting, improve public safety, and provide energy efficiency. Additionally, we highlight the key considerations for successful implementation, including connectivity, data management, and integration

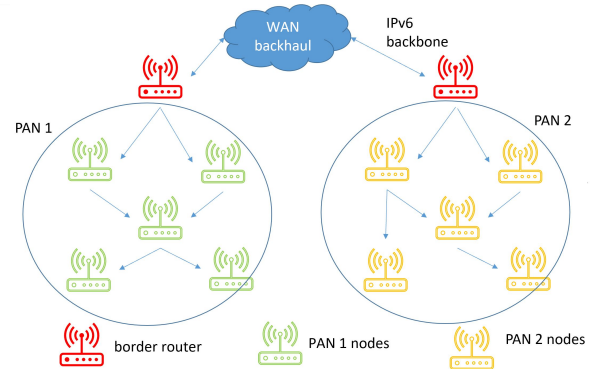


Fig. 1. A typical Wi-SUN mesh network with two Personal Area Networks (PANs).

with existing urban infrastructure. Overall, this paper can serve as a valuable resource for policymakers, urban planners, and researchers interested in the deployment of smart street lighting systems.

Wireless Smart Ubiquitous Network (Wi-SUN) uses a mesh topology that can self heal, where each device can act as a data forwarder for other devices in the network. Wi-SUN is a good choice for communication using street lights owing to its license-free sub-GHz frequency range. It provides signal penetration through obstacles like buildings and trees catering to its long range capability [10], [11]. In this paper, we present the fabrication and deployment of 30 smart street lights systems in the IIIT-Hyderabad campus. These street lights are controlled through Wi-SUN based mesh network protocol, and provide cloud connectivity to any Wi-SUN device in their vicinity. The data from this network is posted and retrieved from oneM2M, a middleware platform. We also present the Relative Signal Strength Index (RSSI) levels reported by the nodes in the network.

II. RELATED WORKS

Saifuzzaman *et al.* developed a street lighting solution, with 3 stages, either ON/OFF or dim, where the luminous control decisions were based upon the data provided from a Light Dependent Resistor (LDR) fixed atop the pole [12]. For connectivity, a cloud-based system dependent on Wi-Fi was used, which may hinder the scalability of such a

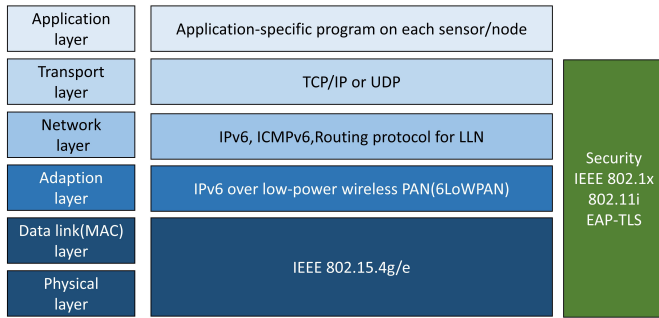


Fig. 2. OSI layer structure of the Wi-SUN network protocol.

model. Chen *et al.* deployed a smart street light solution with communication of multiple poles on NB-IoT protocol with data being posted on the cloud [13]. Mathew *et al.* installed an IoT-enabled street light management system with LED lamps and an additional calibrated DHT 11 temperature/humidity sensor using LoRaWan network [14]. A lack of such works on the Wi-SUN protocol and the potential of such a deployment are the main motivations for this paper.

III. WI-SUN ARCHITECTURE

The Wi-SUN standard specifies three different PHYs, Multi Rate (MR) - Orthogonal Frequency Division Multiplexing (OFDM), Frequency Shift Keying (FSK), Quadrature Phase Shift Keying (QPSK) and with data rates ranging from 50 kbps to 300 kbps [15]. The network layer uses RPL (Routing Protocol for Low-Power and Lossy Networks) that is optimized for multiple hops, long range and low power under the MAC layer uses Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) [16], [17].

The RPL protocol defines the mechanism for forming the Destination Oriented Directed Acyclic Graph (DODAG). The algorithm works with metrics and constraints to compute the most optimized path for sending data packets from the source to the destination. The DODAG has a root node (border router) such that all the edge nodes have their path oriented towards the root node. Once the DODAG is formed, if a node is shut down, the DODAG initiates a local repair to heal and changes itself accordingly. In some cases, it may initiate a global repair and the entire DODAG forms all over again. However, this leads to a long network reconfiguration time which is considered one of the drawbacks of the Wi-SUN network protocol [18]. An example Wi-SUN mesh network is shown in Fig. 1 and its OSI layers detailed in Fig. 2.

Wi-SUN Personal Area Network (PAN) links the cluster of smart devices deployed to the cloud by enabling interoperable, multi-service and secure IPv6 communication. Once the Wi-SUN devices are powered, they have the ability to self organize into a mesh network and transmit data via multi-hop routing which enlarges the range of the entire network by using other nodes as relays. Each node in the Wi-Sun PAN assumes one of three roles [19]:

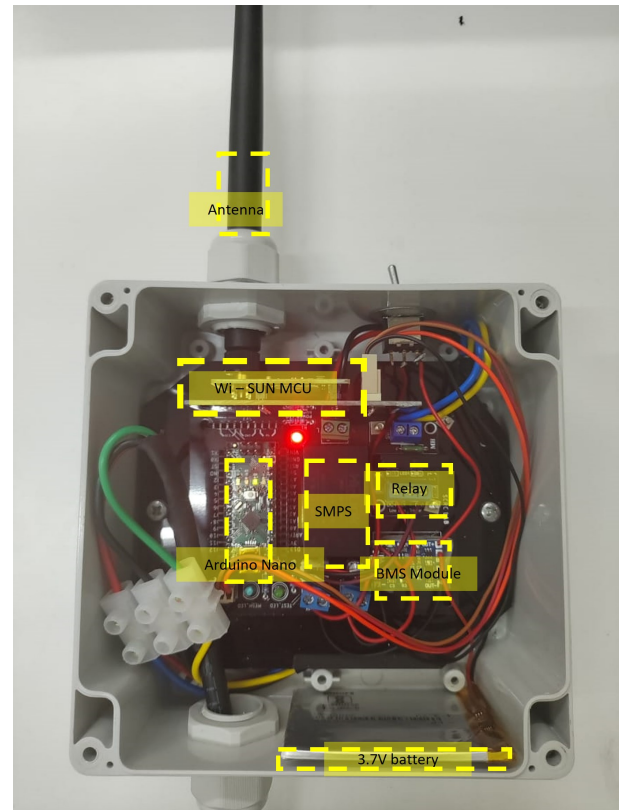


Fig. 3. The internal circuitry for the Wi-SUN node.

- 1) *Border Router* behaves as a gateway between the internet and the Wi-SUN PAN. It maintains and upgrades the source routing table for all the nodes within the PAN.
- 2) *Router nodes* provide both up-link and down-link data packet forwarding and behave as intermediate nodes for the mesh network.
- 3) *Leaf nodes* have the ability to discover and join the PAN and send/receive packets and act as the edge.

IV. SYSTEM ARCHITECTURE

We used an integrated radio transceiver module based on the EFR32MG12 microcontroller from Silicon Laboratories Inc., as part of the Silicon Labs Wi-SUN Wireless Starter Kit. The module was programmed to transmit at 868 MHz with a 200 kHz channel bandwidth at a data rate of 150 kbps and a modulation index of 0.5 using a 3-FSK modulation scheme. The Wi-SUN transceiver was part of the system consisting of an Arduino Nano board, a solid state relay to control the street light, a switch mode power supply (SMPS) for in-situ AC to DC conversion, a battery module comprising of a battery and a TP4056 battery management system in case AC power is lost. A double layered PCB (110 mm×50 mm) was designed to house all the components, and was placed in an IP65 waterproof casing. The antenna was connected outside the box to avoid signal attenuation from the box itself. An image of the circuit box with its components is shown in Fig. 3.



Fig. 4. a) One of the smart lamp posts deployed in the IIIT-H campus b) The design of the fabricated smart lamp post. Several compartments has been provided at the bottom and midway to house nodes with sensors. All measurements are in inches.

V. RESULTS

A smart street lamp was specifically fabricated to contain a compartment for this box as shown in Fig. 4. However, this is not a requirement for Wi-SUN based street lighting solutions. In this study, several Wi-SUN nodes were integrated onto existing infrastructure. The pole was provided with a single-phase AC supply, which was used to power the communication circuitry using the SMPS based AC to DC converter as well as the LED light. The Wi-SUN transceiver receives a string signal from the border router via UDP over Wi-SUN protocol, which it then forwards to the microcontroller, which decodes the incoming signal and triggers the relay accordingly. Once the trigger signal has been sent, the microcontroller, with a five second delay, confirms through a luminous sensor if the status of the light matches with the expectation and sends an acknowledgement signal back to the border router. This closed-loop feedback can help with the maintenance in case the light malfunctions. The acknowledgement packet contains the unique identifier IP of that specific node, the previous state of the light, the current state of the light, and the timestamp of the last change. In case of a power failure, to avoid a reformation of the Wi-SUN mesh network, which is time consuming, the node is powered through the battery until the AC power is restored.

To provide cloud connectivity and to control the network, a border router node is designed and connected to the cloud server using a Raspberry Pi controller board. The server pulls up the sunset and sunrise time from the internet and sends the node the data packets which contains the strings for turning “ON” or “OFF” the lights. This switching can also be manually overridden to directly send the packets from the border router in case of an unexpected event.

To test the robustness of the system, the border router

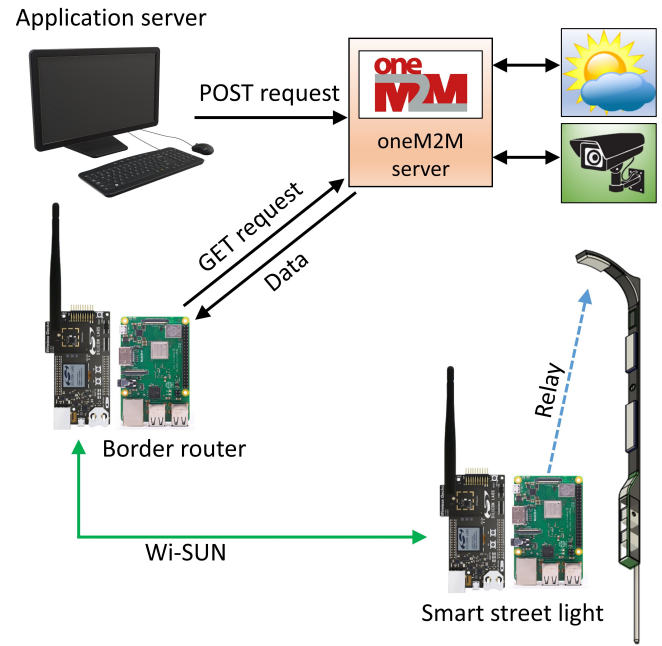


Fig. 5. The data flow diagram for the Wi-SUN smart street lights integrated with the oneM2M platform.

periodically initiates a test, to fetch the RSSI values of the nodes as received by the node from its adjacent connecting node in the mesh. Each exchanged packet consists of an ICMPv6 (Internet Control Message Protocol for IPv6) payload corresponding to 2 bytes of dummy data. Every transmission in the Wi-SUN mesh is done at 19 dBm of power. The received RSSI values are noted and averaged out for ten data packets and a round trip latency from the border router to each node and back is also obtained. At the server end, the Wi-SUN network is integrated into an open-source (OM2M) implementation of the oneM2M middleware platform as a separate <container> within an application entity (<AE>). The latter denoting the prospective “smart street light” vertical. OM2M is an open-source service platform compliant with the oneM2M standards [20]–[22]. The desired status (ON/OFF) of the light is posted to the appropriate <container> as a <contentInstance>. The border router polls the OM2M server periodically using the inbuilt latest data API, receiving the desired status of each smart light. This data is forwarded to the appropriate nodes over Wi-SUN, where the microcontroller powers the light ON/OFF through a relay. The data flow diagram is detailed in Fig. 5.

We created a dashboard to visualize the deployment locations and the status of the lights (Fig. 6). The dashboard was created using Grafana visualization platform, and the data was sourced from the OM2M server. The dashboard also provides information about the RSSI and latency values for the network. We observed that the RSSI values of the nodes were between -38 to -70 dB (Fig. 7), depending on the distance and obstacles, and the round trip latency varied from 0.8 to 3 s.

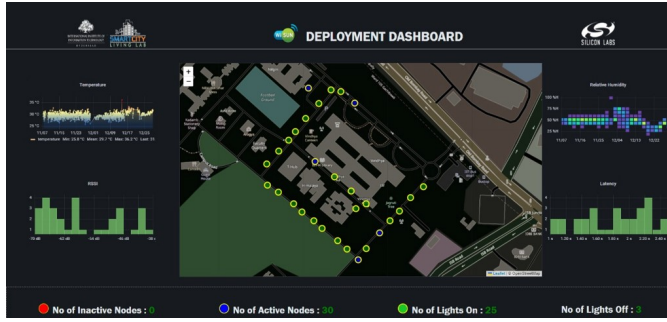


Fig. 6. The dashboard of the Wi-SUN deployment shows the status of each node and the histogram of RSSI and latency values.

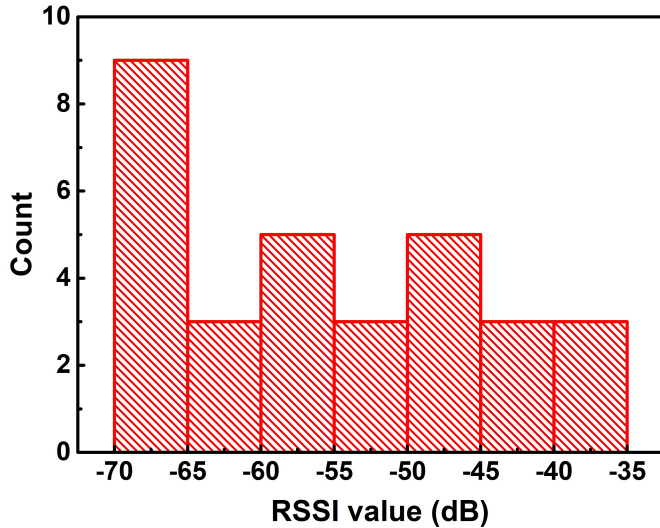


Fig. 7. Histogram of the snapshot of RSSI values for all the nodes deployed in the network.

VI. CONCLUSION

As more and more smart cities evolve, it is imperative to have such scaled deployments in dense urban landscapes. The deployment in this work can serve as a test-bed to collect information about the robustness, durability and reliability of the Wi-SUN protocol before deploying them in a larger area. Some other modalities can also be tried, for example, to switch the lights based on activity using a Passive Infrared (PIR) sensor, which can provide further opportunities for energy saving. oneM2M integration also provides unique opportunities in terms of applications and control. For example, a digital luminescence sensor can be deployed on the rooftop building which posts the data on oneM2M. These values can be fetched by the street light nodes connected to the Wi-SUN border router gateway, to make a decision whether the lamp should be off or at a specific brightness depending on the local weather conditions. Further, information related to pedestrians or large crowds can be obtained from the oneM2M platform, which can then be used for light switching.

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Smart Street Lighting Systems for Sustainable Smart Cities: A Conceptual Framework

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Abstract— The rapid growth of urban population has significantly complicated urban planning and management. This paper examines the connected street lighting network as an innovative infrastructure solution for smart cities by integrating cutting-edge technologies into widespread street lighting network systems. Through a comprehensive literature review and expert consultancy, the study categorizes the predominant components of smart street lighting (SSL) and evaluates their potential contribution to sustainability and urban digital transformation. Among global researchers, China, Spain, and India stand out as the leading countries in this field. Furthermore, it is enhanced that the proposed connected lighting network not only increases the energy efficiency of cities and reduces their carbon footprint, but also contributes significantly to the improvement of human and ecological quality of life and the effective monitoring of various urban planning services. The study inferred that a connected street lighting system affects all six key smart city dimensions with the most substantial contribution (30%) to the smart living dimension. Additionally, the system aligns with nine of the 17 Sustainable Development Goals (SDGs), most prominently supporting SDG 11 (Sustainable Cities and Communities).

Keywords— Sustainable smart cities, sustainable development goals, connected street lighting, smart lamppost.

I. INTRODUCTION

In today's world, rapid and poorly planned urbanization observed globally poses a significant risk to sustainable development. The United Nations (UN) projected that approximately 60% of people will reside in cities by 2030, potentially increasing to 75% by 2050 [1]. Increasing growth implies many challenges, including climate change, energy scarcity, global warming, and incremental environmental pollution [2]. Given these compelling issues, the evolution of urban and rural areas into smart and sustainable living places is no longer an optional strategy but a necessity. [3].

In 1974, the city of Los Angeles pioneered the systematic collection of comprehensive urban data aimed at improving the quality of life of its residents. This initiative, known as 'The State of the City a Cluster Analysis of Los Angeles', constituted an early instance of what could be considered the basic principles of a smart city, without directly using the term smart city at the time. The concept of a 'smart city' implies improving citizens' quality of life, promoting economic development, resolving transportation and congestion problems through effective regulations, providing a clean and sustainable environment, offering transparent, interactive, inclusive and accessible governance, and integrating digitalization into research and development processes [4]. Smart city is the effective integration of physical, digital and human systems in the built of environment to deliver a sustainable, sufficient and inclusive future for its citizens. Sustainable smart city is the incorporates with energy

consumption, waste management, climate change, and the circular economy into city management There are many definitions and evaluation methods in the literature [5]. The brief explanations and key factors of each dimension is presented in Table I.

TABLE I. OVERVIEW OF SMART CITY DIMENSIONS

SMART CITY DIMENSIONS	EXPLANATION & KEY FACTORS
Smart Economy	- Highly efficient and technologically advanced economy - Urban infrastructure management, entrepreneurship and innovation, economic and tourism growth
Smart Environment	- Natural conditions of living in the city, pollution and resource management and environmental protection - Sustainable urban planning, energy storage systems, water, waste, pollution, infrastructure planning, air quality detection, climate change, green energy
Smart Mobility	- Availability and accessibility of intelligent mobility services - E-mobility, accident detection, international accessibility, sustainable, innovative, and safe transport systems
Smart Governance	- Effective and efficient public administration, quality of public services, and the participation of residents in making decisions - Integrated resource management, decision-making participation, transparency & open data, public & social services, e-government
Smart Living	- Considering the health, safety, culture, and living conditions of the citizens - Touristic attractiveness, safety, and health conditions, quality of life and well-being, urban emergency service, facilities for culture, and education and leisure
Smart People	- Qualifications and education of the city's inhabitants, as well as social interactions related to integration and public life - Flexibility, open-mindedness, inclusive society, social and ethnic plurality

In the era where the current global focus is on energy management, approximately 40% of electricity consumed by municipalities is dedicated to street lighting. Globally, streetlights account for roughly 3% of total electricity consumption, a seemingly low percentage that represents a massive global energy demand and underscores the significant impact of urban lighting infrastructure [6].

Given their widespread infrastructure throughout the city, streetlights serve as a backbone for the establishment of smart cities [7]. Connected streetlights, also referred to as intelligent or networked street lighting, represent a significant technological enhancement in urban infrastructure. These advanced lighting systems are equipped with sensors, wireless connectivity, and other cutting-edge technologies, allowing them to interact intelligently with their surroundings [8]. These sensors can track various parameters, including air quality,

noise, temperature, water level and quality, light pollution level, traffic flow, available parking places, and fundamentally pedestrian and vehicle movements. These data enable city officials and policymakers to make informed decisions to enhance several city services' efficiency and liveability.

Several research findings indicate that smart street lighting (SSL) systems contribute to achieving several Sustainable Development Goals (SDGs) [9] adopted by the UN in 2015, including SDG 11 (Sustainable Cities and Communities) [10], SDG 7 (Affordable and Clean Energy) [11], SDG 3 (Good Health and Well-being), and SDG 13 (Climate Action) [12], SDG 15 (Life on Land) [13] and SDG 12 (Responsible Production and Consumption) [12].

The paper categorizes SSL components into four key areas: Light-Emitting Diode (LED) technology, renewable energy-powered lighting, wired/wireless communication technologies, Internet of Things (IoT) integration, and other integrated urban services such as emergency buttons and digital broadband. The study subsequently explores the 13 impacts of smart street lighting on creating a sustainable urban environment.

A conceptual framework is presented to illustrate the comprehensive overview in Section 2. Section 3 details statistical analyses conducted through an extensive literature review. Section 4 discusses the challenges, limitations, and provides a concluding summary.

II. METHODOLOGY

A. Flowchart of the Research

The study's flowchart is illustrated in Fig. 1. The research question was elaborately determined. Subsequently, a comprehensive keyword query was developed, integrating sub-keywords from three essential concepts of the paper: *street lighting*, *smart cities*, and *SDGs*. Utilizing the SCOPUS database, 556 articles were retrieved at first with this search strategy. Through an elimination process, 83 articles were ultimately identified as meeting the research target.

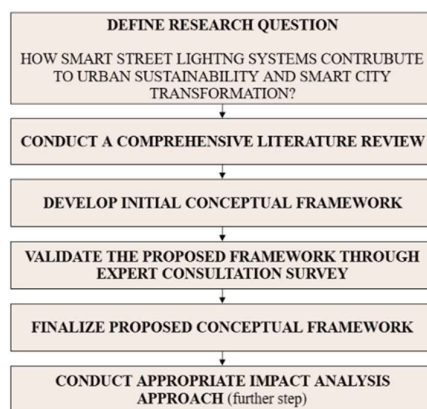


Fig 1. Research flowchart

B. Conceptual Framework

The research develops a comprehensive thematic framework in Fig. 2 mapping smart street lighting's four key components, their 13 fundamental smart and sustainable impacts, and affected smart city dimensions and SDGs. A brief overview of the SSL components is provided below.

LED Technology: The use of LED lamps presents a remarkable advantage over traditional lighting technologies by

significantly reducing energy consumption [14]. LED lamps, when used with the right brightness and colour levels and at the right timing, not only minimize light pollution and CO₂ emissions, contributing positively to the environment and ecosystem, but also offer substantial long-term economic benefits for cities [15]. The capability to remotely control LED lights enhances the quality of life by allowing adjustments to light brightness and colour in accordance with people's routines. Furthermore, light shows and events arranged with LED lights (e.g. "GLOW Eindhoven" in the Netherlands) supports citizen interaction, touristic circulation and the city's economy.

Renewable Energy Powered Lighting: Uncontrolled and non-renewable energy consumption are the primary causes of global issues such as energy crises, environmental pollution, and climate change [6]. By 2030, greenhouse gas emissions are expected to decrease by 37.8%, falling short of the anticipated average reduction target of 46.6% [16]. Transitioning to renewable energy sources, such as solar, wind, and even biomass, for electricity generation presents a viable solution to reduce overall *electricity consumption* and increase *environmental sustainability* [17].

Wired or Wireless Communication Technologies: The connections can be achieved via wireless technologies such as Zigbee, 5G, LoRa, Narrowband IoT (NB-IoT), Light Fidelity (Li-Fi), and global positioning system (GPS), as well as through wired lines like Power Line Communication, ensuring data communication and transfer between lights, sensors, and central control system for optimal operation (e.g. fault detection) [18]. Remote monitoring and management of street lighting can be achieved through these communication technologies. Adaptive street lighting systems intelligently adjust light intensity based on a variety of factors, such as astronomical time, the presence of pedestrians or vehicles, weather conditions, natural light levels, traffic density, and location-specific needs, including those for entertainment venues or university facades [19]. Future SSL systems are anticipated to leverage artificial intelligence techniques for predictive traffic analysis, with the primary objective of optimizing system efficiency and reducing operational load.

IoT Integration: Smart streetlights can be integrated with the IoT, enabling them to communicate with other devices and systems, thereby creating an interconnected infrastructure network [20]. Light sensors (e.g. Light Dependent Resistor (LDR)) sensing the intensity of light and sends the signal to microcontroller for turning on the street lightning system; motion (e.g. infrared (IR) sensor) and sound sensors to detect the pedestrian or vehicle; environmental sensors to measure environmental limitations; vibration sensors to anticipate natural disasters like earthquakes and floods and GPS sensors are some of crucial sensors among IoT devices [15,21].

Other Integrated Urban Services: Streetlamps not only serve as a lighting source but can also function as a charging stations for various electronic devices, including mobile phones and laptops. Installing EV charging units on appropriate lampposts is one way to reduce the cost requirement for a new charging installation [16,22]. Integrating additional components such as emergency call or panic buttons, digital information/advertisement boards enhances citizen interaction, improves safety and security, and contributes to a more sustainable and attractive urban environment [23].

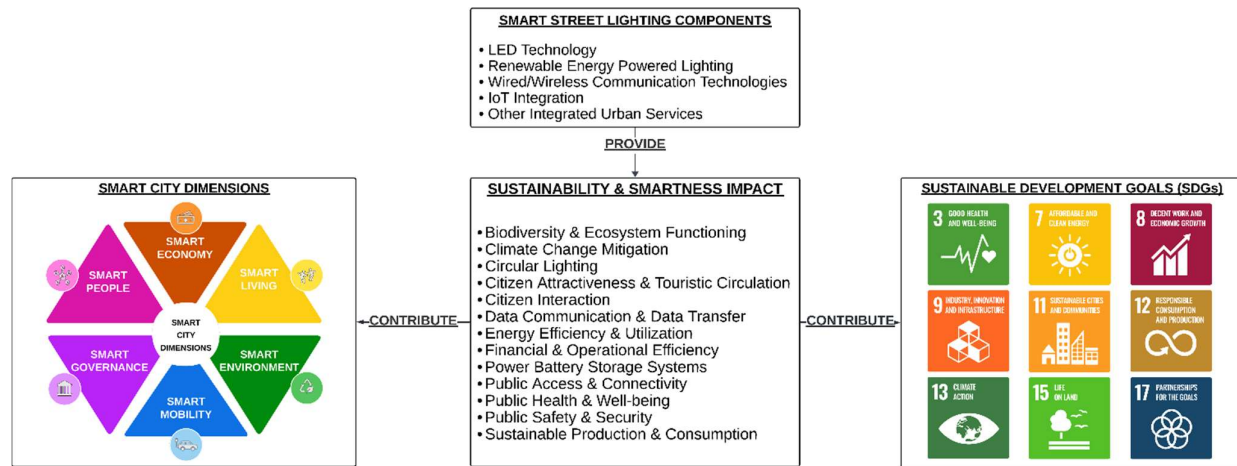


Fig 2. Thematic structure of the conceptual framework

III. RESULTS

This section presents the results of the statistical analyses conducted using the data obtained from the comprehensive literature review. The trend presented in Fig. 3 illustrates a growing focus on street lighting research, with an continuously increasing number of studies conducted each year.

Based on the smart city dimensions outlined in Table I, 83 articles are categorized. As shown in Fig. 4, SSL systems impact all smart city dimensions at varying rates, with the smart economy and smart environment dimensions receiving the most significant contributions. For smart mobility dimension, traffic density can be measured using a traffic sensor installed on streetlights, with real-time data flow and monitoring facilitated through connected systems. By dynamically adjusting light levels based on traffic data and a set schedule, from the environment perspective, energy efficiency is enhanced, CO₂ emissions are reduced. Based on SCOPUS-indexed research, Fig. 5 reveals that SSL infrastructure most significantly supports SDG 11, contributing approximately 30% towards creating safe, accessible, and sustainable urban systems. Although the impact level is lower than others, controlling nighttime lighting through adaptive systems in sustainable urban environments contributes to SDG 15 by mitigating the harmful effects of light pollution on humans, flora, and fauna and protecting the loss of biodiversity [13]. Additionally, urban lighting serves as a tool for security and surveillance, enhancing the city's aesthetics, promoting sustainable tourism (SDG 8), and supporting economic growth (SDG 9) and increasing accessibility and profitability of Information Communication Technology (ICT) in urban infrastructure (SDG 17).

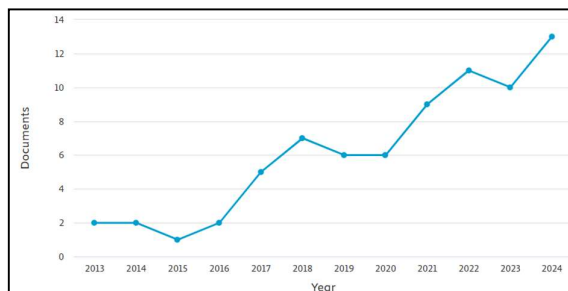


Fig 3. A yearly trend analysis of selected articles

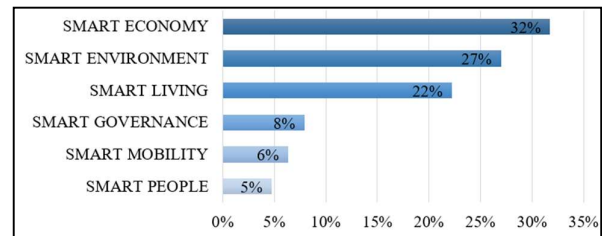


Fig 4. SSL impact distribution across smart city dimensions

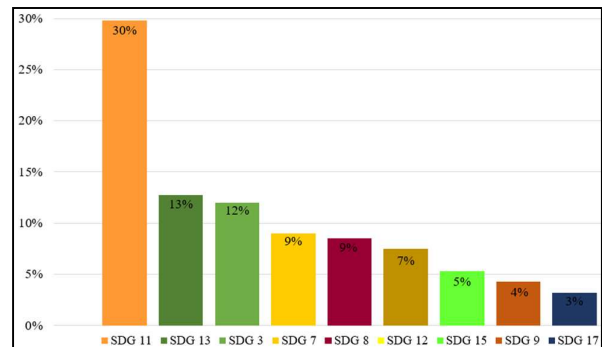


Fig 5. SSL impact distribution across related SDGs

IV. DISCUSSION AND CONCLUSION

Street lights have become essential for distributing and monitoring various network systems from a central hub due to their strategic location, reliable power supply, and integrated communication infrastructure. However, several challenges remain for advancing SSL systems, including identifying gaps in existing infrastructure, the need for advanced technical and security training while integrating diverse sensors and actuators, the deployment of complex technologies, data protection, and cybersecurity concerns [24].

Experts also discuss that solar-powered street lighting in city centers might be economically impractical due to high installation costs and limited power generation. It is more suitable for rural areas where low connection networks exist. Therefore, establishing a well-designed lighting system that considers all aspects of sustainability is crucial. While striving for energy efficiency through LED transformation, challenges arise in balancing operational, and energy consumption costs when integrating IoT and other smart systems. LED lamp

selection should prioritize resilience in adverse weather conditions to optimize driver safety and visibility.

The literature review highlights the broad impact of integrating smart systems into street lighting, gathering disciplines including engineering, computer science, energy, and social sciences. Citizen participation and awareness in developing these systems is essential, yet the role of these aspects with smart street lighting remains underexplored. Given that smart lighting systems rely considerably on citizen activity data, this gap is crucial. Urban planners, architects, and engineers must collaborate to design sustainable, smart cities. This framework guides policymakers on integrating street lighting systems with municipal goals, emphasizing their connection to the SDGs for encouraging smart urban environments.

Street lighting is evolving beyond its traditional roles of ensuring safety, security, and visibility for pedestrians and vehicles, becoming a vital element of urban infrastructure that enhances the use of public spaces at night. This transformation not only improves key aspects of urban life but also facilitates the integration of 'smart services'.

This framework provides policymakers with guidance on key integration areas within street lighting systems that align with municipal goals. It emphasizes the connection between smart street lighting and the SDGs, promoting the development of lighting infrastructure that encourages sustainable, smart city environments. The target is to improve the efficiency of urban street lighting in support of sustainable urban development. As the field evolves, further research is anticipated, highlighting the importance of the comprehensive literature review in identifying key study areas, keywords, and methodologies in smart street lighting. Further research intends to clarify the relationships between concepts by using numerical data and appropriate decision making approaches.

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Street Light Intensity Control

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Abstract---Nowadays the usage of light sources has increased drastically. Particularly in the highways, streets when there is no vehicle movement, a huge amount of light source has been wasted. Nearly 20% of electricity is lost due to technical and commercial reasons according to the Ministry of Power. The actual consumption for public lighting in India is 7753 GWh. The core components of this system include Arduino and sensors. This system not only helps conserve energy but also lowers operational expenses and reduces carbon emissions, offering a sustainable and cost-efficient solution for contemporary urban lighting infrastructure. Our project gives the solution for electric power consumption by street light intensity control. Traditional street lights are being replaced by light emitting diodes, which reduces power consumption the prototype uses LDR (light-dependent resistors), infrared sensors (IR) for detecting the motion of objects, ultrasonic sensors for measuring the distance of moving objects, lithium batteries, piezoelectric sensor, solar PV as a source of electricity.

Keywords---*Arduino, LDR, IR sensor, Ultrasonic sensor, Piezoelectric sensor, Solar Street lights.*

I. INTRODUCTION

People are too busy these days to turn out the lights when they're not needed. Currently, street lights are turned on in the evening just before dusk and turned off the next morning after there is enough light. The main aim is to control the light intensity in streets and highways at night times. In late night lots of energy is wasted when there is no vehicle movement and less density areas. We can control the high light intensity to low light intensity. So, we can save the power consumption of electricity. We use solar PV as a source of electricity. LDR (light-dependent resistor) is used to detect the darkness of the surroundings while an IR sensor is used to detect the speed of the moving object. A street lighting system with conventional

street lights has been replaced by light emitting diodes which reduce power consumption by using LDR.

II. LITERATURE REVIEW

To discern between day and night, there is a light-dependent resistor [1] (LDR) and an ultrasonic sensor for motion detection. The object glows dimly while there is no movement detected by the ultrasonic sensor, then brilliantly when someone walks by. Upon exiting the sensor's range of view, the light automatically [2] decreases in brightness. The accuracy of the system's movement detection is constrained, though. Although the project's setup and beginning expenditures could be considerable, total prices might be decreased through mass manufacturing and technological breakthroughs.

The detection mechanism of the system makes use of infrared emissions. The infrared sensors receiver receives the infrared radiation that the transmitter's infrared radiation reflects off of an item. The object is detected and measured using the signal that was received. The input voltage is raised by a DC-DC converter called the LUO Converter. With the help of a PIC microprocessor, a LUO converter, and an infrared sensor, this system effectively controls street lighting. The duty cycle determines the intensity of the light, and the PIC Controller, which is a member of the Microchip technology family, produces the PWM signal for the power converter.

This converter lacks an inductor, avoiding channel inductance and resulting in high voltage and productivity. Most converters typically use an inductor. The adaptive street lighting system dims when no movement is detected and brightens when there is movement [3]. Unlike traditional, static lighting or pre-set dimmable road lighting, this system adjusts based on traffic and crowd data. Light sensors included in the design are useful in wet environments. The system allows remote management of multiple streetlights. It operates on an RTC, with lights at 100%

intensity during peak traffic hours and lower intensity at other times. This project offers a safe, practical, and affordable way to save energy [5], addressing the issues of wasted energy from burning lamps and conserving electricity. Statistical data indicates a potential saving of about 35% in the electrical energy used by roadways[9][12].

The LUO converter's primary function is to transform low voltage sources into usable voltage sources. Solar energy, which provides low voltage, is maintained by the LUO converter. The PIC Controller uses the output voltage of the LUO converter for control. The LUO converter offers outstanding efficiency, ease of management, and reduced switching losses. The proposed SCDS DC-DC converter design achieves high voltage gain with a small duty cycle and low voltage stress on MOSFETs and diodes [6], reducing power regulation losses. This design supports both continuous and discontinuous conduction mode operations.

III.PROPOSED TECHNOLOGY

The proposed cure for street lights is to do away with the need to manually turn them on and off. We utilised a light-dependent resistor, or LDR, to accomplish this, which activates the light in response to changes in the amount of ambient light. To detect human movement and vehicle speed, respectively, infrared and ultrasonic sensors[7] are employed. During the wet season, solar photovoltaic cells cannot provide enough energy. Therefore, we have turned to piezoelectric sensors, which generate power in response to vibrations caused by moving autos.

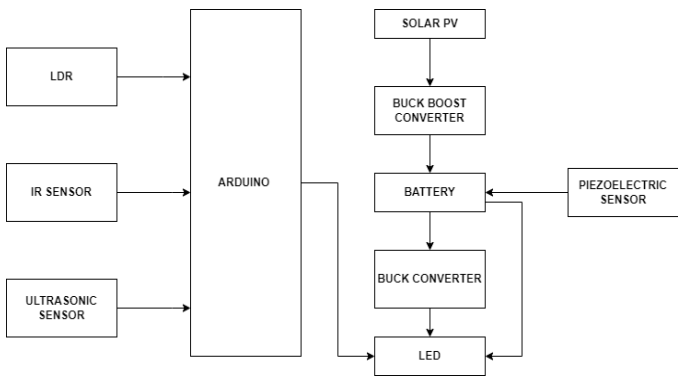


Fig.1 Block Diagram

Two converters make up the suggested methodology, as seen in fig-1. One for working in constant voltage and constant current mode, and another for charging the batteries .

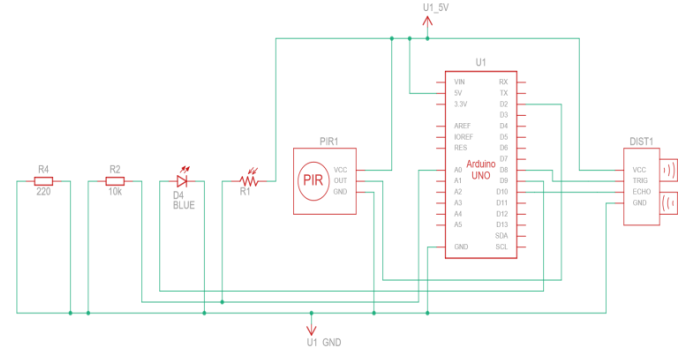


Fig 1.1

The architecture diagram featuring the pin representation of the Arduino and the connections between the various components is displayed in Figure 1.1.

IV. EXECUTION

Light Emitting Diode (LED) lighting takes the role of conventional lights. Install photovoltaic solar panels [13] to power the street lights. Rainy days prevent us from producing enough energy. Thus, we employ the piezoelectric sensor, a kind of sensor that, when subjected to mechanical pressure or vibration, produces an electrical charge. A Buck-Boost converter is a DC-DC converter that has an output voltage magnitude that can be more or lower than the input voltage magnitude. Buck-boost converter operating in both directions for solar panels and battery energy storage systems panels are powered by available light. Through the use of a bidirectional DC-DC converter, the PV array will charge the battery and provide electricity when its irradiance reaches a certain level during that time. install Light Dependent Resistors (LDRs) to measure ambient light levels, Infrared (IR) sensors to detect the motion of objects, and Ultrasonic sensors to measure the distance of moving objects. We use IR sensors to detect the motion of objects (such as vehicles or pedestrians).

V. TECHNOLOGIES USED

- a) Tinker Cad: An online, browser-based 3D modelling application used for electronic simulation and prototyping.
- b) Programming Languages: It had been necessary to use the C language to construct different project components. We utilize C code in our project to connect the components.
- c) Arduino: An open-source hardware and software platform. Arduino [4] boards are based on the AT mega line of microcontrollers, which can be programmed for various functions, making them the project's control centre.
- d) LDR (Light-Dependent Resistor): A type of photosensitive electrical component, also known as a photo resistor, that changes resistance when exposed to light.
- e) Ultrasonic Sensor: An apparatus for measuring distance by converting ultrasonic sound waves reflected from an object into

an electrical signal. The ultrasonic sensor is used in this paper up to a 100-meter distance.

f) IR Sensor (Infrared Sensor): Detects infrared radiation [8]. When an object approaches, the sensor's LED emits infrared light, which is reflected and picked up by the receiver, commonly used in obstacle detection systems.

g) Buck Converter: A DC-to-DC converter that steps down voltage while stepping up current from its input to its output, also known as a step-down converter.

h) Solar PV (Photovoltaic): Photovoltaic panels, usually installed on the lighting structure or integrated into the pole itself, provide the energy for solar street lights, which are elevated light sources.

VI. RESULTS AND DISCUSSION

Tinker cad software has been used to model and simulate the suggested system. Figure-2 displays the suggested simulation diagram; Figure-3 shows the results of the simulation and the numerous system connections that are mentioned. We can alter the range of each sensor in C++ code based on the application and requirement. The simulation provides the results of an IR sensor, an ultrasonic sensor, and a light-dependent resistor connected to a bread board. For instance, the ultrasonic distance is 71.18 cm if the LDR value is 54. There is 255 LED brightness.

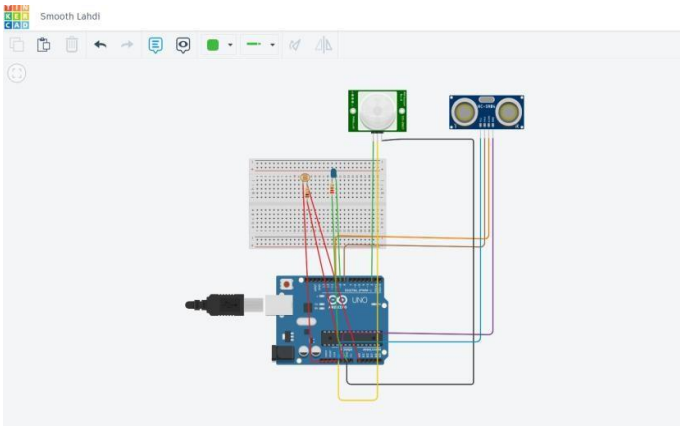


Fig. 2 represents the circuit connection of the simulation.

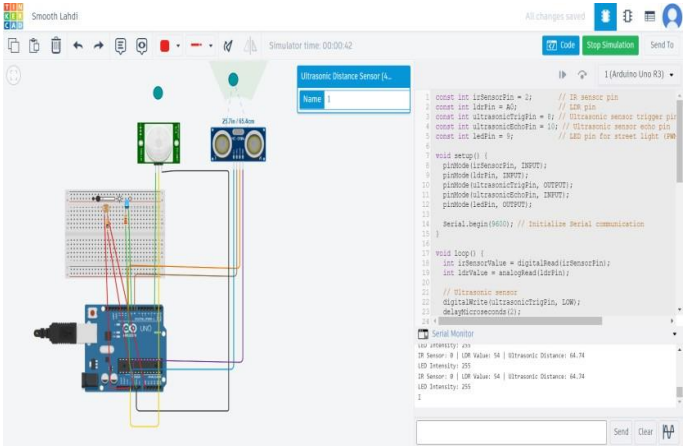


Fig.3 Simulation output

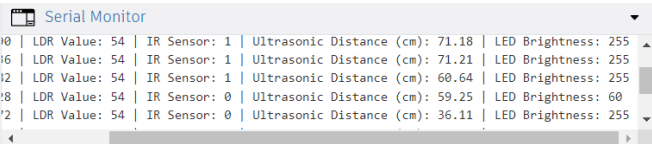


Fig.4.1

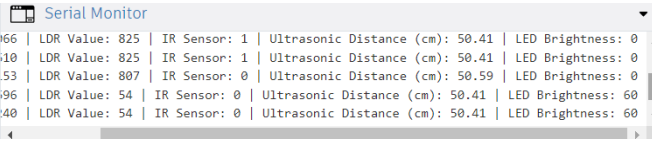


Fig.4.2

Fig.4.1 & Fig.4.2 Represents the serial monitor output from Tinkercad

VII. FUTURE WORK

Install energy monitoring devices like smart meters or energy monitors at the desired locations to measure energy consumption. These devices may vary depending on your specific needs and the scale of monitoring. Set up a data processing system to clean and organize the collected energy consumption data. You may need to convert raw data into usable formats, calculate energy consumption values, and possibly perform data filtering or aggregation. Then connect to a cloud service provider like Azure or Google Cloud. The initiative is part of the Indian government's smart city plan to update the city's infrastructure and street lights. Future needs may require modifications to this project. Even with this suggested approach, electricity usage can be decreased by using solar-powered lighting. In later iterations, this project might have capabilities for detecting accidents and fire incidents. This discovery will be useful for tracking systems in automobiles in the future.

VIII. CONCLUSION

This paper has explained about the solar based street light intensity control by using LDR, ultrasonic sensor, IR sensor and also, we have used piezoelectric sensor. Thus, we have used LDR, ultrasonic sensor and IR sensor and simulated them using the software Tinkercad. We can cut down on the amount of energy wasted by using street lights at night when fewer people and cars are around. The prototype model has to be given the exact solution for the wastage of power and light pollution during the late night in highways and large cities.

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Streetlight System Information Modelling and Management for Energy Reduction

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Abstract—Due to the growth of urbanization, streetlight systems are one of the most crucial devices related to city life. Most of the existing streetlight systems in Malaysia are conventional in terms of energy efficiency and cost reduction. This project focuses on using streetlight system information modeling to create correlations between consumed energy and actual utilization of the streetlight system with real-time data collection in Universiti Teknologi MARA Cawangan Pulau Pinang (UiTM CPP). The proposed technique used the most advanced method known as Artificial Neural Network to predict the total maximum demand for the streetlight system. The data analysis compares the High-Pressure Sodium (HPS) and Light Emitting Diode (LED) streetlight systems. A few comparisons, a total of maximum demand, cost analysis, and luminous flux, have been made. The data comparison between two different streetlight systems makes the proposed replacing the HPS streetlight systems with LED streetlight systems are reasonable.

Keywords—Streetlight System, Energy Efficiency, Artificial Neural Network, HPS, LED

I. INTRODUCTION

Streetlight is very important to all pedestrians and all types of transportation on the road, and it is a night-time safety for the road users and around the buildings. Without streetlight systems, criminal cases will increase, and also the case of the accident or collision of transportation increases higher [1]. The improvement of streetlight systems will result in social monitoring and fewer cases of criminal activities [2]. Besides, streetlight systems are outdoor lighting with high power consumption due to extended operating hours [3]. Streetlight systems cover from 60% to 80% of total electricity consumption by a city, leading to the high cost of maintaining public lighting [4].

The estimation of lighting alone, which refers to streetlight systems, had been estimated to be 19% of the total global electricity consumption [2]. Reducing the lighting energy consumption by 40% may lead to a significant electrical reduction from 19% to 11%. The Advantage of energy reduction of streetlights also may minimize carbon emission. There are several types of lamps used in streetlight systems. These types of lamps are Metal Halide lamps, Low-Pressure Sodium (LPS) lamps, High-Pressure Sodium (HPS) lamps,

Mercury Vapour lamps, High-Intensity Pressure (HID) lamps, and Light Emitting Diode (LED) lamps [2].

Most streetlight systems in Malaysia still use the HPS streetlight system compared to the LED streetlight system. LED lamps are not a new technology; however, they are used widely in many related products and adopted in our daily lives. The LED technology is not entirely accepted now to replace the HPS streetlight system used throughout the country [13]. This project will study these two types of streetlight systems to compare the system performance. The LED streetlight system with an extended operating life span is up to 50,000 hours with very minimal power consumption, making it more worthy of replacing the HPS streetlight system with a shorter lifetime and consuming high-power consumption [5].

Installation of street lighting must provide sufficient visual information to proceed with a safety guide and sense of security for road users [14]. The two aspects that ensure the possibility of street lighting installation is satisfied are the reliability of perception and the degree of visual comfort [14]. The influencing reliability of perception is divided into three parts: average road surface luminance, overall uniformity, and threshold increment [14]. For average road surface luminance (L_{av}), the suitability for road surface luminance is to improve the sensitivity of road users. Next is the overall uniformity (U_o), where the formula is given as below [14]:-

$$U_o = L_{min}/L_{av} \quad (1)$$

The difference sensitivity between the minimum and average road surface luminance must be defined to ensure all the road spots are secured. The threshold increment (TI) is the extra contrast required to see an object when there is glare, relative to the original contrast [14]. The next luminance concept, the parameter influencing visual comfort, is average road surface luminance, longitudinal uniformity, glare control mark, and visual guidance [14]. The principle of road lighting has four factors in lighting design: luminous flux, luminous intensity, illuminance, and luminance. All the basic concepts in lighting design will be calculated by using a calculation based on the type and design of the street light.

These days, most researchers apply the Artificial Neural Network (ANN) technique through time series data analysis, specifically in the electrical power system. The neural network inspires by the human nervous system, where the number of processing units comprises. A load forecast plays a vital role in estimating the future load requirements, and the data obtained can be trained to classify and predict future events [7]. The prediction that is too high or too low can result in revenue loss in the future. This research uses ANN to predict electrical load demand. Two modifications for the backpropagation method uses to improve the error of selected activation and improved error by using mean square error [8]. The ANN uses to predict the maximum demand of the HPS streetlight system and LED streetlight system and compare the actual reading of maximum demand. The possibility of the prediction data comes with great accuracy. The proposed ANN model train with the historical load data of streetlights [9].

A study in the year 2008 presents an implementation of a backpropagation neural network method to improve electricity demand forecasting where demand depends heavily on different independent variables such as weather, temperature, holidays, weekdays even strikes [8]. The research uses ANN to predict electrical load demand where two modifications have been made were the backpropagation method to improve the error of selected activation and the new and improved error by using mean square error [8]. There is no assumed model structure to describe a system. Instead, the network is adjusted or trained so that a specific input results as a specific target output [8].

The other paper presents a performance prediction of a distributed PV generation system using ANN and Mesh Networks. In this research, the method of ANN is used for correlating the climatological parameters by using data of each technology in solar panel for prediction of DC output of an inverter [10]. In the study case by N. Srinivasan and C. Lakshmi, ANN is used to predict the stock market. The paper has studied an efficient algorithm for predicting the stock-based values used in all aspects [15]. The neural network is inspired by the human nervous system, where the number of processing units has been comprised. The connection toward the output layers having connections and implementing this three-layer structure forward the perception network.

UiTM CPP uses the HPS streetlight system with timer operation. As we know, the HPS is a lamp that consumes a lot of energy, and it causes higher maximum demand. This streetlight system uses a timer where it will operate within 12 hours where it turns on at 7.00 pm, and will turn off at 7.00 am. Thus, this study will analyse the maximum demand between two types of streetlights, HPS and LED, and the prediction of cost analysis based on the data collection. In addition, this study can relate the amount of energy used daily by streetlights. From the data, an analysis of energy consumed by streetlights was carried out. The finding from this study has proven that the current streetlight system uses a lot of energy. Thus, this research creates awareness to consumers about wastage that the current streetlight system has done.

Fig. 1 shows a system overview of streetlight information modelling in UiTM CPP. The data logger is to measure the energy consumption of the streetlight system in UiTM CPP. The data logger is a device that presents data where it is connected to desktop software, and it is used to record the correct data every time with a quick guided and graphical user

interface. The data logger had been installed at the Main Switch Board (MSB) of Kolej Kristal in the electrical room for the HPS streetlight system and a feeder pillar at Unit Kesihatan for the LED streetlight system.

II. METHODOLOGY

A. System Operation

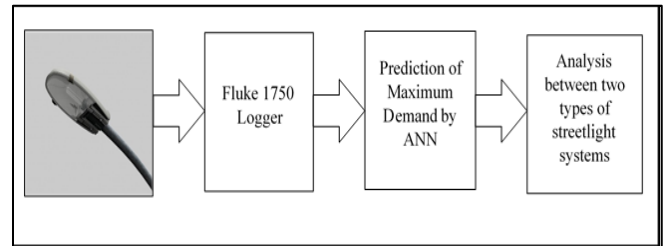


Fig. 1. Block Diagram of Streetlight Information Modelling.

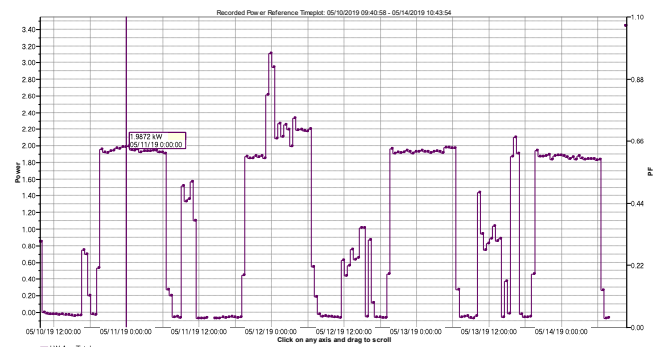


Fig. 2. Power Consumption of HPS During ON

Fig. 2 shows the actual data for power consumption of HPS streetlights from the data logger placed at MSB of Kolej Kristal. It shows the power consumption reading of the HPS streetlight during ON condition. From the graph, it can be seen that the power consumption is relatively stable for the first day of reading. During off, which is in the daytime, the system is not stable where there is power consumption at noon. The timer of the street light may not function in a good situation.

The graph also shows that the power consumption on the second day has a peak reading that goes until 3.107kW, where the actual reading is only about 1.95kW. The reading is very high compared to actual utilization, where one lamp is 150W, and 13 poles of lamps are about 1.95kW. The peak condition will cause a high peak of maximum demand and charge a high cost. The graph concluded that the system of HPS streetlight is not stable and it causes high power consumption and the cost may increase.

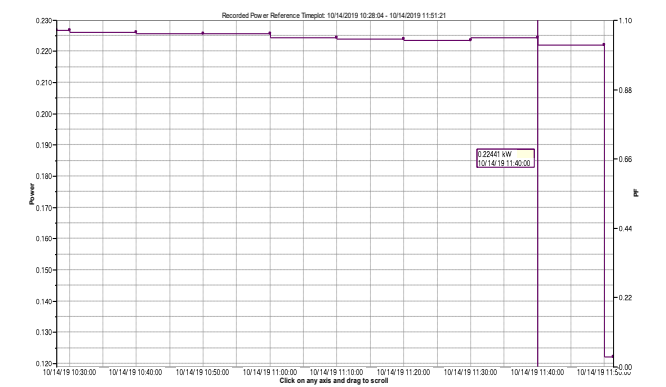


Fig. 3. Power Consumption of LED During ON

On the other hand, Fig. 3 shows the actual power consumption data for LED streetlights from the data logger placed at the feeder pillar of Unit Kesihatan. From the graph, it can be seen that the power consumption reading of LED streetlights is more stable. It shows the power consumption of a feeder pillar which is for eight poles is around 0.23kW. Reading for the actual data is lower than actual utilization, which is for one lamp is 40W. It shows the LED streetlight used very low power consumption compared to the HPS streetlight system.

The results obtained had been imported to MATLAB software. In MATLAB software, the maximum demand prediction will be simulated using Artificial Neural Network (ANN). The prediction of maximum demand by ANN based on duration (time), voltage (V), current (A), power factor (pf), real power (W), apparent power (VA), reactive power (VAR), and frequency (Hz) of both streetlight which is HPS, and LED data logger has measured that in a certain period of time. The comparison graph of the actual maximum demand and predicted maximum demand will be present to show the comparison.

Then, the analysis between two types of streetlight systems has been made. A few comparisons, total maximum demand, cost analysis, and luminous flux, have been made. In addition, the total maximum demand had been analysed between consumed energy and actual utilization. Moreover, the cost analysis is based on TNB tariff and maintenance cost based on the research paper. Lastly, the luminous flux was measured by using a lux meter.

Fig. 4 shows the overall flowchart of the research methodology. Two different types and total of streetlights in a small area are at Kolej Kristal and Unit Kesihatan. Firstly, the calculation of total connected load and maximum demand has been calculated theoretically. Then, the energy consumed has been measured at the feeder pillar at both locations using a logger. The data then be imported to MATLAB software to perform maximum demand prediction by using the ANN method. Finally, a comparison graph from the prediction result has been plotted to compare the actual data of maximum demand consumption with the prediction of maximum demand.

A multilayer feed-forward ANN model with supervised learning has been implemented to predict the hourly peak load demand for any particular day. The proposed ANN model has a single hidden layer where the repetition in convergence results in minimum percentage prediction error by varying the number of hidden layer neurons. Besides, the ANN approach is a generic technique for mapping the relationships between inputs and outputs and requires less expertise and experimentation than traditional modelling.

The maximum demand of both streetlights systems was predicted using ANN to compare the actual reading of maximum demand. The possibility of the prediction data comes with excellent accuracy that has been obtained. The ANN model is trained with the historical load data of both streetlights systems, HPS and LED streetlights systems. The input and output parameters have been imported from the fluke 1750 logger, which are the duration (time), voltage (V), current (A), power factor (PF), real power (W), apparent power (VA), reactive power (VAR) and frequency (Hz). The network will be trained and validated. If the result is satisfied, a comparison graph will be plotted to compare actual and

prediction data. If the result is still not satisfied, all the input will be modified. Lastly, the result obtained has been analysed in the result section.

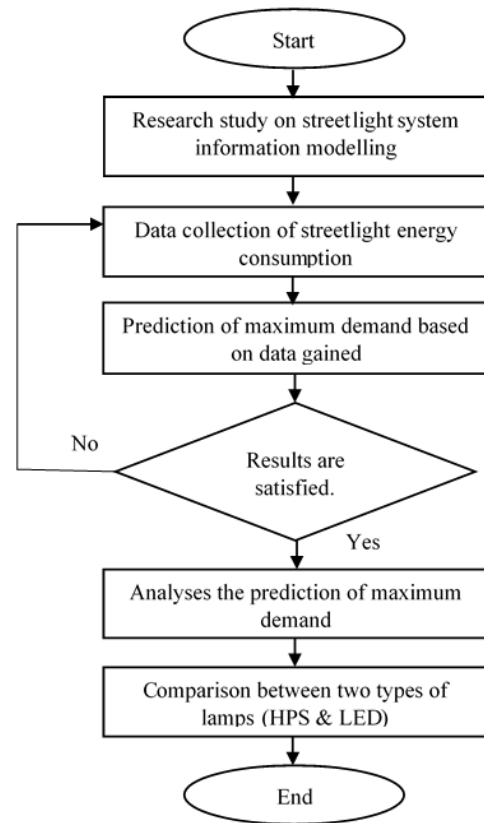


Fig. 4. Overall Flowchart of Research Methodology

B. Artificial Neural Network (ANN) System Operation

Fig. 5 shows the structure of the Artificial Neural Network (ANN). A multilayer feed-forward ANN model with supervised learning will be implemented to predict the hourly peak load demand for any particular day [9]. The proposed ANN model has a single hidden layer where the repetition in convergence results in minimum percentage prediction error by varying the number of hidden layer neurons. Besides, the ANN approach is a generic technique for mapping the relationships between inputs and outputs and requires less expertise and experimentation than traditional modelling [11].

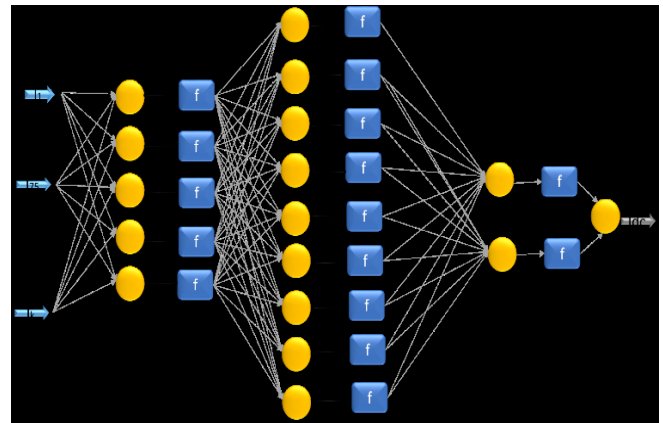


Fig. 5. Structure of the Artificial Neural Network [10]

In this research, eight inputs have been interpreted, which are duration (time), voltage (V), current (A), power factor (pf), real power (W), apparent power (VA), reactive power (VAR) and frequency (Hz). Then from the input, it has made prediction maximum demand based on the input. The single hidden layer will train the output to predict maximum demand with minimum percentage error. The network is adjusted or trained so that the specific input results as a specific target output.

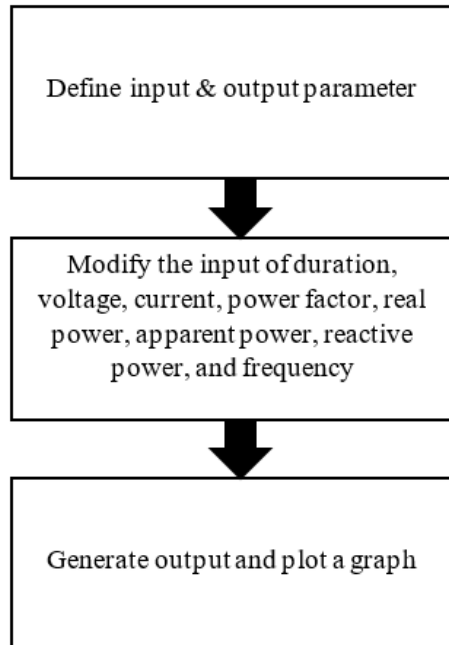


Fig. 6. Artificial Neural Network (ANN) System Operation

The maximum demand of both types of streetlights was predicted by using ANN to compare the actual reading of maximum demand. The possibility of the prediction data comes with reasonable accuracy that has been obtained. The ANN model is trained with a historical load of streetlights, HPS, and LED streetlights, with different data collected by the data logger.

Then, the input and output parameters will be imported from the data logger, which are the duration (time), voltage (V), current (A), power factor (pf), real power (W), apparent power (VA), reactive power (VAR) and frequency (Hz). After complying with the data for training and validation, the coding of prediction using ANN will be specified in an editor. Then, the network will be trained and validated. If the result has been satisfied, a graph will be plotted to compare actual data and prediction data. If the result is still not satisfied, all the input will be modified, as shown in Fig. 6. Lastly, the result obtained has been analysed in the result section.

III. RESULT AND DISCUSSION

A. High-Pressure Sodium (HPS) Streetlight

The actual data from the data logger shows for HPS streetlight consumption, and it shows the data of voltage, current, power consumption, and power factor. Based on the calculation, the total maximum demand has been proved by the actual data from the logger. Table I and II and Fig. 7 show the plot of actual maximum demand data and prediction of maximum demand by ANN. The comparison of actual results with the experimental visualized is in the graph. The

prediction values of maximum demand are ahead of the actual maximum demand values. There is a minimum error percentage between actual maximum demand and predicted maximum demand.

TABLE I. DATA OF HPS STREETLIGHT

Time	Vrms			Irms		
	AB	BC	CA	A	B	C
19:00:00	408.748	409.707	409.251	0.992	2.292	0.252
20:00:00	409.965	411.698	410.711	5.273	5.121	0.272
21:00:00	410.702	412.667	411.560	5.204	5.176	0.265
22:00:00	411.070	413.024	411.834	5.329	5.213	0.268
23:00:00	408.332	410.519	409.126	5.231	5.146	0.281
00:00:00	410.307	412.281	410.853	5.262	5.145	0.279
01:00:00	411.576	413.289	412.188	5.250	5.218	0.293
02:00:00	412.192	413.991	412.932	5.090	5.208	0.284
03:00:00	413.674	415.389	414.405	5.291	5.204	0.291
04:00:00	409.548	411.116	410.129	5.228	5.201	0.281
05:00:00	412.355	413.864	412.991	5.064	5.333	0.266
06:00:00	408.808	410.561	409.646	5.139	5.307	0.267
07:00:00	409.937	411.608	410.685	0.999	2.252	0.250

kW	kVA	kVAR	pf
1.96	1.10	1.10	0.07
1.94	3.53	2.95	0.55
1.91	3.52	2.96	0.54
1.94	3.57	3.00	0.54
1.89	3.50	2.95	0.54
1.92	3.51	2.93	0.55
1.95	3.52	2.93	0.55
1.92	3.47	2.89	0.55
1.96	3.57	2.99	0.55
1.93	3.50	2.92	0.55
1.98	3.49	2.88	0.57
1.97	3.48	2.87	0.57
1.91	1.08	1.08	0.11

$$1 \text{ lamp} = 150\text{W}$$

$$1 \text{ feeder pillar} = 13 \text{ lamps}$$

$$\text{TCL} = \text{Connected load} \times \text{unit of lamps}$$

$$= 150\text{W} \times 13$$

$$= 1.95\text{kW}$$

$$\text{MD} = \text{TCL} \times \text{Diversity Factor (DF)}$$

$$[\text{DF} = 1 \text{ (outdoor lighting)}]$$

$$= 1.95 \text{ kW} \times 1$$

$$= 1.95 \text{ kW}$$

TABLE II. DATA OF ACTUAL MD AND PREDICTION MD OF HPS STREETLIGHT

Time	Real MD (kW) HPS	Prediction MD (kW) HPS
19:00:00	1.9643	1.9554
20:00:00	1.9475	1.9382
21:00:00	1.9426	1.9322
22:00:00	1.9611	1.9605
23:00:00	1.984	1.9947
0:00:00	1.979	1.9846
1:00:00	1.9503	1.9379
2:00:00	1.928	1.9378
3:00:00	1.9215	1.9361

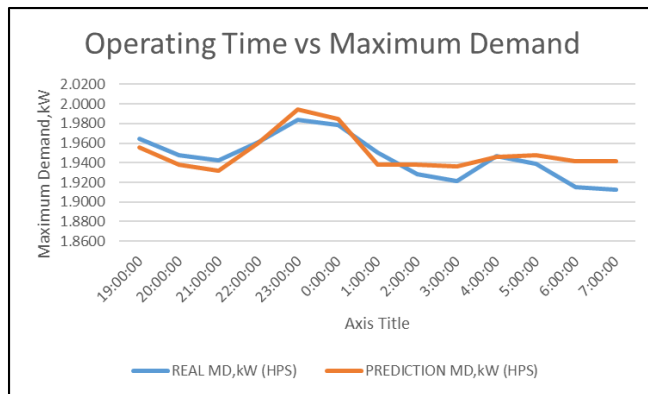


Fig. 7. The Operating Time vs. Maximum Demand Chart of HPS Streetlight

B. Light Emitting Diode (LED) Streetlight

The actual data from the data logger shows LED streetlight consumption, and it shows the data of voltage, current, power consumption, and power factor. By referring to calculation, the total maximum demand has been proved, which is 0.32kW. The actual data from the logger shows the actual maximum demand is lower than the actual utilization, which is around 0.23kW. Table III and IV and Fig. 8 show the plot of actual maximum demand data and prediction of maximum demand by ANN. The graph in Fig. 8 shows the actual maximum demand is ahead for 40 minutes earlier. While for 40 minutes and above, the predicted maximum demand is ahead of the actual maximum demand data. It can be concluded that the prediction of maximum demand is ahead of actual maximum demand.

TABLE III. DATA OF LED STREETLIGHT

Time	Volt	Volt	Volt	Cur	Cur	Cur
	Avg	Avg	Avg	Avg	Avg	Avg
10:30	237.92	238.63	238.88	0.363	0.501	0.532
10:40	237.68	238.30	238.62	0.364	0.501	0.531
10:50	237.42	237.97	238.44	0.364	0.501	0.530
11:00	238.66	239.23	239.62	0.362	0.499	0.526
11:10	236.95	237.50	237.86	0.364	0.501	0.527
11:20	236.75	237.37	237.79	0.363	0.501	0.527
11:30	236.39	237.05	237.46	0.364	0.501	0.527
11:40	238.12	238.70	239.13	0.361	0.498	0.523
11:50	236.59	239.33	239.70	0.360	0.493	0.522

		kW	kVA	kVAR	pf
A	Min	0.00	0.00	0.00	--
	Avg	0.05	0.09	0.07	0.63
	Max	0.06	0.09	0.07	--
B	Min	0.00	0.03	0.03	--
	Avg	0.09	0.12	0.08	0.72
	Max	0.09	0.12	0.08	--
C	Min	0.08	0.12	0.09	--
	Avg	0.08	0.13	0.09	0.66
	Max	0.08	0.13	0.09	--
Total	Avg	0.22	0.38	0.31	0.58

1 lamp = 40W

1 feeder pillar = 8 lamps

TCL = Connected load x unit of lamps

= 40W x 8

= 0.32kW

MD = TCL x Diversity Factor (DF)

[DF = 1 (outdoor lighting)]

= 0.32 kW x 1

= 0.32 kW

TABLE IV. DATA OF ACTUAL MD AND PREDICTION MD OF LED STREETLIGHT

Time	Real MD (kW) LED	Prediction MD (kW) LED
10:30:00	0.2300	0.227
10:40:00	0.2300	0.2254
10:50:00	0.2300	0.2279
11:00:00	0.2300	0.2290
11:10:00	0.2200	0.2291
11:20:00	0.2200	0.2252
11:30:00	0.2200	0.2255
11:40:00	0.2200	0.2255
11:50:00	0.2200	0.2280

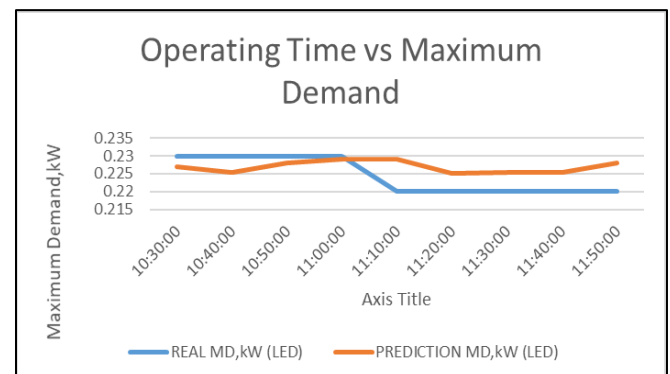


Fig. 8. The Operating Time vs. Maximum Demand Chart of HPS Streetlight

C. Comparison of HPS and LED Streetlight

TABLE V. DATA OF ACTUAL MD AND PREDICTION MD OF HPS AND LED STREETLIGHT

Real MD (kW) HPS	Prediction MD (kW) HPS	Real MD (kW) LED	Prediction MD (kW) LED
1.9426	1.9322	0.2300	0.2270
1.9611	1.9605	0.2300	0.2254
1.9840	1.9947	0.2300	0.2279
1.9790	1.9846	0.2300	0.2290
1.9503	1.9379	0.2200	0.2291
1.9280	1.9378	0.2200	0.2252
1.9215	1.9361	0.2200	0.2255
1.9470	1.9462	0.2200	0.2255
1.9385	1.9477	0.2200	0.2280

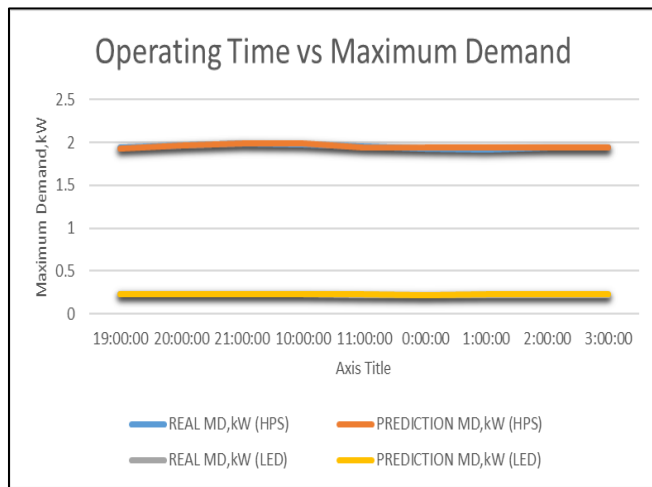


Fig. 9. The Actual and Predicted Maximum Demand of HPS and LED Streetlight

Fig. 9 shows the actual and predicted maximum demand between HPS and LED streetlights by referring to the data in Table V. The maximum demand for the HPS streetlight is ahead of the LED streetlight. It shows a significant difference in power consumption between these two types of streetlights. As stated in the calculation, eight lamps of LED streetlights had been declared because the result obtained by the logger is for one feeder pillar, which is eight lamps to compare actual and predicted maximum demand.

TABLE VI. TARIFF C1 – MEDIUM VOLTAGE GENERAL COMMERCIAL TARIFF [12].

For each kilowatt of maximum demand per month	30.3 RM/kW
For all kWh	36.5 Sen/kWh

By referring to TNB tariff C1 in Table VI, the cost analysis between two types of streetlights has been made for eight lamps of the streetlight. The eight lamps of HPS and LED streetlights were analysed by referring to Table VI. The number of lamps should be evenly equal to make it easier to compare.

TABLE VII. DATA OF ACTUAL MD AND PREDICTION MD OF HPS AND LED STREETLIGHT

High-Pressure (HPS) Streetlight	Sodium	Light Emitting Diode (LED) Streetlight
1 lamp = 150W 8 lamps = 1.2kW		1 lamp = 40W 8 lamps = 0.32kW
MD (1 day) = 1.2kW x 1 x 12 hours = 14.4kWh		MD (1 day) = 0.32kW x 1 x 12 hours = 3.84kWh
MD (30 days) = 14.4kW x 30 days = 432kWh		MD (30 days) = 3.84kW x 30 days = 115.2kWh
Cost for a month: -		Cost for a month: -
For each kilowatt of maximum demand per month:		For each kilowatt of maximum demand per month:
= 30.3 RM/kW x 14.4kW = RM 436.32		= 30.3 RM/kW x 3.84kW = RM 116.352
For all kWh: = 36.5 Sen/kWh x 432kWh = RM 157.68		For all kWh: = 36.5 Sen/kWh x 115.2kWh = RM 42.048
Total = RM 594.00		Total = RM 158.40
For a year = RM594.00 x 12 months = RM 7128.00		For a year = RM158.40 x 12 months = RM 1900.80

Table VII shows that the cost analysis for a month of HPS streetlight is RM594.00 while the LED streetlight is RM158.40. For a month, the difference in the cost is RM435.60. The HPS streetlights cost around RM7128 for a year, while the LED streetlights only cost RM1900.80. Moreover, for a year, the difference is about RM5227.20. It is proved that LED streetlights consume low power consumption at a low cost of billing charges.

TABLE VIII. LUMINOUS FLUX OF HPS AND LED STREETLIGHT

Distance from the pole of streetlight (meter)	Luminous flux of High-Pressure Sodium streetlight (lux)	Luminous flux of Light Emitting Diode streetlight (lux)
1	26	13
2	29	19
3	30	34
4	34	45
5	39	47
6	39	49
7	37	46

Finally, a comparison of luminous flux has been made, as shown in Table VIII. The luminous flux was taken by using a lux meter. The first distance, which is 1 meter from the pole, shows the difference is about 13 lux between the two types of streetlights where the HPS streetlight is higher than the LED streetlights. Besides, further distance from the pole shows that the LED streetlights are the higher value of the lux. The result in Table VIII proved that the luminous flux of the LED streetlights has higher lux compared to the HPS streetlights.

Moreover, having the higher lux, the LED streetlights have a wide range of colour temperature of 2200K-6000K compared to the HPS streetlights where the colour temperature is around 2200K only [2]. It has proved that LED streetlights have better lighting performance compared to HPS streetlights.

IV. CONCLUSION AND RECOMMENDATION

The conclusion that can be made for this project, the consumed energy by streetlight systems in a part of the area in UiTM CPP is relatively high based on data collected. The result by logger obtained has been visualized in the results section where the objective of creating correlations of streetlight system in UiTM CPP has been achieved.

The Artificial Neural Network (ANN) is a suitable method to predict the total maximum demand in a short period for this streetlight system information modelling and management for energy reduction. From the result, the prediction values of maximum demand are ahead of the actual maximum demand values. The second objective is to predict the total maximum demand by using ANN. Next, the usage of HPS streetlights consumed a lot of energy; thus, replacing HPS streetlights with LED is reasonable based on the results obtained, which are the comparison of energy consumed, cost analysis, and luminous flux.

For recommendation in the future, the information modelling of the streetlight systems should consider all the streetlights in the UiTM CPP area. By implementing this system, there will be a vast cost down. This study only takes a small case, which is one feeder pillar for each type of streetlight. In the future, the data collection should consider for overall streetlight systems in UiTM CPP.

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The Smart Street Lighting System Based on NB-IoT

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Abstract—With the development of communication technology, in order to meet the smart city's demand for intelligent infrastructure, the smart street lighting system based on NB-IoT is an effective method to realize the goal. This paper elaborates on the smart street lighting system's architecture, including the perception and control layer, the transport layer, the platform layer and the application layer. In addition, the actual system is built to verify. This system uses STM32 to build the perception and controller, and relies on China Telecom's NB-IoT communication network for the data transmission, finally using the IoT management platform-Huawei OceanConnect- to implement the connection management, the device management, the data calculation and the remote control functions. This system verifies the feasibility of the smart street lighting system based on NB-IoT technology, which provides the reference for the promotion and application. It is expected to provide intelligent, diversified, efficient and economical services for the smart cities.

Keywords—smart street lighting system, NB-IoT, IoT, smart city

I. INTRODUCTION

The National “13th Five-Year Plan” explicitly proposes construction of the “Smart City”, which focuses on intelligent infrastructure, facilitation of public services and refined social governance, making full use of modern information technology and big data[1]. The smart city is a solution for the urban development problems such as urban traffic congestion, excessive city consumption, urban environmental pollution, which is new concepts and models for the sustainable urban development [2-3]. The street light is one of the city's basic public facilities, which is widely distributed in cities. Additionally, it is with precise

geographical position, complete power supply network and reliable communication. The street lighting system is an excellent entrance to the smart city's internet of things, which can be used as a vehicle for the perception of the city's internet of things[4]. Therefore, the developed smart street lighting system is a typical application of the smart city. The intelligent operation and management of the street light is an effective means to realize smart city. The single control and intelligent adjustment strategy of the street light by IoT technology can realize the refined, intelligent and visual management of the street light, which can achieve a good results in energy saving, efficient operation and diversified services.

Compared with the traditional street light, smart street light has advanced communication technology, control technology and flexible management. The control mode of traditional street light adopts a group control mode through a fixed on/off light time or overall control based on the area, and it does not have single light control function. Moreover, the cloud server cannot collect real-time status information of the light poles, and it cannot monitor whether a single light is faulty, which is difficult to achieve refined management. However, the smart street light is equipped with a node controller on every light post, and all the street lights are connected to the network[5]. The status information is uploaded to the cloud server in real time. The cloud management system can realize independent control for the each street light according to the status of the light poles, which can reduce energy wastage due to manual mistake[6] and achieve efficient and remote management. In addition, compared with the traditional street light based on time and region, the smart street light has a more flexible control strategy, which can adjust the brightness of street light according to illumination, season and basic traffic conditions[7-8]. Moreover, based on the single illumination of the traditional street light, the modern smart street lights are integrated into urban management service such as wireless communication, video surveillance, information release, environmental monitoring, automobile charging

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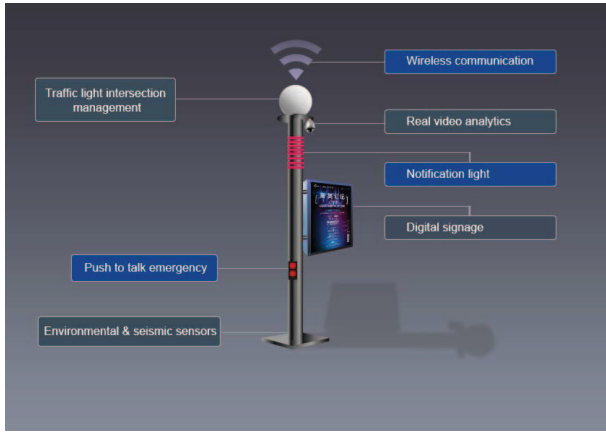


Fig.1. The modern smart street light

piles and other functions. The modern smart street light is shown in Fig. 1.

In recent years, the domestic pilot application in smart cities such as Beijing, Shanghai, Nanjing and Shenzhen in recent years have not only effectively improved the management and control level of urban lighting, further saved lighting electricity, but also provided important technical support for the informatization and intelligence of urban management and service [9-11]. This article mainly focuses on the design and implementation of smart lighting subsystems of smart street light. The research on the integrated smart street lighting system will be detailedly introduced in future articles.

As a typical application of IoT, the smart street lighting system is generally divided into the perception layer, the transmission layer, the platform layer and the application layer. The transmission layer is the key to the application of the smart street light, so the choice of the communication method is crucial [12-13]. At present, the communication methods used by smart street light mainly include PLC, Zigbee, LoRa and NB-IoT. The smart street light system based on NB-IoT provides an end-to-end, efficient, flexible and secure management solution. The NB-IoT network based on licensed spectrum has strong anti-jamming capability, high data security, convenient technical service, complete infrastructure and easy promotion, which provides a reliable communication technology solution for the construction of the smart street lighting system. In addition, NB-IoT technology has the advantages of wide coverage, mass connectivity and high reliability [14], which can meet the requirements of wide deployment, large number of rods and high data reliability for urban street lights. At the same time, the unified IoT platform provided by telecom operator can solve the compatibility issues of different terminal vendor protocols, which easily makes the terminal connect to the cloud platform and achieve the large-scale connection of the smart street light.

The organization of the paper is as follows: Section II introduces the overall framework of the smart street lighting system based on NB-IoT, and describes in detail the perception layer, transmission layer, platform layer and application layer. Section III mainly elaborates the actual smart street lighting system and relevant verification. Finally, the conclusions are drawn, and the future research directions are given in Section IV.

II. THE SMART STREET LIGHTING SYSTEM

The smart street lighting system based on NB-IoT technology applies the high-security, wide-coverage, large-connection and lower power wide-area communication technology to achieve real-time acquisition of the street light information, equipment condition monitoring, control command delivery and other functions, which realizes the refined management, the efficient maintenance, the convenience services of the street light.

A. The system architecture

The architecture of smart street light system based on NB-IoT is shown in Fig. 2. It mainly includes the perception layer, transmission layer, platform layer and application layer.

In the perception layer, there is the single light monitoring and controller with NB-IoT module and various sensors, which can be used to control the opening and closing of the street light and collect street light information, then communicating with the wireless network through the NB-IoT module.

In the transmission layer, it adopts the advanced NB-IoT communication technology, which has the perfect network infrastructure. Additionally, the telecommunications operators operate to ensure network stability and data security.

In the platform layer, it can provide access management, device management, application enablement and service analysis capabilities for smart street light [15]. That can achieve the unified access and management of different types of terminal based on unified protocol and interfaces.

In the application layer, it is the smart street lighting integrated management platform deployed in the cloud. It can realize the street light inspection, fault active alarm, free grouping, remote data query and control functions based on the big data and cloud computing technology.

B. The single light monitoring and controller

The single streetlight monitor and controller is the terminal node of the smart street lighting system, which can collect the real-time status information about the operation of

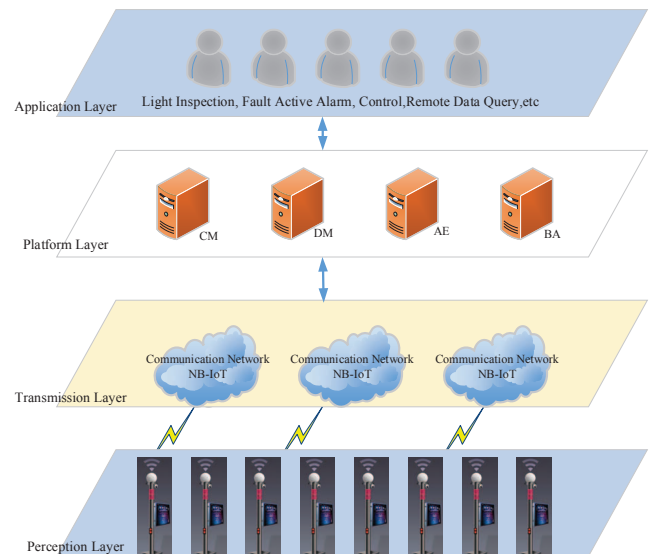


Fig.2. The architecture of the smart street lighting system

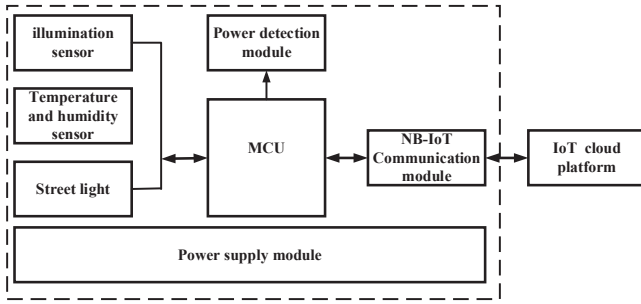


Fig.3. The architecture of terminal

the street light and send it to the cloud management platform. It also receive the control instructions from the management platform and perform the corresponding control operations in order to achieve the single street light or a group of street lights for switch control and lighting brightness adjustment.

Generally, the single light monitor and controller is mainly composed of various types of sensors(light sensor, temperature and humidity sensor), power detection module, microprocessor, NB-IoT wireless communication and power supply module[16]. The monitoring and controller acquires the temperature of the street light and the ambient light illumination. The power detection module is used to monitor the power supply status and the working power of the street light. The above information is sent from the NB-IoT module to the smart street light integrated management platform via the operator network. The platform can performs data storage, data calculation, data analysis and intelligent management of street lights. The architecture of terminal is shown in Fig.3.

C. NB-IoT network

The system is based on the NB-IoT network of telecom operators, featuring low speed, low power consumption, low cost and wide coverage. NB-IoT adopts the narrow-band communication technology. The narrow-band signal has high power spectral density, which can significantly improve the signal's anti-interference ability[17]. NB-IoT supports stand-alone, Guard-band and in-band deployment. The characteristics of NB-IoT technology is shown in table I. It has strong compatibility with existing mobile networks. In addition, NB-IoT optimizes the existing 4G network[18] to achieve cost savings and lower terminal power consumption. The NB-IoT network infrastructure is perfect, which can reduce network construction and maintenance costs.

TABLE I. THE CHARACTERISTICS OF NB-IoT TECHNOLOGY

Hierarchy	Technical features	
Physical layer	Uplink	BPSK or QPSK modulation
		S Single subcarriers, subcarrier spacing are 3.75kHz and 15kHz, Transmission rate is 160kbit/s~200kbit/s
	Downlink	D Multiple subcarriers, subcarrier spacing is 15kHz, transmission rate is 160kbit/s~250kbit/s
		M OFDMA, Subcarrier spacing is 15kHz, transmission rate is 160~250kbit/s
Above the physical layer	LTE-based protocols	
Core Network	Based on S1 interface	

The NB-IoT network has the following major features to meet the network performance requirements of the smart street lighting system.

- The broad coverage. At the same frequency band, NB-IoT gains 20 dB more than the existing network, and the coverage area increases by 100 times.
- The mass connection. NB-IoT can support 100,000 connections in one sector.
- The low power consumption. NB-IoT terminal module can be up to 10 years in standby time.
- The low cost. The cost can be controlled below 5 USD after the commercialization of the module.

The NB-IoT network consists of NB-IoT terminals , NB-IoT base station, NB-IoT packet core network, NB-IoT connection management platform and vertical industry center, as shown in the Fig. 4.

D. IoT platform

IoT platform has the functions of connection management(CM), device management(DM), application enable(AE) and business analysis(BA). It supports rapid access to a variety of IoT terminal and rapid integration of applications in various industries, so that users can focus on business application and innovation.CM module provides connection status query and management functions such as connection awareness, connection diagnosis and connection control. DM module performs functions such as remote monitoring, setting adjustment, software upgrade, system upgrade and life cycle management on the terminal. AE module is a PaaS platform that provides application development and unified data storage. BA module provide the flexible and efficient data management, including big data analytics and machine learning capabilities, which can classify, analyze and provide visual analysis results for data and generate forecasts based on historical data. It provides users with predictive, cognitive or complex analytical business logic to meet the development needs of industry applications. The architecture of IoT platform is shown in Fig.5.

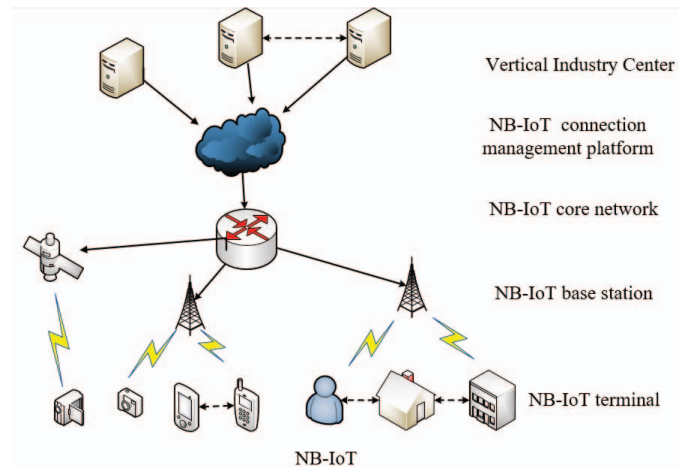


Fig.4. The network architecture of NB-IoT

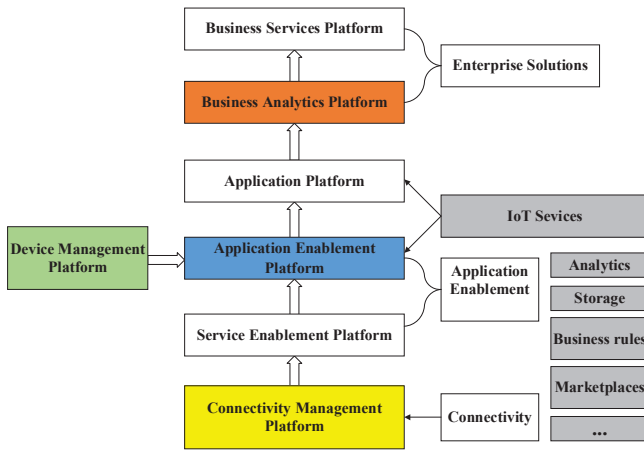


Fig.5. The architecture of IoT platform

E. The smart streetlight management platform

The smart streetlight management platform establishes connection with the monitor and controller of the single street light through the NB-IoT network and IoT platform of telecom operator. It calculates and analyzes the real-time data of the street lights, and intelligently controls the street lights such as adjustment control and fault detection according to management control strategies. In addition, the integrated management platform can use GIS to accurately locate the street lights, make statistics of the status, power consumption and other information of street lights, and save data in the cloud. Also, the management platform can achieve street light inspection, fault active alarm, remote meter reading and other functions. It can upgrade single lamp monitor and controller software flexibly.

III. SYSTEM SETUP AND VERIFICATION

The system adopts China Telecom's NB-IoT network. The single light acquisition and control terminal takes the STM32L431 as the main control MCU. The STM32L431 uses the Cortex®-M4 as the core with ultra-low power consumption, which has rich hardware resources and peripheral interfaces. The terminal uses TAOS TSL2561 for light intensity monitoring and the SHT21 to monitor the ambient temperature in real time. The acquisition and Control Terminal Integrates Shanghai Mobile's BC95-HB Low-Power NB-IoT Module and the module's receiving sensitivity is -129dBm. The signal of based on the 850MHz frequency band telecommunications NB-IoT network has strong penetrability and a great advantage in coverage, which can meet the reliability and security of data transmission in urban environments. The module embeds network service protocol stacks such as UDP, CoAP and LwM2M. It is commonly used in smart cities, wireless meter reading and environmental monitoring.

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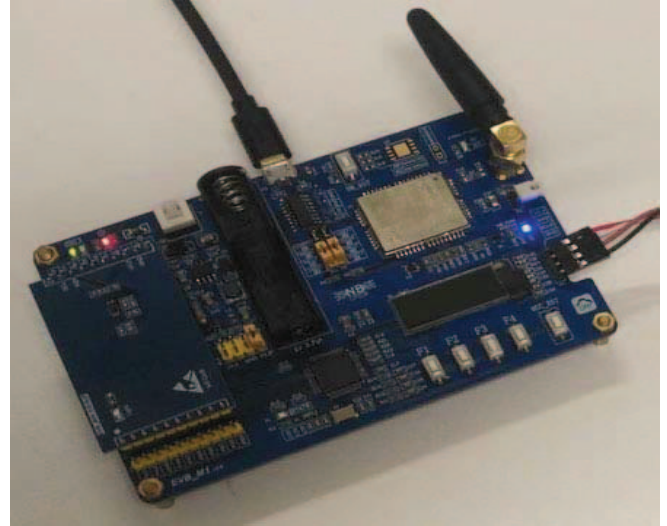
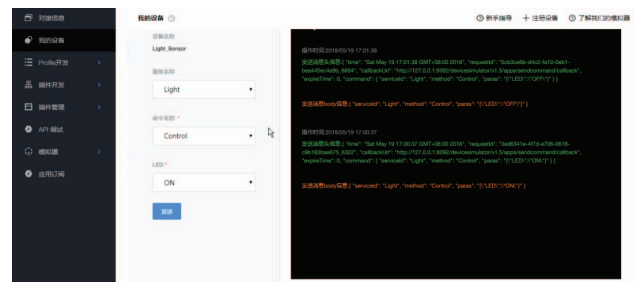
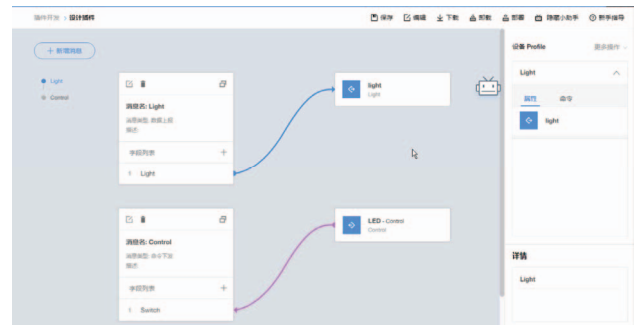


Fig.6. The single light monitoring and control terminal



(a)The light control



(b) The design of plug-in components

设备	数据项	数据
Light	[{"light": "200"}]	200
Light	[{"light": "200"}]	200
Light	[{"light": "200"}]	200
Light	[{"light": "200"}]	200
Light	[{"light": "200"}]	200
Light	[{"light": "200"}]	200
Light	[{"light": "200"}]	200
Light	[{"light": "200"}]	200

(c)The historical data

Fig.7. The Oceanconnect platform

This system uses Huawei's open ecosystem management platform-OceanConnect- based on IoT, cloud computing and big data technologies. OceanConnect provides more than 100 kinds of open APIs and serialized Agents to achieve two-way communication between data acquisition and command delivery between devices and platforms[19]. The NB-IoT module on the terminal side communicates with the OceanConnect management platform through the CoAP protocol. The payload in CoAP packets is packaged using CJSON. The part of the Oceanconnect platform is shown in Fig.7.

IV. CONCLUSION AND DISCUSSION

This paper describes the design of the smart street lighting system based on NB-IoT technology, and analyzes the smart street lighting system structure, NB-IoT technology, IoT platform and so on. The verification shows that the smart street lighting system, combined with the NB-IoT network, IoT platform, and cloud computing of telecom operators, can realize single light monitoring and other functions by constructing a variety of management platforms, which meets the needs of IoT intelligence and interconnected. In addition, based on the basic platform of the smart street lights, the system management can use the basic data to carry out street lamp failure alarms, remote meter reading, group management, map guidance, street lamp inspection, and visual analysis. This paper does not introduce the traditional control strategy of the smart street light. The actual application of the control strategy needs to be further tested. The new control strategy needs further research in order to obtain the optimal control strategy and minimize the city energy consumption. This article only introduces the smart lighting subsystem of the smart street light. But the modern smart street light system also integrates the micro base station, multimedia screen, information exchange screen, charging pile and other carriers. Next, we will study the street lamp poles and integrated management platform to maximize the role of the smart street lighting system as a basic network platform for smart cities. Finally it provides diversified, intelligent, economical and efficient services for smart cities.

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The Smart Street Lighting System Based on NB-IoT

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Abstract—With the development of communication technology, in order to meet the smart city's demand for intelligent infrastructure, the smart street lighting system based on NB-IoT is an effective method to realize the goal. This paper elaborates on the smart street lighting system's architecture, including the perception and control layer, the transport layer, the platform layer and the application layer. In addition, the actual system is built to verify. This system uses STM32 to build the perception and controller, and relies on China Telecom's NB-IoT communication network for the data transmission, finally using the IoT management platform-Huawei OceanConnect- to implement the connection management, the device management, the data calculation and the remote control functions. This system verifies the feasibility of the smart street lighting system based on NB-IoT technology, which provides the reference for the promotion and application. It is expected to provide intelligent, diversified, efficient and economical services for the smart cities.

Keywords—smart street lighting system, NB-IoT, IoT, smart city

I. INTRODUCTION

The National “13th Five-Year Plan” explicitly proposes construction of the “Smart City”, which focuses on intelligent infrastructure, facilitation of public services and refined social governance, making full use of modern information technology and big data[1]. The smart city is a solution for the urban development problems such as urban traffic congestion, excessive city consumption, urban environmental pollution, which is new concepts and models for the sustainable urban development [2-3]. The street light is one of the city's basic public facilities, which is widely distributed in cities. Additionally, it is with precise

geographical position, complete power supply network and reliable communication. The street lighting system is an excellent entrance to the smart city's internet of things, which can be used as a vehicle for the perception of the city's internet of things[4]. Therefore, the developed smart street lighting system is a typical application of the smart city. The intelligent operation and management of the street light is an effective means to realize smart city. The single control and intelligent adjustment strategy of the street light by IoT technology can realize the refined, intelligent and visual management of the street light, which can achieve a good results in energy saving, efficient operation and diversified services.

Compared with the traditional street light, smart street light has advanced communication technology, control technology and flexible management. The control mode of traditional street light adopts a group control mode through a fixed on/off light time or overall control based on the area, and it does not have single light control function. Moreover, the cloud server cannot collect real-time status information of the light poles, and it cannot monitor whether a single light is faulty, which is difficult to achieve refined management. However, the smart street light is equipped with a node controller on every light post, and all the street lights are connected to the network[5]. The status information is uploaded to the cloud server in real time. The cloud management system can realize independent control for the each street light according to the status of the light poles, which can reduce energy wastage due to manual mistake[6] and achieve efficient and remote management. In addition, compared with the traditional street light based on time and region, the smart street light has a more flexible control strategy, which can adjust the brightness of street light according to illumination, season and basic traffic conditions[7-8]. Moreover, based on the single illumination of the traditional street light, the modern smart street lights are integrated into urban management service such as wireless communication, video surveillance, information release, environmental monitoring, automobile charging

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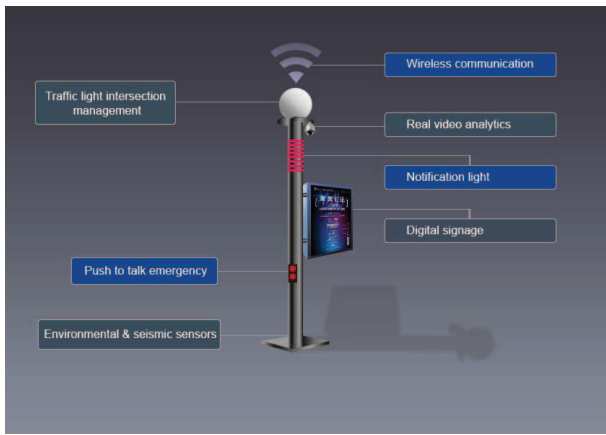


Fig.1. The modern smart street light

piles and other functions. The modern smart street light is shown in Fig. 1.

In recent years, the domestic pilot application in smart cities such as Beijing, Shanghai, Nanjing and Shenzhen in recent years have not only effectively improved the management and control level of urban lighting, further saved lighting electricity, but also provided important technical support for the informatization and intelligence of urban management and service [9-11]. This article mainly focuses on the design and implementation of smart lighting subsystems of smart street light. The research on the integrated smart street lighting system will be detailedly introduced in future articles.

As a typical application of IoT, the smart street lighting system is generally divided into the perception layer, the transmission layer, the platform layer and the application layer. The transmission layer is the key to the application of the smart street light, so the choice of the communication method is crucial [12-13]. At present, the communication methods used by smart street light mainly include PLC, Zigbee, LoRa and NB-IoT. The smart street light system based on NB-IoT provides an end-to-end, efficient, flexible and secure management solution. The NB-IoT network based on licensed spectrum has strong anti-jamming capability, high data security, convenient technical service, complete infrastructure and easy promotion, which provides a reliable communication technology solution for the construction of the smart street lighting system. In addition, NB-IoT technology has the advantages of wide coverage, mass connectivity and high reliability [14], which can meet the requirements of wide deployment, large number of rods and high data reliability for urban street lights. At the same time, the unified IoT platform provided by telecom operator can solve the compatibility issues of different terminal vendor protocols, which easily makes the terminal connect to the cloud platform and achieve the large-scale connection of the smart street light.

The organization of the paper is as follows: Section II introduces the overall framework of the smart street lighting system based on NB-IoT, and describes in detail the perception layer, transmission layer, platform layer and application layer. Section III mainly elaborates the actual smart street lighting system and relevant verification. Finally, the conclusions are drawn, and the future research directions are given in Section IV.

II. THE SMART STREET LIGHTING SYSTEM

The smart street lighting system based on NB-IoT technology applies the high-security, wide-coverage, large-connection and lower power wide-area communication technology to achieve real-time acquisition of the street light information, equipment condition monitoring, control command delivery and other functions, which realizes the refined management, the efficient maintenance, the convenience services of the street light.

A. The system architecture

The architecture of smart street light system based on NB-IoT is shown in Fig. 2. It mainly includes the perception layer, transmission layer, platform layer and application layer.

In the perception layer, there is the single light monitoring and controller with NB-IoT module and various sensors, which can be used to control the opening and closing of the street light and collect street light information, then communicating with the wireless network through the NB-IoT module.

In the transmission layer, it adopts the advanced NB-IoT communication technology, which has the perfect network infrastructure. Additionally, the telecommunications operators operate to ensure network stability and data security.

In the platform layer, it can provide access management, device management, application enablement and service analysis capabilities for smart street light [15]. That can achieve the unified access and management of different types of terminal based on unified protocol and interfaces.

In the application layer, it is the smart street lighting integrated management platform deployed in the cloud. It can realize the street light inspection, fault active alarm, free grouping, remote data query and control functions based on the big data and cloud computing technology.

B. The single light monitoring and controller

The single streetlight monitor and controller is the terminal node of the smart street lighting system, which can collect the real-time status information about the operation of

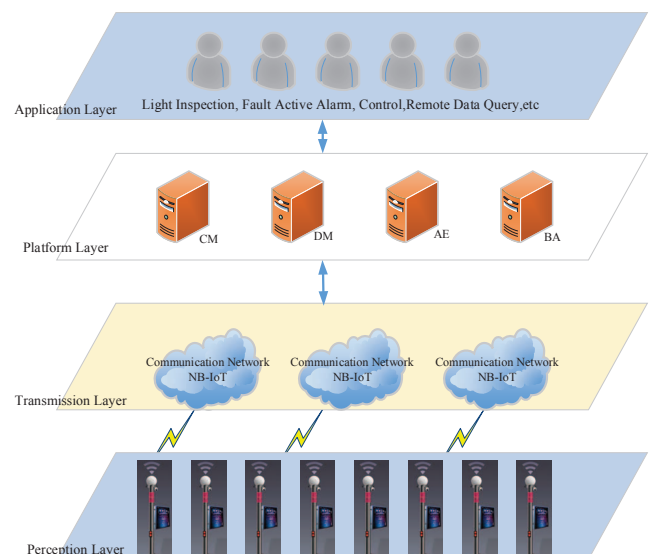


Fig.2. The architecture of the smart street lighting system

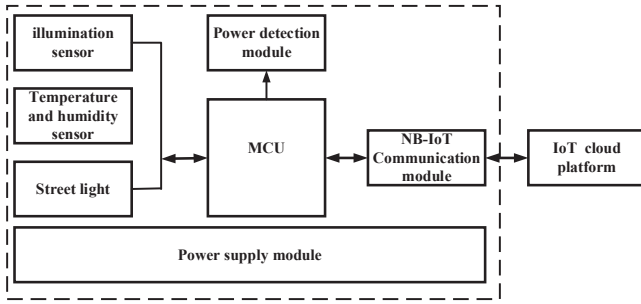


Fig.3. The architecture of terminal

the street light and send it to the cloud management platform. It also receive the control instructions from the management platform and perform the corresponding control operations in order to achieve the single street light or a group of street lights for switch control and lighting brightness adjustment.

Generally, the single light monitor and controller is mainly composed of various types of sensors(light sensor, temperature and humidity sensor), power detection module, microprocessor, NB-IoT wireless communication and power supply module[16]. The monitoring and controller acquires the temperature of the street light and the ambient light illumination. The power detection module is used to monitor the power supply status and the working power of the street light. The above information is sent from the NB-IoT module to the smart street light integrated management platform via the operator network. The platform can performs data storage, data calculation, data analysis and intelligent management of street lights. The architecture of terminal is shown in Fig.3.

C. NB-IoT network

The system is based on the NB-IoT network of telecom operators, featuring low speed, low power consumption, low cost and wide coverage. NB-IoT adopts the narrow-band communication technology. The narrow-band signal has high power spectral density, which can significantly improve the signal's anti-interference ability[17]. NB-IoT supports stand-alone, Guard-band and in-band deployment. The characteristics of NB-IoT technology is shown in table I. It has strong compatibility with existing mobile networks. In addition, NB-IoT optimizes the existing 4G network[18] to achieve cost savings and lower terminal power consumption. The NB-IoT network infrastructure is perfect, which can reduce network construction and maintenance costs.

TABLE I. THE CHARACTERISTICS OF NB-IoT TECHNOLOGY

Hierarchy	Technical features	
Physical layer	Uplink	BPSK or QPSK modulation
		S Single subcarriers, subcarrier spacing are 3.75kHz and 15kHz, Transmission rate is 160kbit/s~200kbit/s
	Downlink	D Multiple subcarriers, subcarrier spacing is 15kHz, transmission rate is 160kbit/s~250kbit/s
		M OFDMA, Subcarrier spacing is 15kHz, transmission rate is 160~250kbit/s
Above the physical layer	LTE-based protocols	
Core Network	Based on S1 interface	

The NB-IoT network has the following major features to meet the network performance requirements of the smart street lighting system.

- The broad coverage. At the same frequency band, NB-IoT gains 20 dB more than the existing network, and the coverage area increases by 100 times.
- The mass connection. NB-IoT can support 100,000 connections in one sector.
- The low power consumption. NB-IoT terminal module can be up to 10 years in standby time.
- The low cost. The cost can be controlled below 5 USD after the commercialization of the module.

The NB-IoT network consists of NB-IoT terminals , NB-IoT base station, NB-IoT packet core network, NB-IoT connection management platform and vertical industry center, as shown in the Fig. 4.

D. IoT platform

IoT platform has the functions of connection management(CM), device management(DM), application enable(AE) and business analysis(BA). It supports rapid access to a variety of IoT terminal and rapid integration of applications in various industries, so that users can focus on business application and innovation. CM module provides connection status query and management functions such as connection awareness, connection diagnosis and connection control. DM module performs functions such as remote monitoring, setting adjustment, software upgrade, system upgrade and life cycle management on the terminal. AE module is a PaaS platform that provides application development and unified data storage. BA module provide the flexible and efficient data management, including big data analytics and machine learning capabilities, which can classify, analyze and provide visual analysis results for data and generate forecasts based on historical data. It provides users with predictive, cognitive or complex analytical business logic to meet the development needs of industry applications. The architecture of IoT platform is shown in Fig.5.

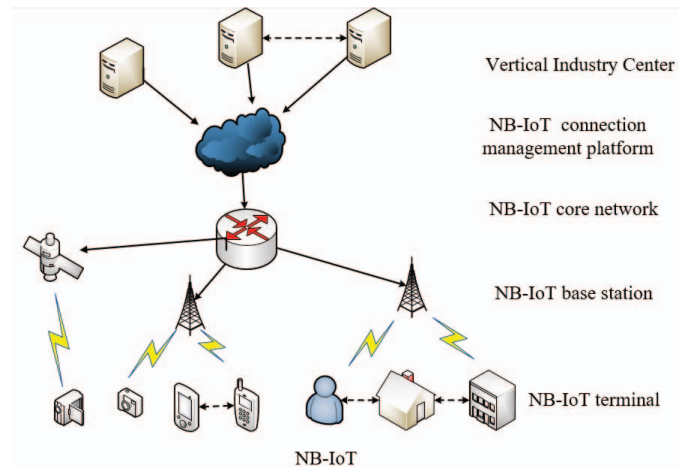


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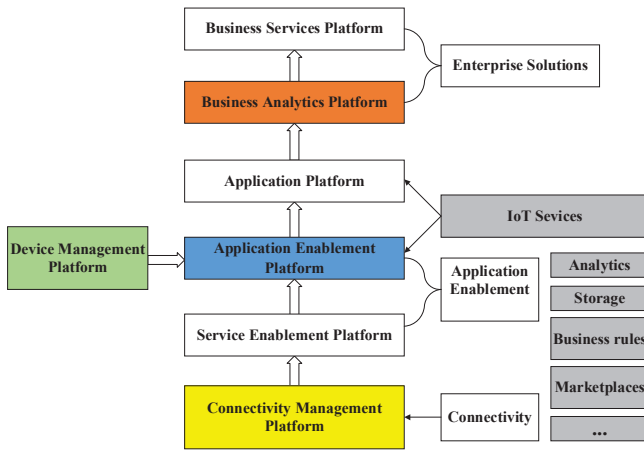


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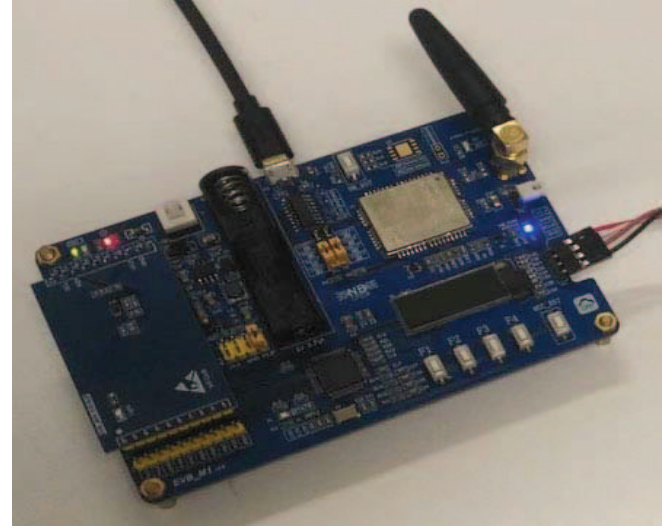
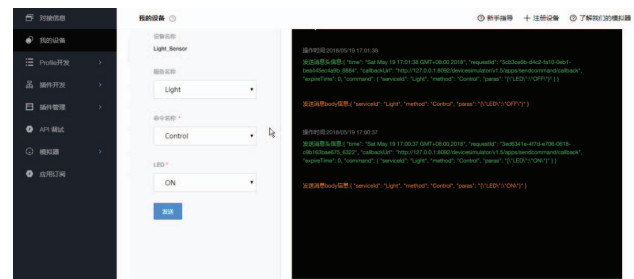
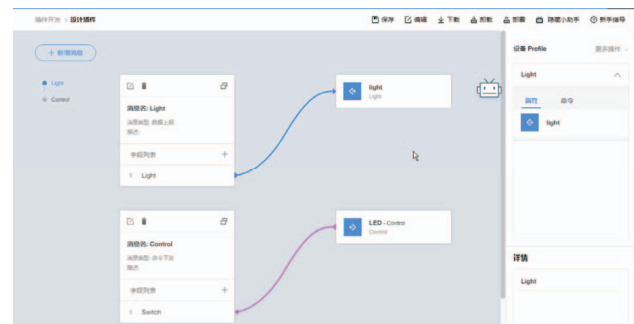


Fig.6. The single light monitoring and control terminal



(a)The light control



(b) The design of plug-in components

Device ID	Device Name	Device Type
Light	[\"light1-2017\"]	2018/05/16 11:17:40
Light	[\"light1-2017\"]	2018/05/16 11:17:40
Light	[\"light1-2017\"]	2018/05/16 11:17:40
Light	[\"light1-2017\"]	2018/05/16 11:17:40
Light	[\"light1-2017\"]	2018/05/16 11:17:40
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